# Home Network & Media Wall Plan

Revised Oct 20, 2026 · @emma · version 27 (revision 27: the dining cabinet is the plan, made for six with the chairs inside it; your rear speakers), drawn to your measurements

**What changed in v27: the dining cabinet is the plan, made for six.** You asked to use the cabinet as the main table, for six, with the chairs inside it. It now leads section 5: one oak cabinet under your shelf, a 180 × 90 table that swings down out of its bay, **six in comfort**, three a side, with the coffee table left where it is, and **all six chairs inside it** (four on the floor, two on the shelf, the seat pads beside them). The halves are shorter than v26's (90 instead of 112.5), so each is lighter and there's room over the folded table for a **linen flap**. **Step 11 of the build guide** is now the cabinet's build, in sixteen stages with every size, with two new drawings: the carcass from the front, and the table's halves from underneath. The buy list carries the cabinet; the flip-top stays in section 5 as the alternative. **Your rear speakers** stand behind the sofa (section 9; your photo will confirm the stands). The brick under the stairs is drawn painted, as it has a wall in front of it. On the plan page the 3D room opens on the cabinet, the flip-top's chair cupboard can come off the wall in Edit room, and Edit room clips things to walls more easily: a quarter turn keeps a piece against its wall, and **Back to the wall** pushes it there.
**v26: option C, the dining cabinet, and up to eight at the table.** A second way to build the stairs wall, beside the flip-top (which stays the plan): **one oak cabinet** under your shelf, with the table standing up in one bay and all eight chairs folded in the other, so nothing shows when it's packed away. Open, the table is 180 × 90: **eight at a squeeze, six in comfort**, the most this room takes (section 5 shows why a ninth won't fit). It costs more than the flip-top and the chair cupboard; it isn't in the totals unless you choose it. On the plan page, the 3D room's Tools panel is tidier (its title and buttons stay put, every tab opens at the top, and on a laptop it docks to the right with the room beside it); a **Hide labels** button takes the numbered stickers off the room; and Look and Extras have more to try.
**v25: Demma Home Improvements, and three lighting presets.** The plan page is a new app: your room in 3D is the home screen, with numbered stickers on it (tap one for its key facts, to open its chapter, or to fly there), one **Tools** button for every 3D control, and the plan, the build steps, the buy list, the ideas and the questions sliding up over the room. Questions are answered with a tap or a photo. **The lights** have three presets instead of six buttons: **Bright**, **Cosy** and **Film** (section 8). Dinner is now a rule (the table open lifts bar 2 over it) and daytime a switch, so they work with any preset; step 12 sets the three in the app. The plan itself doesn't change.
**v24: less to read, and a cleaner page.** The round numbers are out of the text (the history stays here and in the appendix), long captions fold behind **More**, and the notes on how the media wall's layout was chosen moved to the appendix. The plan page has a new look: each section in its own card, a contents list down the side on a computer (a row of chips on a phone) that follows where you are, a floating tab bar, clearer tables, and a footer with the downloads. The plan itself doesn't change.
**v23: more ways to build the chairs in, and the room as it really is. Chairs built into the table** (section 5): as well as the chair cupboard, four chairs can hang **behind the folded top** on a deeper backboard, out of sight, and a **fold-down oak seat** can go in the gap at the stair end, a fifth place that's always there. Both are options for you to choose, and both are in the fit proof. **Under the stairs** is a brick side wall with plasterboard beneath, as you said: drawn in brick, so chairs five and six go in the garage. **Your sofa's ottoman**, square, blue corduroy, the size of one of its seats, is on the rug beside the coffee table (and moves in Edit room). **Your sockets** are drawn where they are: the single in the corner in front of the left Xbox game drawer (the one the plan makes a double), a double behind the sofa's short side, below and left of the PC, which now powers the rear speakers (the 12-way strip is off the buy list), and a double behind the plant stand in the window corner, for your lamp and the grow lights. The floating **banister end at the top of the stairs** is gone. The plan page has a **new colour** and four **page looks** (Card Room, Oak & linen, Sofa blue, Plain) and larger text to choose from; the questions are a list now, with no answer boxes; and the 3D room has **six more suggestions** to switch on or off (twelve in all).

**What changed in v18: your plant stand and your record shelf, and a room you can rearrange. Your new plant stand** (98 × 29.5 × 158: two five-tier towers and an open middle) and **your vinyl stand** (70 × 32 × 78: two face-out bays on top, wire dividers below) are in, at the sizes on the listings you sent (v18.1), with two layouts to choose from (section 4): **A, recommended**, the plant stand in the window's west corner, where it gets daylight from the window beside it, and the vinyl stand on the chimney breast under the “now playing” ledge, a step from the turntable; or **B**, the other way round. Your red lamp moves to the sofa's end either way. **The PC's clear side** now shows its graphics card, cooler, memory and fans. In the 3D room, **Edit room** lets you drag things around like a dolls' house, turn them, recolour them, take them out and add new ones (an armchair, a bookcase, a floor plant and more), and paint a wall, the ceiling or the skirting by tapping it; the menus are shorter for it. **HD with shadows** no longer asks more of the graphics card than it has, and if a phone does give up, the room comes back in Fast. More ideas for later (appendix). Fit proof 129 checks, none fail.
**v17: the room finished round your notes. The sofa's open end** is now a proper ottoman end: the seat wraps round to the back as one long cushion and the frame stops 20 short, and the sofa stands 13 cm off the window wall, as in your photo. **The window** has its real deep sill, 22 cm into the reveal, with your plants along it (3D room), and the **curtains** now hang clear of it. **Even light by default:** every spot head points straight down, evenly spread, and a new **General** scene comes first; tilt a head only if its pool lands badly. Your **red lamp's top (green) head** lights the wall with your shelf. **The flip-top** is drawn as built: pegged breadboard ends and three brass hinges. **The stairs:** one wall all the way up in one colour, the living room ceiling's edge where it cuts off the view, and the banister on the room side only, as you said. **The PC** is drawn with its insides, and lights up in the 3D room (the "PC lit" button); the plan adds a **USB hub** by the sofa, a **remote power button** and a **dust filter** (section 6), and checks the soundbar against the TV's remote (section 3). Your **OBEGRÄNSAD** (905.262.63) is confirmed at IKEA's 52 × 36 × 6. The build guide PDF now **stands on its own**: each step has its own cut list and pictures. New **ideas**, off the plan and the costs, are in their own tab and in the 3D room's buttons, all off until you turn them on (appendix). Fit proof 129 checks, none fail.
**v16: the whole wall, redesigned round everything we now know. No dead space in the record bay:** with LPs only, bay 3 splits in two. Under the turntable, two **snug LP crates** (a 33 cm trough, the width record shops use) still hold about 180 LPs face-out; beside them a **slim column** of three drawers: **two face-out game drawers** (about 44 DVD-size cases, flipped like the records) and a drawer for the **pads and remotes** with a charging lead, so nothing lies on the worktop (section 4). **The table by the stairs is now the default**, so the chimney breast is free: it takes a **“now playing” oak ledge** beside the turntable for the sleeve of what's on and the next two, under the two framed LPs. **Smart colour strips** in the upright grooves and under the side-wall run, in the same app as the bars' bulbs, so one scene sets the whole wall. On your shelf by the stairs: your **IKEA OBEGRÄNSAD**, black, its 16 × 16 white LEDs drawn showing a pixel music note; **the laser is there but off**; the **plant hangs from the shelf's window end**; and the **hollow pipe boxing** in the window corner (about 23 × 22, floor to ceiling) is drawn in, the shelf stopping just short of it. **Your hats** are on your red lamp, as in your photo: the graduation cap on top of the pole, the navy cap with its poem hung below the top head. **The sofa's end towards the TV is open**: no arm, one long seat cushion wrapping round to the back, an ottoman-style end. Fit proof 129 checks, none fail. Earlier rounds are in the appendix.

# 1 · The plan on a page

Drawn to your measurements, in Card Room Green: the alcove with the turntable zone on the right and bay 3 split into the slim game column and the LP crates; the chimney breast free, with its “now playing” ledge (the table is by the stairs, behind this view); both spot bars. Full-resolution renders come with this plan.

**£3,438–6,733**
the whole plan, item by item in section 11; no electrician on top

**88 of 129**
fit checks pass outright; 31 are tight, 10 need a check on site (section 3)

**291 · 6**
LPs (191 in the wall, 100 in your vinyl stand) · seats at the table, the chairs kept inside it

| Area | Decision |
|---|---|
| Alcove | The TV between two oak uprights, the left one now a corner post on the side wall; on the right, the turntable on the worktop and the Series X on a shelf at 134; below, three 60 cm bays: the Xbox consoles, the PlayStations with the sub, and two snug LP crates beside a slim column of game and pad drawers. |
| Side wall | One oak run from the corner: a 7 cm ledge with four more LPs that sweeps out into the PC's shelf above the sofa's open end, a smart colour strip underneath, clear of the TV's sightlines. |
| Dining | The dining cabinet under your shelf by the stairs: one oak cabinet with the table and the six chairs inside it. The table swings down and opens to 180 × 90, six in comfort; packed away it's a flush run of oak. The chimney breast keeps a “now playing” ledge and two framed LPs. The flip-top (by the stairs or on the breast) is the alternative (section 5). |
| Records and games | About 291 LPs: 191 in the wall, every one face-out (180 in the LP crates, 8 on the two ledges, 3 on the “now playing” ledge), and 100 in your vinyl stand (50 face-out on top, 50 spine-up between its wires). Games: about 92 face-out or spine-up in the game drawers, and 32 PS1 games in the plinth drawer. |
| Heights | Worktop and table 76 · TV centre 116.5 · Series X shelf 134 · ledges and PC shelf 155.4 · high shelf 196, 39 under the 235 ceiling. |
| Colour and light | Card Room Green No. 79, colour-drenched. The ceiling LED strip, smart colour strips in the uprights and under the side run, a bias light, and two Wolston-style spot bars with smart colour bulbs on your two existing ceiling points; one app, one scene for the lot. |
| Power and network | Your own job: the side-wall socket becomes a double, left for the 10-gang (everything modern, the PC included), right for the retro set, switched at the wall. Deco A in bedroom 3, Cat6 down the airing cupboard. |

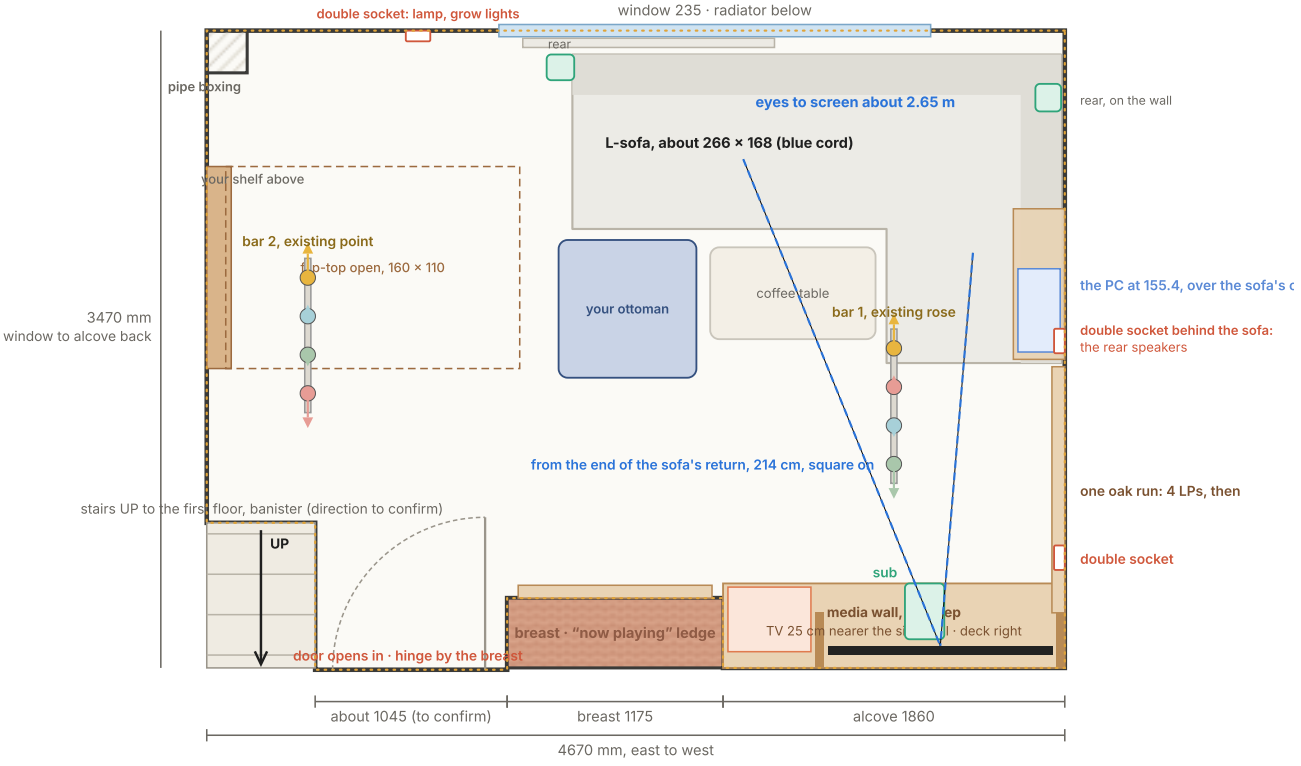
**To confirm (section 12):** the stairs wall's build · that you're swapping the light fittings yourselves · the sofa's return length · the shelf's height · the turntable's model · the fan stand's model · the rear speakers' stands · the ottoman's height · record counts · game counts · extra wired ports · the 10-gang · console model numbers · the pendant point · the window corner at floor level · the RetroTINK's size.

# 2 · The room

Everything is drawn from your hand-measured sketch; where the agent's floor plan disagrees, your tape wins. The L-sofa is drawn from your photo[24]: its long side ends in line with the breast's reveal (about 266 cm); the return along the side wall is a typical 168 until you measure it. Its end towards the TV is an **ottoman end** (no arm; the seat wraps round, the back stops 20 short) and it stands 13 cm off the window wall, clear of the curtains. The window has a **deep sill**, 22 cm into the reveal at 88 from the floor, with your plants along it.

### The living room from above, as you measured it

To scale, north at the top. Media wall in the alcove, one oak run on the side wall over the sofa, the flip-top table by the stairs (the default), two spot bars




Ceiling 2350 mm. Sofa from your notes (long side to 40 cm short of the window's end; return 168 assumed). Both spot bars run from the TV towards the window.

The whole room in 3D, from above the window side, the dining cabinet packed away by the stairs: your blue sofa with its open end, the rug, the bars running towards the window, the “now playing” ledge on the breast. On the plan page you can walk round it, switch the table's wall, slide every light, colour the smart bulbs, paint the walls, ceiling and skirting separately, and save any view in HD.

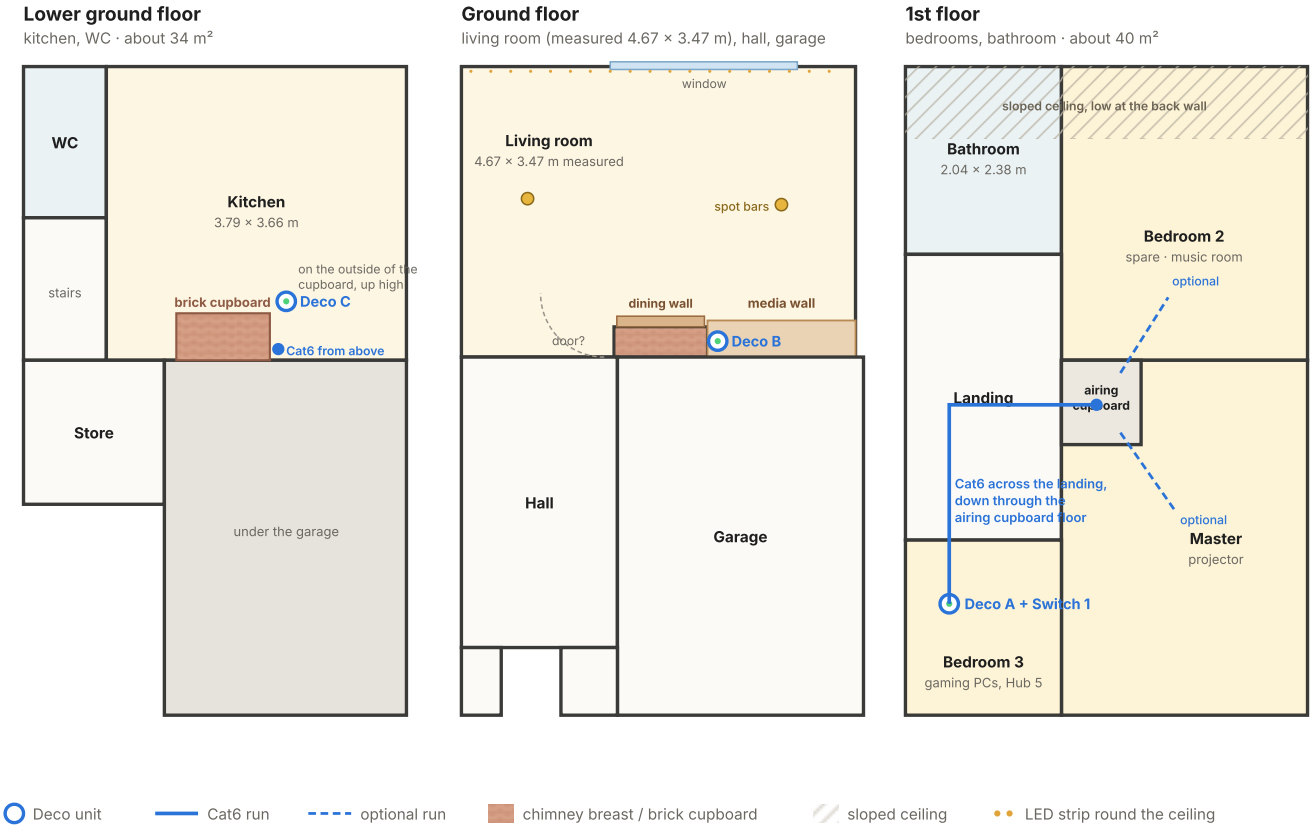
| From your sketch | mm | Notes |
|---|---|---|
| Room, east to west along the window | 4670 | The media wall is the south wall, shared with the hall and garage |
| Window wall to the alcove's back wall | 3470 | To the chimney breast's face: 3090 |
| Ceiling | 2350 | |
| TV alcove, east wall to the breast | 1860 × 380 | Floor to ceiling |
| Chimney breast | 1175 | Projects 380 on the alcove side, 390 on the west side: the dining wall |
| Window and radiator (north wall) | 2350 · 1370 | Window starts 730 from the east wall; radiator ends 130 inside its west end; the glass from 880 to 2120 up, set 220 back in its reveal over a deep sill (from your note; the depth is drawn, please check it) |
| Skirting | 60 × 20 | White now; Card Room Green eggshell in the plan |
| West end: recess, door, stair box | 935 / 1045 / 660 / 820 | Drawn roughly: please confirm which is which |
| Stairs, beside the door | 820 wide | They go **up** to the first floor (you confirmed), behind a wooden banister; drawn rising away from the room until you send a photo |

**From the sofa:** from the middle of the long side your eyes are about 2.65 m from the screen, which a 55-inch fills at about 26°: right for TV and games. With the TV 25.25 cm nearer the side wall, it now sits square on to the end of the sofa's return (214 cm away), and only about 14° off-centre from the middle of the long side. The screen's centre at 116.5 cm is 1–2.5° above a seated eye line. Both spot bars hang 22.4 cm, so the lowest shade is at 212.6 cm: no headroom worries.

## The house

### The house: where the Decos and Cat6 runs go

Deco A and Switch 1 move to bedroom 3 with the Hub 5. Both Cat6 runs drop through the airing cupboard, right over the media wall's corner




Outline traced from the agent's plan (north, the window side, at the top). The living room's alcove, chimney breast and windows are placed from your measurements.

Deco A and Switch 1 sit with the Hub 5 and the gaming PCs in bedroom 3; both Cat6 runs go across the landing and drop through the airing cupboard, which sits over the alcove's corner (section 9).

# 3 · The media wall

Numbered view: the game drawer and the lower LP crate are pulled out 30 cm, with a view of each from above. Dashed lines show what's hidden: gold for data, audio and low-voltage leads, coral for mains. The sofa isn't drawn so the base stays in view; the room plan shows it.

| | |
|---|---|
| **OAK JOINERY** | |
| **1** | Solid oak uprights, 4.5 × 30, worktop to ceiling (159 long) |
| **2** | The left upright is a corner post against the side wall |
| **3** | Flush wedged tenons carry the Series X shelf (134) |
| **4** | High shelf, top 196: books up to 30 cm |
| **5** | Record ledge above the TV: 4 LPs face-out |
| **6** | The oak run wraps the corner: 4 more LPs face-out |
| **7** | LED groove in each upright: a smart colour strip washes the wall |
| **8** | High shelf, right: a small framed sleeve |
| **ON THE WORKTOP** | |
| **9** | LG OLED C6 55in · centre 116.5 · now 25 cm nearer the side wall |
| **10** | Soundbar · eARC, optical and AUX inputs |
| **11** | Turntable (AT-LP120X size allowed) · lid clear to 130 |
| **12** | Xbox Series X on its fan stand · 15 cm air each side |
| **13** | Phono preamp, hung under the shelf at the back |
| **14** | Solid oak worktop 186 × 46 × 3, top at 76 |
| **IN THE ALCOVE BASE** | |
| **15** | Oak face frame; its bottom rail hides the 9 cm plinth |
| **16** | Xbox One + HDMI switch · its brick flat at the back |
| **17** | Xbox 360 + RetroTINK · its brick flat at the back |
| **18** | Game drawer: 2 face-out bins, spine-up rows behind |
| **19** | PS3 + controller basket · bay 2 top row |
| **20** | PS2 · bay 2, beside the sub |
| **21** | PS1 · under the PS2 |
| **22** | Upper record tray: an LP crate, ~90 face-out, under the turntable |
| **23** | Lower LP crate: the same, shown pulled out |
| **24** | Slim drawer: pads and remotes, a charging lead |
| **25** | Two slim game drawers: ~22 games each, face-out |
| **26** | Hidden slim drawer in the plinth, finger pull |
| **27** | Sub · bay 2, under the TV, on an isolation riser |

| | |
|---|---|
| **THE CHIMNEY BREAST (THE TABLE IS BY THE STAIRS)** | |
| **28** | “Now playing” oak ledge: what's on and the next two |
| **29** | Two framed LPs above |
| **THE SIDE WALL** | |
| **30** | PC on the oak run, above the sofa's open end |
| **31** | Brush grommet: the PC's leads drop into the wall |
| **32** | Pad charging dock |
| **33** | PC cables chased into the wall: no spine, no trunking |
| **34** | Keyboard tray under the shelf, rolls out |
| **POWER, NETWORK AND LIGHT** | |
| **35** | Deco B: leads in the groove behind the right upright |
| **36** | Double socket: its right switch is the retro switch |
| **37** | 10-gang extension, hidden behind the TV |
| **38** | LED strip round the ceiling edge (as now) |
| **39** | Smart spot bar on the existing rose, TV to window |
| **40** | Second smart spot bar, existing point by the door |
| **41** | Stairs UP to the first floor, beside the door (direction to confirm) |

### Built to your alcove: 186 wide, 38 deep, 235 to the ceiling

- **Base:** 3 cm filler, three 60 cm bays, 3 cm filler. Bays 1 and 2 are IKEA BESTÅ 60×40×64 frames on a 9 cm plinth (tops at 73, back panels off); bay 3 is an 18 mm ply carcass 60 × 40 × 73 from the floor, split by a 12 mm divider into the slim game column (two face-out game drawers and the pads drawer) and the LP column (two face-out LP crates) under the turntable. One oak face frame runs across all three; its bottom rail hides the plinth and a hidden drawer under bay 2 with a finger pull (no push-to-open at floor level).
- **Depth:** 40 cm frames, a 4 cm cable channel behind and the 2 cm face frame make 46 cm from the back wall, 8 cm proud of the chimney breast.
- **Worktop:** solid oak, 186 × 46 × 3, top at 76, with a brush-lined slot behind the TV and a grommet behind the deck.
- **Oak uprights:** 4.5 × 30 cm, 159 cm from the worktop to the ceiling. The left one stands against the side wall at 0–4.5, a corner post that frames the TV and carries the wrap ledge. The right one is at 131–135.5. That leaves 126.5 cm between them, 2.15 cm clear of the TV each side.

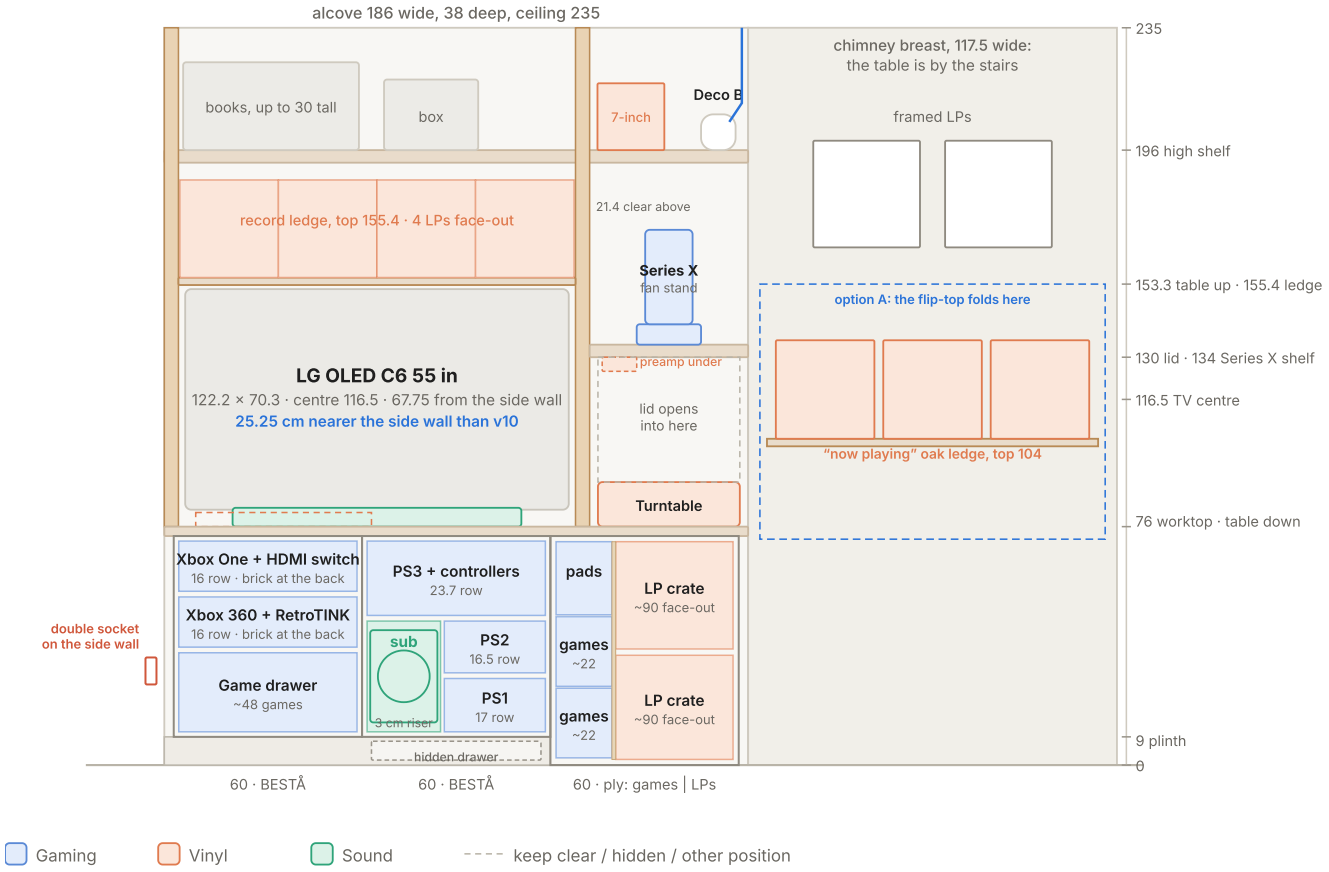
- **TV:** centre 67.75 cm from the side wall (it was 93), on LG's stand set 2 cm off the back wall: bottom 81.4, centre 116.5, top 151.7. The soundbar sits centred under it at the worktop's front edge.
- **The turntable zone:** 50.5 cm between the right upright and the breast. The deck sits on the worktop with 2.65 cm each side and its lid opens to about 121. Above it is an oak shelf at 134, through-tenoned into the upright, carrying the Series X on its fan stand; the phono preamp hangs underneath at the back.
- **Shelves above:** the record ledge at 155.4 between the uprights, four LPs face-out; the high shelf at 196, wall to wall in two lengths, with books, a box, a print, a small framed sleeve and Deco B.
- **Bay 2 with the sub:** the PS3 keeps the top row. Below its shelf the bay splits: on the left the sub's column, 23.4 cm wide, then a 12 mm ply divider, then 32.2 cm for the PS2 over the PS1 on a part shelf. The sub stands on a 3 cm isolation riser with 1 cm each side and 2.9 above. Its front is flush with the face frame, so its driver faces the room and its rear port has 15.6 cm to the wall.
- **Power:** the side-wall socket in front of bay 1 becomes a double. Its two leads enter through a notch at the left end of the face frame's bottom rail and run back through the plinth to the channel.

### The media wall, looked over again

- **The TV's remote:** the soundbar's top is 0.6 above the screen's bottom edge, right where LG puts the remote's receiver. Sit the bar 2 cm proud of the stand's foot so the beam from the sofa clears it; LG's Magic Remote works by radio once paired, so it doesn't mind either way (a check in the fit proof).
- **Heat:** every console bay is open front and back, the bricks lie flat at the back with nothing on top, and the Series X has 15 cm each side and 21 above. Nothing changed in v17.
- **One remote for most of it:** HDMI-CEC (SIMPLINK) lets the TV's remote work the soundbar and the modern consoles (step 13); the HDMI switch has its own remote for the retro side.

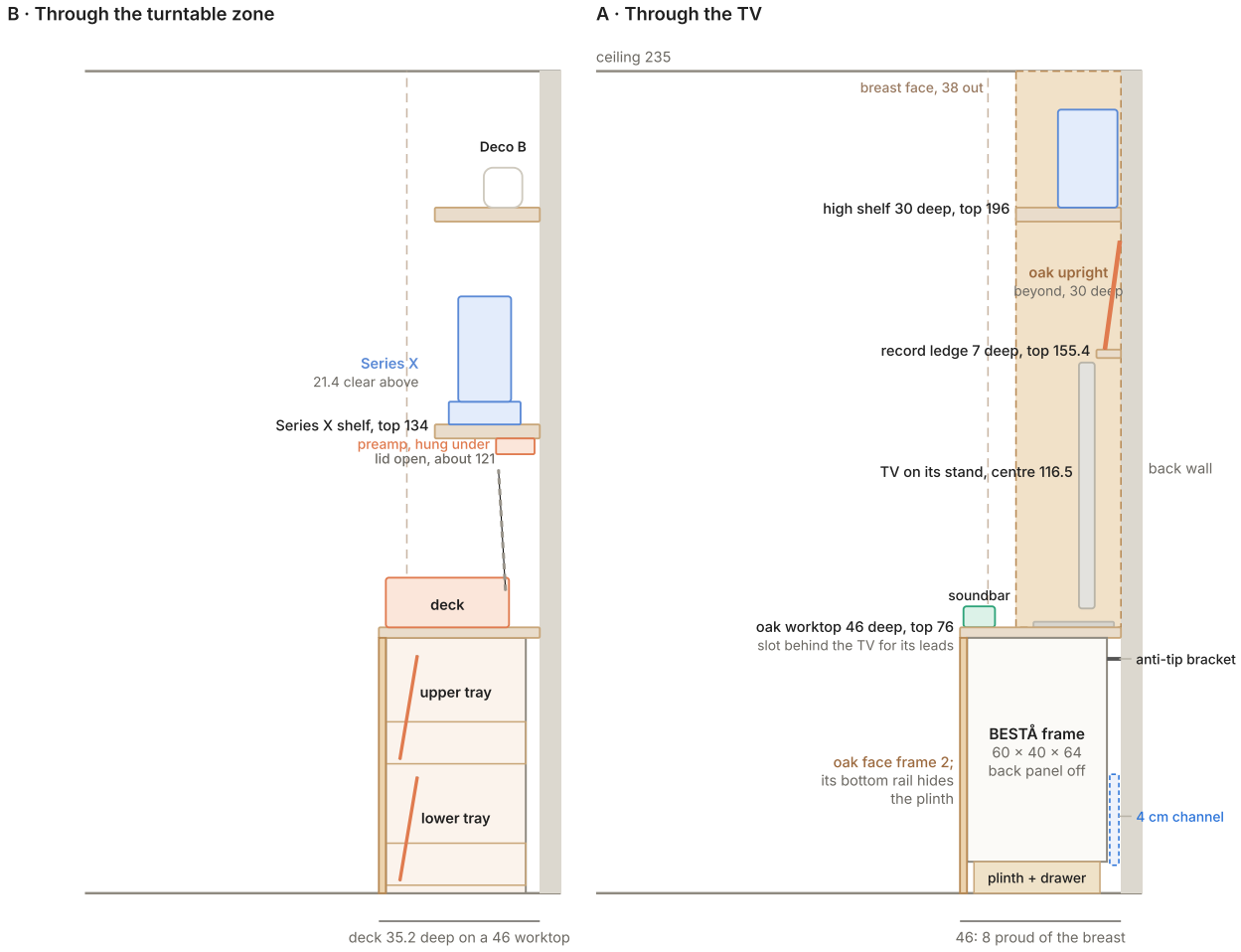
### The media wall and the chimney breast from the front, to scale

The TV between oak uprights nearer the side wall, the turntable zone on its right, three bays below (bay 3: games beside the LP crates); the “now playing” ledg



### How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open




## The sub in the media wall

You asked for the sub back in the unit, with the Xbox bricks moved behind. It fits, and it sits in the best spot for it: under the TV, beside the soundbar, so the bass comes from the same place as the picture. Here's how the base makes room, and what was checked.

- **The bricks go to the back.** The Xbox One's brick (17 × 7.5 × 4.8[11]) and the Xbox 360's (allowed 21 × 10 × 6[16]) lie flat at the back of their shelves, by the channel's power leads. Microsoft asks for a ventilated space, flat on the rubber pads, with nothing on top[26]; the bays are open front and back.
- **That frees bay 1's fronts** beside the two Xboxes, so the HDMI switch (beside the Xbox One, where its remote reaches it) and the RetroTINK (beside the Xbox 360) move up from bay 2.
- **Bay 2's lower half becomes two columns:** the sub on the left, under the TV, and the PS2 over the PS1 on the right. The sub is 21.4 × 30.4 × 29.4[3]; its column is 23.4 wide and 35.3 tall.
- **The port:** the sub's front sits flush with the face frame, so its rear port is 15.6 cm from the wall through the open back and the channel. That's inside the 15–30 cm advised for a rear port[25], at the low end, so the channel behind bay 2 stays clear: Switch 2 and the retro 6-way move behind bay 1.
- **Vibration, the one thing to test:** Microsoft advises keeping the Xbox 360 away from subwoofers[27], and the deck shares the worktop. The 360 is a whole bay away, the sub stands on an isolation riser without touching the divider or the frame, and the deck has isolation feet. Once it's built, play a disc game and a record at film volume. If anything skips, turn the sub down for that source; if it still does, the sub can go back under the dining wall.
- **The dining wall is clear underneath:** no cover to make, and nothing to move for eight at the table.

## The fit proof: every item against its published size

Each zone lists what goes in it, its size and its source (section 12), the space, what's left and a verdict. 88 of the 129 checks fit outright and none fail; 31 are tight, most of them snug by design; 10 need a check on site (the TV's plugs, the RetroTINK's size, the sub's vibration, the soundbar and the TV's remote eye, the stairs wall, your shelf's height and, for option A, the flip-top's end on the breast).

### The TV between the uprights

126.5 between the oak uprights, the left one now a corner post on the side wall; the ledge's underside at 153

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| LG OLED C6 55 in, on its stand | 122.2 × 70.3 × 4.5 panel; 75.7 tall on the stand[2] | 126.5 wide | 2.15 each side | **Fits** |
| Top of the screen | 151.7 (bottom 81.4, centre 116.5)[2] | ledge at 153 | 1.3 | **Fits** |
| LG stand base | 47 × 23[2] | worktop 46 deep | set 2 cm off the wall: 11 cm in front of it for the soundbar's leads | **Fits** |
| Soundbar, Hisense AX5100G, centred under the TV | 92 × 9 × 6 tall[3] | x 21.75–113.75, top 82 | the sightline to the screen's bottom passes 83.54–84.17 over it | **Fits** |
| The TV's remote receiver, bottom centre of the screen's frame | screen bottom 81.4[2] | soundbar top 82, in front of the stand's foot | 0.6 over the frame's edge: sit the bar on the worktop 2 cm proud of the stand's foot, not against it, so the remote's beam from the sofa clears it; if a button stutters, pair LG's Magic Remote (it talks to the TV by radio) | **Check** |
| The TV's HDMI plugs, with the screen 2.15 from the corner post | C6 sockets face sideways on the back panel[2] | behind the screen | plugs stay behind the panel; if a stiff lead pushes out, use 90° HDMI adapters | **Check** |

### The turntable zone, right of the TV

50.5 wide × 46 deep, worktop 76 to the Series X shelf's underside at 130

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Turntable, AT-LP120X size (worst case) | 45.2 × 35.2 × 14.2 with its lid[8] | 50.5 wide | 2.65 each side | **Fits, tight** |
| Its depth, with room behind for the lid's hinges | 35.2, front 2 cm in from the worktop's edge[8] | 46 deep, to the painted wall | 8.8 behind it | **Fits** |
| Lid open | cover 44.1 × 34.6; standing up, its top at about 121[17] | shelf underside 130 | about 9 | **Fits** |
| Phono preamp, hung under the shelf at the back | about 11 × 11 × 4.5 | 125.5–130, at the back | 4.4 over the open lid's edge; lift the lid gently | **Fits, tight** |
| Series X fan stand on the shelf | 20.5 × 20.5 (assumed; model to confirm) | 50.5 × 30 shelf | 15 each side | **Fits** |
| Xbox Series X, standing | 15.1 × 15.1 × 30.1 tall[4] | top at 170.6 | 21.4 clear above (Microsoft's minimum is 4); 15 each side, inside the 10–15 it recommends | **Fits** |
| Deco B on the high shelf, right | 11 × 11 × 11.4[7] | 50.5 wide, 39 to the ceiling | beside the framed 7-inch, 27 clear above | **Fits** |

### Shelves above the TV

ledge top 155.4, 7 deep; high shelf top 196, 30 deep; ceiling 235

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Four LPs on the record ledge | 4 × 31.4 = 125.6 wide [13] | 126.5 long | 0.9: snug, as drawn | **Fits** |
| Ledge LPs leaning back 8° | tops at 186.5 [13] | high shelf underside 192 | 5.5 to lift one out over the 1.5 lip | **Fits** |
| Books on the high shelf | up to 30 tall | 39 to the ceiling | 9 over the tallest | **Fits** |
| Oak uprights | 159 long (worktop to ceiling) [1] | 235 ceiling | buy 165 and scribe to the ceiling | **Fits** |

### Bay 1, Xbox (BESTÅ frame on the 9 cm plinth)

56.8 wide inside, 40 deep; the bricks lie at the back of their rows

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Xbox One, original, + HDMI switch | 33.3 × 27.4 × 7.9; switch 12.7 × 5.9 × 2.5 [12,10] | 16 tall row | 8.1 above; the switch in the 23.5 beside it, at the front for its remote | **Fits** |
| Xbox 360, original, + RetroTINK-6X CE | 30.9 × 25.8 × 8.3; RetroTINK allowed 23 × 23 × 8 (no size published) [12,9] | 16 tall row | 7.7 above; 25.9 beside it for the RetroTINK: measure it when it arrives | **Check** |
| Both bricks, flat at the back of their rows | Xbox One 17 × 7.5 × 4.8; Xbox 360 allowed 21 × 10 × 6 [11,16,26] | behind the switch and the RetroTINK, by the channel's power leads | 23 + 10 = 33 of the 40 depth for the RetroTINK and the 360's brick; on their rubber pads, nothing on top, open to the channel; 2 cm or more from the consoles' rear vents | **Fits** |
| Game drawer | box 54.3 × 35 on 350 mm runners; cases 19 × 13.5 × 1.4 [15,13] | 25.2 tall row, 40 deep | about 3 above the leaning cases; 5 behind the box | **Fits** |
| Depth with the leads | console 27.4 + HDMI plug about 5 [12] | 40 deep, back open | about 7.6, then the 4 cm channel | **Fits** |

### Bay 2, PlayStation and the sub (BESTÅ frame on the plinth)

56.8 wide inside, 40 deep; under the PS3's shelf, the sub's column 23.4 wide, a 12 mm ply divider, then 32.2 for the PS2 over the PS1

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| PS3, original, + controller basket | 32.5 × 27.4 × 9.8 [12] | 23.7 tall row, full width | 13.9 above; 24.3 beside for the basket | **Fits** |
| The sub in its column | 21.4 × 30.4 × 29.4 tall, on a 3 isolation riser [3] | 23.4 wide × 35.3 tall | 1 each side, 2.9 above; it mustn't touch the divider or the frame | **Fits, tight** |
| The sub's driver and port | front-firing driver, rear port; its front flush with the face frame [3,25] | 15.6 from the port to the wall, through the open back and the channel | inside the 15–30 advised for a rear port, at its low end: keep the channel behind bay 2 clear (Switch 2 and the retro 6-way go behind bay 1) | **Fits, tight** |
| PS2, original, over the PS1 | 30.2 × 18.3 × 7.8 · 27 × 18.8 × 6 [12] | 32.2 wide; 16.5 and 17 rows on a part shelf | PS2 1 each side, 8.7 above; PS1 2.6 each side, 11 above; both open front and back | **Fits, tight** |
| Vibration: the consoles and the deck share the base with the sub | Microsoft: keep the Xbox 360 away from high-vibrating sources such as subwoofers [27] | the sub on its riser, clear of the divider and the frame; the deck on isolation feet | the 360 is a bay away, the PS2 and PS1 a divider away: play a disc game and a record at film volume; if anything skips, turn the sub down for that source, or put it back under the dining wall | **Check** |
| Hidden slim drawer in the plinth | box 52 × 34 × 5, 300 mm full-extension runners, a finger-pull groove (no push-to-open at floor level) [15] | 9 cm plinth, 120 wide | about 2 above and below the box; felt its runners, it's under the sub | **Fits** |
| Switch 2 (TL-SG108) and the retro 6-way in the channel | 15.8 × 10.1 × 2.5; 6-way about 34 × 6 × 4 [6] | 4 cm channel, behind bay 1 | on their edges, clear of the sub's port | **Fits, tight** |

### Bay 3: the LP crates and the slim game column

56.4 wide × 40 deep × 69.4 tall inside the ply carcass, split by a 12 mm divider

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Across the bay | slim drawer 15.26 + divider 1.2 + LP crate 34.86, each drawer with 1.27 runners a side [15] | 56.4 inside | exact: 15.26 + 2.54 + 1.2 + 34.86 + 2.54 = 56.4; nothing left over | **Fits** |
| LPs in the crate | 31.4 sleeves face-out, leaning back 10° [13,36] | 33.06 trough | 0.83 each side: snug like a shop crate; outer sleeves up to 32.4 still fit | **Fits, tight** |
| Crate heights | lower sleeves to 34.4, upper to 69.1 [13] | top panel underside 71.2 | 2.6 and 2.1 to lift a sleeve | **Fits** |
| Games in the slim drawers, face-out | DVD-size cases 13.5 wide, 19 tall, leaning back 10° [13] | 14.06 inside (6 mm ply sides) | 0.56 spare: they flip like the records; about 22 a drawer in 40 deep | **Fits, tight** |
| The game drawers' height | cases stand 18.71 on a 1.2 base [13] | 22.8 from one drawer to the next | 2.89 clear over the cases | **Fits** |
| Pads in the top slim drawer | 4 pads on edge, two side by side: 2 × 6.6 = 13.2 across, 16 long [35] | 14.06 wide, 22.1 tall | 0.86 across; 10.6 tall in 22.1; the remotes in front | **Fits, tight** |

### The side wall: one oak run from the corner to the PC

one board line at 155.4, 4 thick, 220 long: 7 deep along the LPs, sweeping out to 28 under the PC

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| The sweep from the ledge to the PC shelf | front edge 7 → 28 over 14 (z -118 to -132) | the last wrap LP ends at z -123.3; the PC starts at z -134 | the LP stands on the start of the curve; 2 between the full depth and the case | **Fits** |
| Four more LPs on the wrap ledge | 4 × 31.4 + gaps = 129.8 [13] | 134 long | 2.7 | **Fits** |
| The PC on its shelf | 23 × 45.3 × 46.6 tall (an average gaming case, e.g. Corsair 4000D) [29] | 28 × 82 | 2.5 off the wall, the brush grommet and pad dock beyond it; height 46.6 under the 79.6 to the ceiling | **Fits** |
| Air above the PC | top at 202 | 235 ceiling | 33 for the top exhaust | **Fits** |
| Shelf over the end of the sofa | underside 151.4 [24] | sofa back 80 | 71.4; the shelf covers only the 28 cm nearest the wall | **Fits** |
| Keyboard tray under the PC shelf | 7 cm housing, tray rolls 25 out on 300 mm runners [15] | sofa back 80 | 64.4 over the sofa's back; pulled out, its front edge is 53 from the wall, still over the return's seat | **Fits** |
| Out of the TV's sightlines | PC about 54° left of the line to the screen from the sofa [1] | — | well clear of the screen's left edge | **Fits** |

### The dining cabinet (the plan)

one oak cabinet 198 × 24 × 189 under your shelf; the table 180 × 90 open, six chairs inside

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| The cabinet under your shelf | 189 tall [28] | shelf's underside about 191 | 2: the shelf stays as it is | Fits |
| Its length | z -86 to -284 [28,1] | the shelf's length; the stair's foot before it stays clear | 2 short of the pipe boxing, 2 prouder than it | Fits |
| The table standing up in its bay | two halves 90 long, hinged at 73.3 [45,43] | 187.2 inside the bay | its top edge at 163.3: 23.9 over it for the linen flap's space | Fits |
| The linen flap over the folded table | an opening 21.1 tall, 91 wide [46] | 21.6 deep inside | the cloth, napkins and candles; a lift-up flap on two stays, flush with the table's underside | Fits |
| Folded, how deep | two aprons 4.6, two tops 2.7 [44] | 23.4 inside the carcass | 14.9 deep folded: 8.5 spare; the room sees the oak underside, flush with the doors | Fits |
| Lowering it and flipping the half | each half about 16 kg [43,30] | a 90 piano hinge at its foot; three flip-top hinges at the joint | it swings down like a drop-front: about 16 kg, two hands; the legs drop and lock as it comes down | Fits |
| The legs | oak 6.5 × 4.5, 68.7 long, a pair at each half's end [44] | four self-locking folding brackets | each pair folds flat inside its half's 4.6 apron; 68.7 of knee room under it | Fits |
| Six chairs in the chair bay: across | two side by side, 2 × 43 [39] | the bay 101.6 wide inside | 15.6 spare | Fits |
| Six chairs: deep and tall | two deep (18), each 89 tall [39] | 21.6 deep inside; the shelf at 93 | 3.6 spare deep; 4 on the floor (3–92), 2 on the shelf from 94.8, the shelf's front half free for the seat pads and the runner | Fits |
| The chair bay's doors | two oak doors, opening past flat [46] | they lie back against the cabinet | the bay opens wide to lift the chairs out | Fits |
| Stair side: three chairs | 58.5 of edge each [23,39] | 175.4 from the far end to the cabinet | 55 is the squeeze, 60 the aim: three sit in comfort | Fits |
| Stair side: the stair's foot and the door | chair backs at z -47.3 [1,28] | the stair box's face at -41; the door's edge reaches -44 | 6.3 from the stair box, 3.3 from the door's swing: tuck the end chair in after a meal | Fits, tight |
| Window side: three chairs | 58.3 of edge each [23,24] | 175 from the sofa's arm (x 268) to the cabinet | three in comfort; backs at z -219.3, 59.2 clear of your plant stand | Fits |
| The far end and the coffee table | the table ends at x 267.6 [23] | the coffee table to x 193 | 74.6 between them: the coffee table stays put; a seventh chair fits at the end (33.6 behind its back) | Fits |
| The ottoman | 75 square [42] | the open table reaches x 267.6; the ottoman to x 275.5 | 7.9 under the table's end: push it to the sofa while you eat | Check |
| The table's side and the sofa | window edge at z -178.3 [24] | the sofa's front at z -201 | 22.7 between them: the table passes the sofa's end | Fits |
| Light over the table | bar 2 on the existing pendant point [28] | point about (412, -143) | over the table; the table rule turns bar 2 up when it's open | Fits |

| | | | | |
|---|---|---|---|---|
| Fixing it | a floor-standing cabinet, screwed to the wall at the top [28] | the west wall's build isn't known yet | it stands on the floor, so the wall only steadies it: two screws into studs or plugs (knock and look first) | **Fits** |

### How many can sit down

every table we've drawn, in this room, with chairs a person wide and backs 41 behind the edge

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| The dining cabinet (the plan) | 6 in comfort [23,39] | 3 on the stair side, 3 on the window side | 58.5 and 58.3 of edge each; a seventh at the far end with a spare chair | **Fits** |
| B: the flip-top by the stairs (the alternative) | seven at a squeeze [23,24] | three a side (53 each) and one at the end | five in comfort: two a side and the end | **Fits, tight** |
| A: the flip-top on the chimney breast | six at a squeeze [23,24] | three a side, 53 each | four in comfort; no end seat: the sofa's in the way | **Fits, tight** |
| A fourth chair a side at the cabinet | eight round it [23,24,1] | 175.4 and 175 of edge | 43.8 each: under the 55 squeeze; v26's longer table did it, with the coffee table moved and all eight chairs in a bigger bay | **Doesn't fit** |
| A longer table in the cabinet | halves over 113.9 [28] | they must stand up under your shelf | 113.9 is the most that folds away; 90 leaves the linen flap's space and keeps each half to about 16 kg | **Fits** |

### The flip-top table under your shelf, by the stairs (B, the alternative)

110 × 160 open, running from the stairs wall into the room, its end at x 296.7

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Open: 110 × 160, still on the wall | the top 110 × 80 on its brackets, the half 110 × 80 on the gate leg [18,30] | nothing comes off the wall; no leaf to store | the table runs 170.3 out from the wall | **Fits** |
| Three flip-top hinges at the joint | 1.27 × 7, set into both edges [30] | a 2.7 edge | 0.72 of oak either side of each hinge; they open to 180°, so the half lies flat on the top to fold | **Fits** |
| The gate leg folds inside the half | legs 73.3 long, hinged 3 in from the far edge [20] | 74.75 of the half for them to lie along | 1.45 spare | **Fits** |
| Folded on the wall | the top, the half and its gate: 13.3 off the wall, top edge 153.3 [19] | a 7.3 hinge rail makes room for the half and gate behind the top | about 7 deeper than a single top; the room still sees the plain oak underside | **Fits** |
| Folding it up | top 17 kg + half 17 kg + gate 2 kg [19,18] | lift-off hinges for a 120 kg door; brackets 500 kg a pair | about 37 kg to swing up: two people, one at each corner; turn buttons hold it | **Fits, tight** |
| Knees by the brackets | each arm 15 in from the side edge, at 70.9 [18,22] | knees at about 60 over a 48 seat | the first chair each side has the arm above its knees, not in them | **Fits** |
| Four: two down each side | 80 of edge each [23] | 60 each | 20 spare each | **Fits** |
| Six: three down each side | 53.33 of edge each [23] | 60 each, 55 at a squeeze | 53.33 each: elbows in | **Fits, tight** |
| The stair side's chairs | backs at z -84 when pulled out [22,28] | the stair box's face at -41 | 43 behind them: tuck them in after a meal so the stair's foot stays clear | **Fits, tight** |
| The window side's chairs | backs at z -276 [22] | the window wall at -309 | 33 behind them; your red lamp moves from the window corner to the sofa's end or by the breast while it's open | **Fits, tight** |
| A seat at its end | one chair at x 255.7–304.7 [22,24] | the sofa's arm ends at x 268, front z -201 | one chair, centred on the table, meets the sofa's front (0.5 over, with the sofa 13 off the wall): tuck it 2 cm further under the table: seven with three a side (a second end chair would hit the arm) | **Fits, tight** |
| Folded, under the shelf | top edge 153.3, 13.3 off the wall [28] | shelf's underside about 191 | 37.7: the shelf and what's on it stay | **Fits** |
| Light over the table | the existing pendant point, for bar 2 [28] | point about (412, -143) | over the open table's stair side: the bar runs along it | **Fits** |
| Your shelf and the pipe boxing | the shelf 200 long [28,1] | the boxing's face at z -286, 22 out | drawn 2 short of it, -43 past the stair box (your photo: about 41) | **Fits** |
| The OBEGRÄNSAD panel on the shelf | 52 × 36 × 6, standing landscape at the stair end [34] | shelf top 196 to the 235 ceiling | 3 over it: this needs the shelf at 198 or lower (measure it); higher, hang the panel on the wall beside the shelf instead | **Check** |
| The window side's chairs and the pipe boxing | backs at z -276 [22,1] | the boxing's face at z -286 (x 445 to the wall) | 10 clear of the boxing | **Fits** |
| Fixing the brackets | 500 kg a pair, into the wall [18] | the west wall's build isn't known yet | solid (plugs and screws) or studwork (fix into studs): knock and look before choosing this site | **Check** |

### The chimney breast, with the table by the stairs

breast 117.5 wide; the “now playing” ledge 105.5 × 7, top 104

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Three LPs on the ledge | 3 × 31.4 = 94.2 [13] | 105.5 long | 2.83 between and at the ends | **Fits** |
| Leaning back 8°, under the framed LPs | tops at 135.09 [13] | frames from 165 | 29.91 between | **Fits** |
| Beside the turntable | ledge 104 up, 7 deep [8] | the deck's top at about 90.2, an arm's reach to the left | sleeve off the ledge, record on, sleeve back: the base stands 8 proud of the breast, the ledge 7 | **Fits** |
| If you choose option A instead | the flip-top folds on the breast [19] | the ledge's place | the ledge comes off its two screws; the frames stay | **Fits** |

### Option A: the flip-top table on the chimney breast

110 × 160 open, running from the breast towards the window, 170.3 out

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Open: 110 × 160, still on the wall | the top 110 × 80 on its brackets, the half 110 × 80 on the gate leg [18,30] | nothing comes off the wall; no leaf to store | the table runs 170.3 out from the wall | **Fits** |
| Three flip-top hinges at the joint | 1.27 × 7, set into both edges [30] | a 2.7 edge | 0.72 of oak either side of each hinge; they open to 180°, so the half lies flat on the top to fold | **Fits** |
| The gate leg folds inside the half | legs 73.3 long, hinged 3 in from the far edge [20] | 74.75 of the half for them to lie along | 1.45 spare | **Fits** |
| Folded on the wall | the top, the half and its gate: 13.3 off the wall, top edge 153.3 [19] | a 7.3 hinge rail makes room for the half and gate behind the top | about 7 deeper than a single top; the room still sees the plain oak underside | **Fits** |
| Folding it up | top 17 kg + half 17 kg + gate 2 kg [19,18] | lift-off hinges for a 120 kg door; brackets 500 kg a pair | about 37 kg to swing up: two people, one at each corner; turn buttons hold it | **Fits, tight** |
| Knees by the brackets | each arm 15 in from the side edge, at 70.9 [18,22] | knees at about 60 over a 48 seat | the first chair each side has the arm above its knees, not in them | **Fits** |
| Four: two down each side | 80 of edge each [23] | 60 each | 20 spare each | **Fits** |
| Six: three down each side | 53.33 of edge each [23] | 60 each, 55 at a squeeze | 53.33 each: elbows in | **Fits, tight** |
| Its end and the sofa | the table ends 170.3 out [24] | the sofa's front at 201 | 30.7 between them: no chair at that end (it would need 41), and the way round is down the sides | **Check** |
| The coffee table | east-side chairs back to x 148.75 [22] | coffee table x 103–193 | for six, slide the coffee table towards the TV before you sit down | **Fits, tight** |
| The door's swing | door 83 wide, opens in, hinged by the breast [1] | its edge sweeps 83 round the hinge | the west side's first chair is 60.03 from the hinge: in its swing, so keep the door shut while you eat (or rehang it to open into the hall) | **Fits, tight** |

### The room

467 × 347, ceiling 235

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Eyes to screen from the sofa | the L-sofa's long side against the window wall, 10 off the radiator [1,24] | 335 window wall to screen | about 2.65 m; a 55-inch fills about 26° | **Fits** |
| From the end of the sofa's return | sitting up at its end, 214 from the screen [24] | straight ahead of the TV | the TV's move puts it square on to this seat | **Fits** |
| TV centre against seated eyes | 116.5 [2] | eyes at 105–112 | about 1–2.5° above the eye line | **Fits** |
| Both spotlight bars under the ceiling | 22.4 drop [14] | 235 ceiling | lowest shade at 212.6 | **Fits** |
| Both bars turned to run TV to window | 84 long along the room [14,28] | bar 2 at 55 from the west wall | bar 1 spans z -62.5 to -146.5; bar 2 now runs along the wall, so its end no longer reaches past the west wall | **Fits** |
| Smart colour bulbs in the bars | 8 × E14: WiZ P45 4.9 W, 4.5 × 10.1; or Hue lustre 5.1 W, 4.5 × 7.7 [14,38] | the bar's 6 W a lamp; 15 cm shades | 1.1 W under; the WiZ's 10.1 sits inside a 15 shade with its holder; if it shows below the rim, the Hue's 7.7 doesn't | **Fits** |

### Your plant stand and your vinyl stand

the window wall's west end, 107 between the curtains' stack and the pipe boxing; the chimney breast, 117.5 wide

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| A: the plant stand in the window corner (recommended) | 98 × 29.5 × 158 [32] | x 338 to 445 | x 342.5–440.5: 4.5 from the curtains, 4.5 from the boxing; daylight from the window beside it; 77 to the ceiling for the top plants | **Fits, tight** |
| A: the vinyl stand on the breast | 70 × 32 × 78 [33] | breast 117.5 wide, the “now playing” ledge's underside at 101.6 | 23.75 each side, 23.6 under the ledge; a step from the turntable | **Fits** |
| A: LPs in its top bays, face-out | 2 bays × 32.3; an LP 31.4 [33,13] | shelf at about 46 | 0.9 spare a bay; the front LP's top at about 77.4, 24.2 under the ledge; about 50 LPs | **Fits, tight** |
| A: LPs on its bottom shelf, between the wires | 4 wires 19.5 tall [33,13] | 70 wide, 38 high | three slots of LPs standing; about 50 more (about 100 in all) | **Fits** |
| B: the vinyl stand in the window corner, as today | 70 wide [33] | 107 between the curtains and the boxing | 37 to spare | **Fits** |
| B: the plant stand on the breast | 98 wide, 158 tall [32] | breast 117.5; the frames from about 147.3 | 9.75 each side; the “now playing” ledge comes off and the two frames go up 30 (or elsewhere): at 158 it stands past the frames' bottoms | **Fits, tight** |
| Your red lamp | 28 base [1] | the window corner is taken in A and B | it moves to the sofa's west end (in the 3D room, drag it anywhere) | **Fits** |

### The flip-top's chair cupboard (B, the alternative)

an oak cupboard built in beside the table, 51 × 22 × 189, between the table's window end and the pipe boxing. Folded chairs 43 × 9 × 89

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Its width, between the table and the pipe boxing | 51 wide, 1.8 sides [28,1] | z -235 to -286 | 47.4 inside for a 43 chair: 2.2 each side | **Fits, tight** |
| Its depth: four chairs, two deep | 2 × 9 folded [39] | 22 deep, the same as the pipe boxing; 19.6 inside behind the doors | 1.6 spare: snug, so they can't rattle; it stands no further out than the boxing beside it | **Fits, tight** |
| Its height: a pair on the floor, a pair on the shelf | each 89 tall [39,28] | 189 tall, the shelf at 93; your shelf's underside at 191 | lower pair 3–92 under the shelf at 93; upper pair 94.8–183.8 under the top at 187.2; the cupboard stops 2 under your shelf | **Fits, tight** |
| Six instead of four | three deep, 3 × 9 [39] | the 31-deep version, 28.6 inside | 1.6 spare; it stands 9 proud of the pipe boxing's face | **Fits, tight** |
| Weight on the wall | four chairs, about 14.44 kg, and an oak carcass of about 18 kg [39] | it stands on the floor, held to the wall at the top | the floor takes the weight; two screws through the top rail into the wall (or studs) stop it tipping | **Fits** |
| The doors | two oak doors, 51 × about 93, hinged on the table's side [1] | they swing towards the table's end | open the doors and take the chairs out before you open the table; folded, the table and the cupboard are one run of oak, nothing left out | **Fits** |
| With the table open: the window side's first chair | the cupboard's face 11.7 past the table's wall end [22,23] | window side 160 long | for four, two a side, nothing changes; for six the window side's three get 49.4 each instead of 53.3: elbows in | **Fits, tight** |
| Open hooks instead (the cheaper option) | two 200 mm storage hooks, two chairs on each [40] | the same gap, no cupboard | £8–15 instead of the cupboard; the chairs stay on show | **Fits** |
| At the stair end instead | folded 43 wide [39,28] | 41 between the shelf's start (z -84) and the table (z -125) | 2 short: too narrow for this chair | **Check** |
| Chairs five and six, with the 22-deep cupboard | two more, folded [39,1] | the garage (under the stairs is a brick side wall, plasterboard beneath: no way in from the room) | the guests' chairs, fetched when six sit down; or build the 31-deep cupboard, or hang four behind the folded top (below), and all of them pack away | **Fits** |

### More ways to keep the chairs with the flip-top (B)

behind the folded top, and the 41 cm gap at the stair end

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Behind the folded top: across | two folded chairs side by side, 86 [39,30] | the top's 110 width | 12 spare each side: out of sight behind it | Fits |
| Behind the folded top: deep | two deep, 18, on hooks [39,40] | a backboard 22 deeper (the hinge rail stands 31.3 off the wall) | 1 spare; the folded table then stands 34 off the wall instead of 13 | Fits, tight |
| Behind the folded top: tall | folded 89 tall [39,30] | the folded top, 80 tall, and a 10 oak upstand under it | the upstand hides their feet, so nothing shows when it's folded | Fits |
| Behind the folded top: the open table | the table 22 further into the room [23,24] | open edge at x 274.7 (now 296.7) | four seats as now; the end seat comes nearer the TV but still clears the sofa's front line; take the chairs off their hooks and sit down | Fits, tight |
| Fold-down seat at the stair end: width | an oak seat 38 wide [41,28] | the 41 gap | 1.5 each side | Fits, tight |
| Fold-down seat: folded | bracket 5 + seat 2.7 [41] | the folded table, 13 off the wall | 7.7 proud: flatter than the table beside it | Fits |
| Fold-down seat: the load | one person, about 100 kg at most [41] | 150 kg a pair, the brackets 30 apart | screwed into the stairs wall's studs or with wall plugs to suit (knock first) | Fits |
| Fold-down seat: open | 33 deep at 46 high [41,23] | at the table's south end, by the wall | a fifth seat for the table, and one that's always there; the stairs' foot stays clear | Fits |

### Your sofa's ottoman

75 × 75, on the rug

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| Beside the coffee table | 75 square [42,24] | coffee table to x 193 | 7.5 between them; 6 off the sofa's front: a footrest for the long side | Fits |
| On the rug | x 200.5–275.5 [42] | rug x 100–330 | inside it, clear of the door's swing and the table's walkway | Fits |

### Your rear speakers (to confirm from your photo)

the AX5100G's rears, 9 × 14.4 × 10.9, on stands behind the sofa

| Item | Size (cm) | Space | Spare (cm) | Result |
|---|---|---|---|---|
| In the gap behind the sofa | 10.9 deep, stands 11 at the base [3,47] | 13 between the sofa's back and the window wall | 2 spare: snug; slide the sofa out a little if the stands' bases are bigger | Fits, tight |
| Above the sofa's back | the speakers 96–110.4 [3,24] | the sofa's back 80 tall | 16 clear above it, at about seated ear height: the sound passes over your heads | Fits |
| The curtains | the pole's line at z -298 [28,47] | the curtains hang in the same gap | drawn, they'll touch the stands: pull the stands to the sofa's ends if they catch (your photo will show) | Check |
| Power and the link cable | the rears are linked by a cable; one plugs in [3] | your double socket behind the sofa's short side (x 0, z -140) | the link runs along the skirting behind the sofa, out of sight | Fits |

## Console sizes, every version

The bays are sized for the largest version of each console. The model number is on the label underneath; a smaller version just leaves more room.

| Console | Version | W × D × H (cm) | Power |
|---|---|---|---|
| PS1 | Original (largest) · PS one | 27.0 × 18.8 × 6.0 · 19.3 × 14.4 × 3.8 | Built in · external adapter |
| PS2 | Original (largest) · Slim | 30.2 × 18.3 × 7.8 · 23.1 × 15.2 × 2.8 | Built in · external brick |
| PS3 | Original (largest) · Slim · Super Slim | 32.5 × 27.4 × 9.8 · 29.0 × 29.0 × 6.5 · 29.0 × 23.0 × 6.0 | Built in |
| Xbox 360 | Original (largest) · S · E | 30.9 × 25.8 × 8.3 · about 27.0 × 26.4 × 7.5 · 25.4 × 26.4 × 6.7 | External brick |
| Xbox One | Original (largest) · S · X | 33.3 × 27.4 × 7.9 · 29.5 × 22.9 × 6.4 · 30.0 × 24.0 × 6.0 | External brick · built in |
| Xbox Series X | Standing | 15.1 × 15.1 footprint, 30.1 tall | Built in |

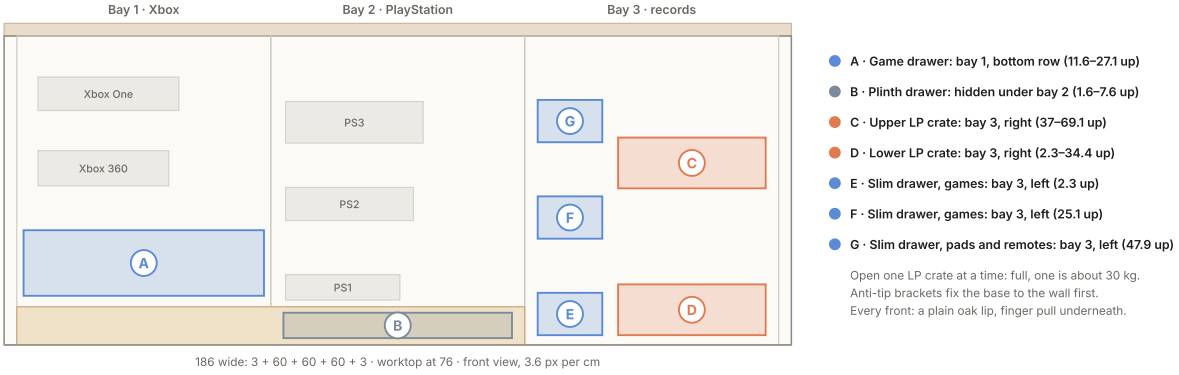
## The drawer layouts

Seven drawers in the alcove base, all drawn to scale from above: **A** the game drawer (bay 1), **B** the hidden plinth drawer under bay 2, **C** and **D** the two face-out LP crates and **E** to **G** the slim drawers beside them (bay 3). Each one shows its compartments, what goes in them and how many, and its runners. Every front is a plain oak lip with a finger pull cut underneath.

**The drawer layouts: where each drawer is and what goes in it**

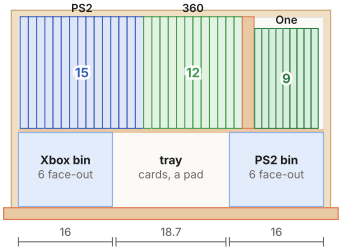
Seven drawers in the alcove base, all to scale: A the game drawer, B the hidden plinth drawer, C and D the LP crates, E to G the slim drawers beside them

**Where each drawer is: the alcove base from the front**



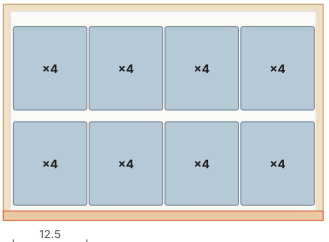

186 wide: 3 + 60 + 60 + 60 + 3 · worktop at 76 · front view, 3.6 px per cm

**A · Game drawer**
box 54.3 × 35 · 350 mm soft-close runners · rolls out 30



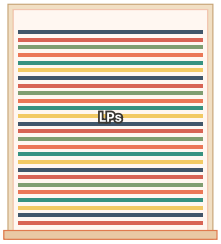

back: 36 spine-up, titles on top (19 cm cases on edge)
front: 12 face-out · about 48 games in all

**B · Hidden plinth drawer**
box 52 × 34 × 5 inside · 300 mm runners · finger pull




32 PS1 games flat: 8 stacks of 4 jewel cases
or remotes, spare pads, cables and manuals

**C and D · LP crates**
box 34.9 × 40 · 400 mm runners




trough 33.1: snug, no dead space
face-out, leaning back 10°
~90 LPs each · two crates

**E, F, G · Slim drawers**
15.3 × 40 · 400 mm

14.1
E, F: ~22 games
each, face-out
G: 4 pads, remotes,
a charging lead

Plans drawn from above at 5 px per cm, front (the room) at the bottom; the elevation at 3.6 px per cm. Sizes: BESTÅ frame 60 × 40 × 64 (IKEA);
game cases 19 × 13.5 × 1.4 and 17 × 13.5 × 1.2; jewel case 14.2 × 12.5 × 1; LP sleeve 31.4; pads 16 × 10.6 × 6.6; runners Emuca 400 mm, Accuride 350 mm (see the fit proof).

Every drawer out, in place: the game drawer, the upper LP crate and the middle slim game drawer 30 cm out, the plinth drawer 25 cm out. In use, open one LP crate at a time (a full one is about 30 kg).

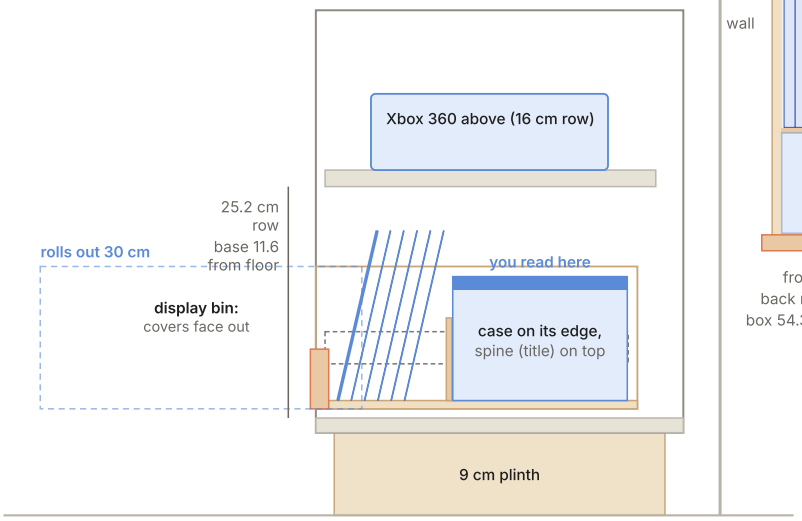
### The game drawer (bay 1, bottom row)

Unchanged. Two display bins at the front show covers face-out even with the drawer shut (Xbox left, PS2 right, a small tray between them); behind them the rest of the games stand on edge with the spine on top, so every title faces up when you roll it out.

### The game drawer: covers on show at the front, every title readable from above at the back

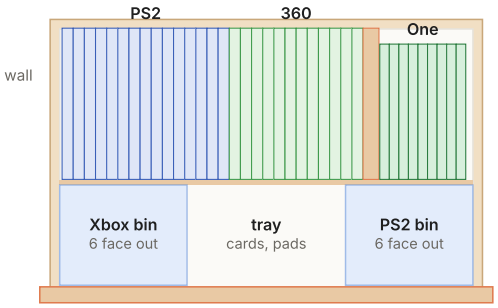
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

**Side section through bay 1**




box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 4 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

**Top view, drawer pulled out**




front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

- **Box:** about 54.3 × 35 cm outside on 350 mm full-extension soft-close runners screwed to the BESTÅ sides; sides 15 cm; 9 mm base lined in felt. About 48 games.
- **Runners:** Accuride DZ3832EC (43 kg) or Screwfix Essentials 964VN (40 kg); not IKEA's BESTÅ runners, which aren't full extension.
- **The bricks:** both Xbox bricks lie flat at the back of their shelves, on their rubber pads, beside the channel's power leads, with nothing on top. Microsoft only asks that the space is ventilated[26], and these bays are open at the front and the back.

# 4 · The records and the turntable

### Your plant stand and your vinyl stand

- **Your plant stand:** 98 × 29.5 × 158. Two slatted towers of five tiers, with an open middle holding two shelves for taller plants.
- **Your vinyl stand:** 70 × 32 × 78. A black steel frame and rustic wood: two face-out bays on top, LPs standing between wire dividers below.
- **Sizes** from the listings you sent[32,33].
- **Two good places:** the window's west corner and the chimney breast (free now the table is by the stairs), so two layouts. In the 3D room, Tools › Layout switches between them; in Edit room you can drag either anywhere.
- **A, recommended: plants in the window corner, records on the breast.**
  - The plant stand goes against the window wall, between the curtains' stack and the pipe boxing: 107 cm for its 98, so 4.5 each side. Measure before it goes in; if it's tight, open the curtains' stack a little towards the window.
  - Its plants get daylight from the window beside them; at 158 it leaves 77 to the ceiling.
  - The vinyl stand goes on the breast, centred, under the “now playing” ledge (23.6 cm clear). Pick a record from its top bays, play it on the deck an arm's reach away, stand its sleeve on the ledge.
  - About 100 more LPs.
- **B: records in the window corner, as today, plants on the breast.**
  - The plants become the breast's display, 9.75 cm clear each side.
  - At 158 tall the stand comes up past the two frames, so the frames go up above it (or elsewhere) and the “now playing” ledge comes off.
- **Either way:**
  - Your red lamp moves from the window corner to the sofa's west end.
  - Drip trays or a waterproof mat under the pots protect the slats and the carpet.
  - Felt pads under the vinyl stand.

Layout A, from the 3D room: your plant stand in the window's west corner, its towers and open middle full of plants.

Layout A: your vinyl stand on the chimney breast, under the “now playing” ledge, beside the turntable.

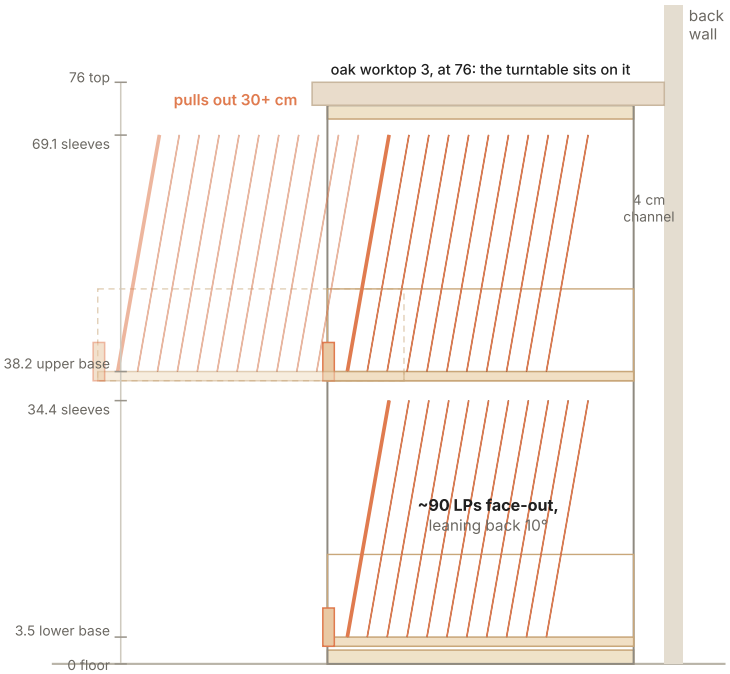
Layout B: the plant stand on the breast instead; the vinyl stand goes back to the window corner.

The vinyl is in one place, in the alcove: the deck on the worktop, the records in the two LP crates right underneath it, the preamp tucked under the shelf above, and the “now playing” ledge on the breast an arm's reach away. Pull a crate, pick a record, play it, and stand its sleeve on the ledge.

### Bay 3: two LP crates under the turntable, a slim game column beside them

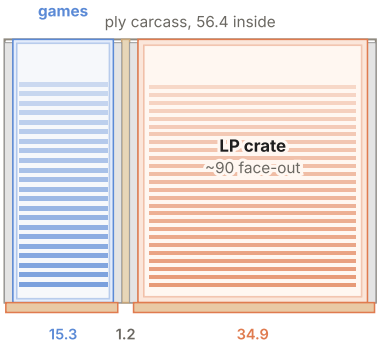
Section left through the LP crates, plan right. Pull a crate and flip through the covers like a record shop browser; the slim drawers flip the same way for games

**Side section through bay 3**




carcass 18 mm birch ply, 60 × 40 × 73, back open to the channel · crates 40 deep on 400 mm runners
12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

**Bay 3 from above**




LP crate 34.9 × 40, a 33.1 trough: LPs snug, no dead space
slim drawers 15.3 × 40, 14.1 inside: DVD-size cases face-out
12 mm ply divider · 12.7 mm runners each side
an LP crate is about 30 kg full: open one at a time

## How bay 3 works

- **Why it changed:** with LPs only, v15's trays had about 10 cm of empty width each side. Now a 12 mm divider splits the bay: the LP crates take only the width LPs need, and the rest becomes a slim column that does a job.
- **The LP crates:** a 33.06 cm trough, the snug width record shops use[36]: the sleeves stand with a little play and you flip them by their top edges. They lean back 10°; the last one leans on a raked oak rest. Outer sleeves up to 32.4 still fit.
- **Capacity:** about 90 LPs per crate; two crates, about 180 LPs, the same as before.
- **The slim column:** three drawers 15.26 wide, 14.06 inside. The bottom two hold DVD-size game cases face-out (13.5 wide), about 22 each, flipped like the records. The top one holds four pads on edge[35], the TV, soundbar and streaming remotes, and a USB-C lead fed from the 10-gang to charge a pad in the drawer.
- **Runners:** 400 mm full extension everywhere; 68–70 kg a pair under the crates. A full crate weighs about 30 kg: open one at a time, and fit the anti-tip brackets before loading.

| Dimension | Figure | How it adds up |
|---|---|---|
| Bay 3 carcass | 60 × 40 × 73 outside, 56.4 × 69.4 inside | 18 mm birch ply; back open to the channel; a 12 mm divider 17.8 in from the left |
| Across the bay | 15.26 + 2.54 + 1.2 + 34.86 + 2.54 = 56.4 | Slim drawer, its runners, the divider, the LP crate, its runners: nothing left over |
| LP crate | 34.86 × 40 outside, 33.06 × 38.2 inside | 9 mm sides and ends; 0.8 either side of a 31.4 sleeve |
| Slim drawers | 15.26 × 40 outside, 14.06 inside | 6 mm sides; bases at 2.3, 25.1 and 47.9 |
| Lower crate · upper crate | Undersides 2.3 · 37.0; sleeves to 34.4 · 69.1 | 2.6 and 2.1 cm finger room; top panel at 71.2 |

### The turntable zone

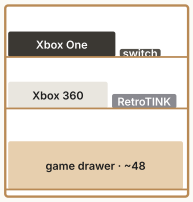
- **The deck:** on isolation feet, 2 cm back from the worktop's front edge, with 8.8 cm behind it for the lid's hinges. The zone takes a deck up to the size of an AT-LP120X (45.2 × 35.2 cm); most decks are smaller.
- **The lid:** AT's cover is 44.1 × 34.6 cm[17], so standing open its top edge is at about 121; the shelf's underside is at 130.
- **Bass:** the sub is in bay 2, in the same base the deck sits on. It stands on an isolation riser, clear of the divider and the frame, and the deck stands on isolation feet. If a record skips or hums at film volume, turn the sub down for vinyl.
- **Signal path:** turntable set to PHONO → the preamp under the shelf → a 2 m RCA-to-3.5 mm lead down the groove behind the right upright and along the back of the worktop to the soundbar's AUX. If the deck has no PHONO/LINE switch, plug it straight into the AUX.
- **Soundbar:** Music mode, bass up, surround, night and dialogue modes off. Always the AUX cable, never Bluetooth.

## Storage options, to scale

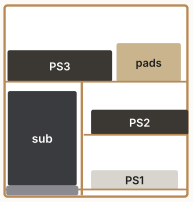
### Storage options: what fits each space, to scale

Front views and views from above of every storage space in the wall, with counts, and where the overflow would go

**Bays 1 and 2: inside each BESTÅ frame**
56.8 wide × 40 deep inside; rows as planned, front view



Bay 1 · Xbox



Bay 2 · PlayStation + sub

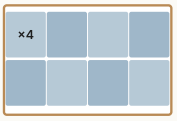
Swap any 16 cm row for two KUGGIS boxes, 26 × 35 × 15, side by side (52 of 56.8 wide).

**Bay 3 from above: LP crate and slim drawer**
56.4 inside: a 15.3 slim drawer, the divider, a 34.9 LP crate

games | divider | LPs: no dead space

**≈ 90 LPs a crate, two crates**
= 180 LPs, all face-out
beside them 44 games face-out
and a drawer for pads and remotes
trough 33.1: thick outer sleeves fit

**Hidden plinth drawer, from above**
box 52 × 34 × 5 inside, under bay 2; finger pull, not push-to-open




**32 PS1 games flat**
jewel cases 14.2 × 12.5 × 1, 8 stacks of 4
or: remotes, spare pads, cables, manuals

**Series X shelf, front view**
50.5 wide, 134 to the high shelf at 192: 58 clear

**15 air each side, 21.4 above**
Microsoft: at least 4 above, 10–15 recommended all round
Under it: the deck, lid opens to 121

**High shelf, front view: as planned, or as LP overflow**
186 long, 30 deep, 39 clear to the ceiling; the right-hand 50.5 keeps Deco B and the framed 7-inch

about 1 m of books, a lidded box, a print

**As planned**

≈ 380 LPs spine-out at about 3.3 mm each; a 31.4 sleeve overhangs the 30 cm shelf by 1.4

**As LP overflow**

**PC shelf, from above**
82 along the wall × 28 out, at 155.4; keyboard tray in a 7 cm housing under it

side wall along the top · alcove to the right

Keyboard tray: 300 mm runners, rolls 25 cm out; the housing stops 12 short of the sofa end so the cables drop past it. Full-size wireless keyboard and mouse.

**If the collection outgrows the wall**
two overflow options, both face-out or spine-out to match

**1 · LPs spine-out on the high shelf: about 380 more**
No building; books move to the second option's top.

**2 · A second record unit, west of the sofa, under the window**
Another LP column like bay 3 (40 × 46 × 73, two crates): 180 LPs.
At x 300–360 it leaves 123 cm to the table set for six (100 to walk).

front view, 60 × 73 (shown at half height)

All to one scale, 2.2 px per cm. Sizes: KUGGIS box 26 × 35 × 15 (IKEA); jewel case 142 × 125 × 10 mm (Wikipedia: Optical disc packaging); consoles, sleeves, Series X and the trays from the fit proof (section 3 of the plan).

## Where every record lives

| Where | What | How many |
|---|---|---|
| Bay 3, the two LP crates | LPs, face-out | About 180 LPs |
| The breast | The “now playing” ledge | 3 LPs, the one that's on and the next two |

| | | |
|---|---|---|
| Above the TV | Oak record ledge, the "now playing" wall | 4 LPs |
| Side wall | The oak run round the corner | 4 LPs |
| Your vinyl stand | LPs, face-out on top, spine-up below | About 100 LPs (50 + 50) |
| **Total** | | **About 291 LPs: 191 in the wall, every one face-out, and 100 in the stand** |

If the collection is bigger than this, tell me the count. The neatest overflow is another LP column like bay 3's (40 wide, two crates, 180 LPs), or LPs spine-out on the high shelf.

# 5 · The dining wall

**The dining cabinet is the plan**: one oak cabinet under your shelf with the table and all six chairs inside it. You asked for "a seating of 6" with "the chairs to exist within the same thing the table is built into". Folded, it's a plain run of oak, 198 × 24 × 189, as deep as the pipe boxing, with the stairs' foot left clear: no table on the wall, no separate cupboard, nothing on show. Open, the table is 180 × 90 and six sit in comfort, three a side.

Packed away, in the 3D room: the table standing up in its bay, the drawer under it and the linen flap over it; the chair bay's two doors beside it.

- **The table bay, at the stair end.** A 180 × 90 solid oak top in two 90 × 90 halves stands up inside it, folded face to face. Its oak underside, framed by its apron, is the bay's front. Under it, a drawer for the mats; over it, a lift-up **linen flap** for the cloth, the napkins and the candles.
- **The chair bay, at the window end.** Behind two doors, the six FRÖSVI chairs stand folded: four on the floor, two side by side and two deep, and two on the shelf, with the six seat pads stacked in front of them.
- **Why 180, not v26's 225:** six fit in comfort at 58 cm of edge each without moving the coffee table, each half is lighter (about 16 kg), the chair bay holds exactly the six, and the shorter halves leave room over the folded table for the linen flap.

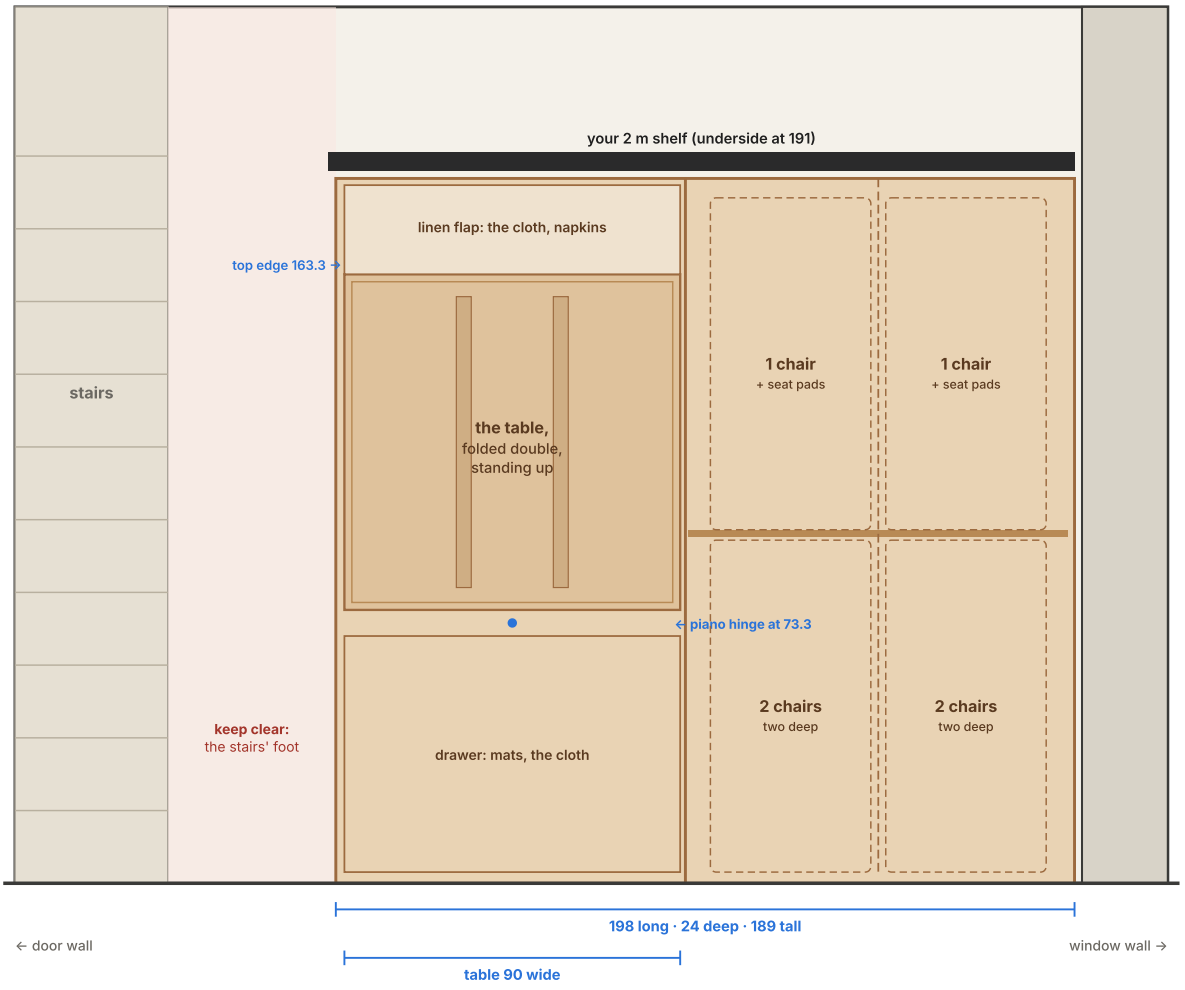
### Using it

1. **Opening:** clear the floor in front of it and push the ottoman to the sofa. Pull the knob at the top of the folded table and lower it halfway; unfold the inner half's two legs until each clicks[44], then lower it onto them. It turns on the piano hinge at its foot[43].
2. Lift the outer half up and over on its flip-top hinges[30], unfold its two legs until they click, and lower it. The table is now 180 long, its top at 76, level with the media wall's worktop.
3. Open the chair bay's doors right back (they open past flat[46]) and take the chairs out: the top two first, then the four on the floor. Seat pads on.
4. **Folding away:** the same the other way round: chairs and pads back in, the outer half's legs folded and the half flipped back onto the inner one, the inner legs folded, then swing it up until the catches take it. Two hands; never with anything on it.

The idea is a proven one: fold-down tables built into wall units are made this way[45]. The cabinet stands on the floor and is screwed to the wall at the top so it can't tip.

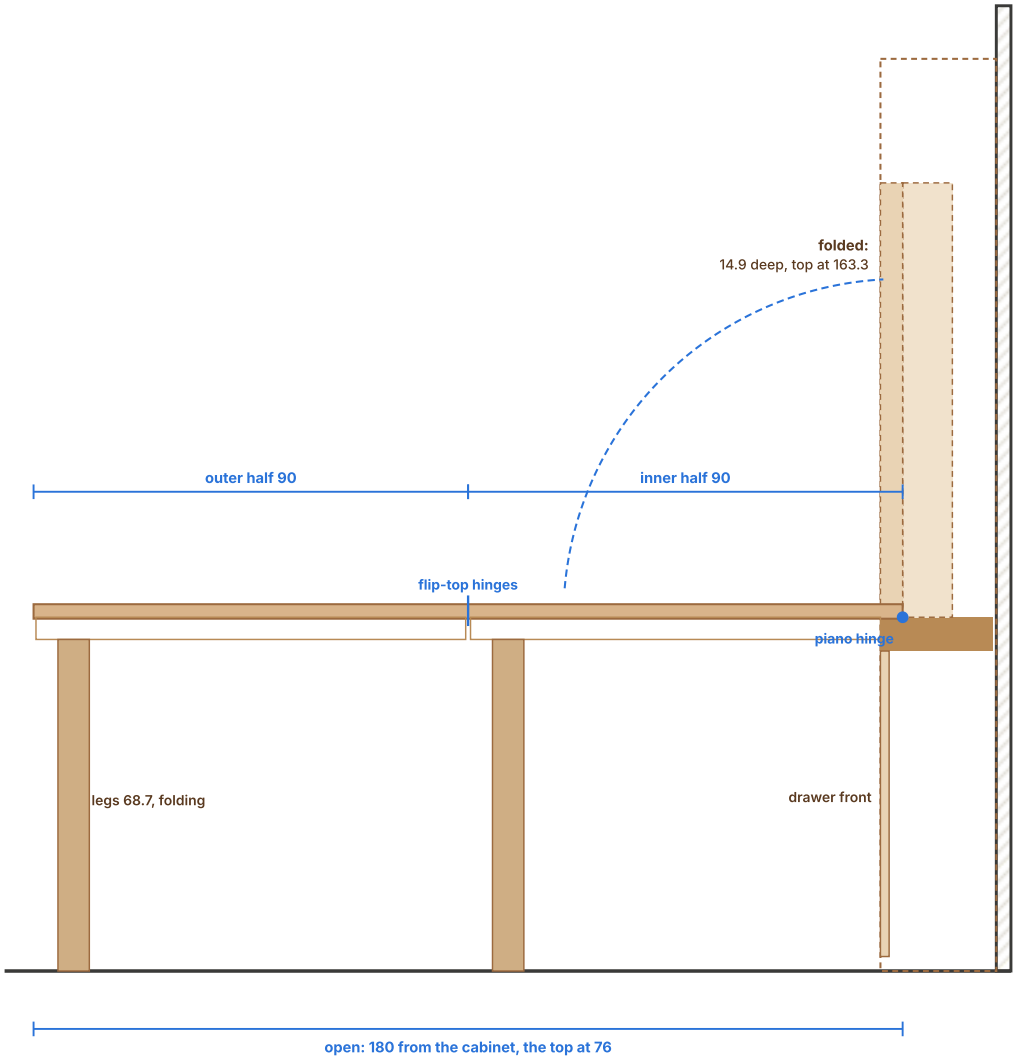
## The dining cabinet, packed away

The stairs wall from the room, to scale (1 cm = 2.55 px). One oak cabinet: the table stands up in its bay, the six chairs behind the doors (drawn see-through).



### The dining cabinet: how the table comes down

Cut through the table bay, the wall on the right. Folded (dashed): the two halves face to face, standing up. Open: swing it down, flip the outer half over.



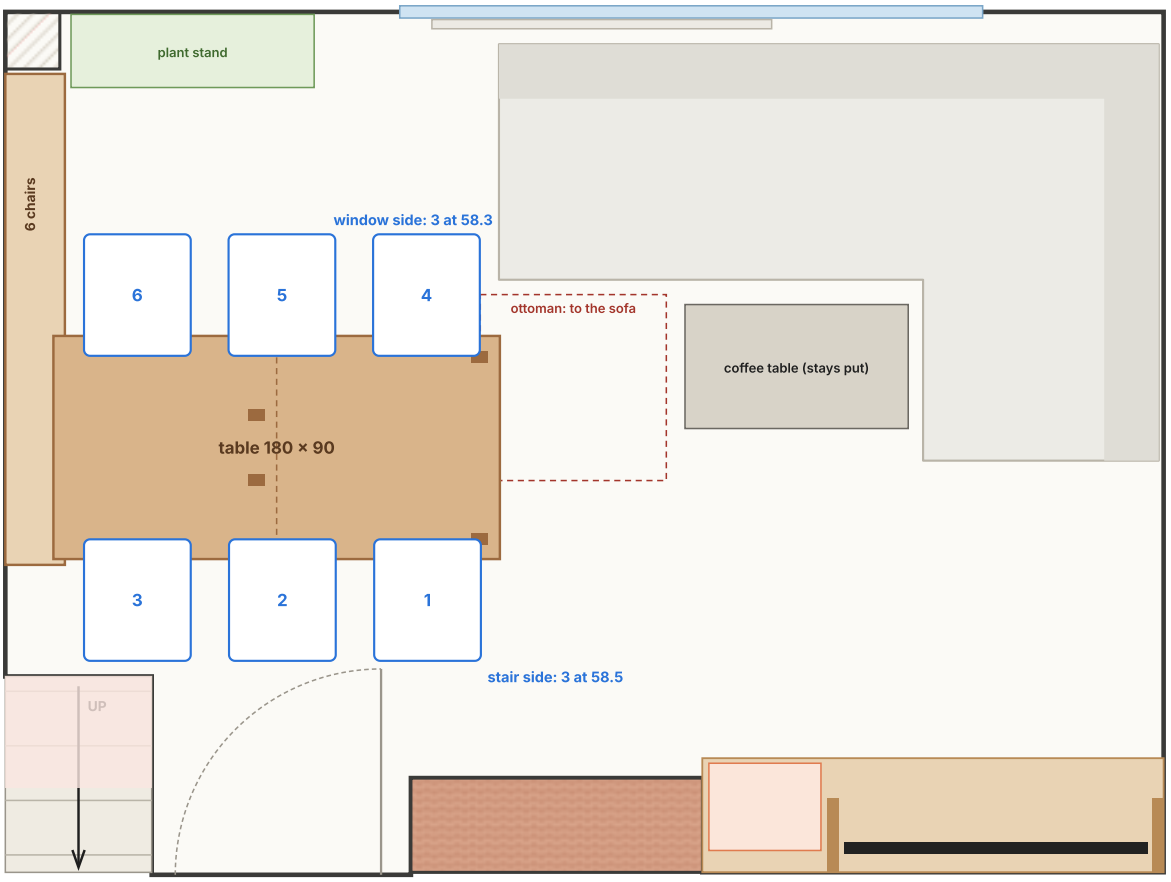
The chair bay open: four folded chairs on the floor, two side by side and two deep; two on the shelf, the seat pads in front of them.

## How many sit down

**Six, all in comfort:** three on the stair side (58.5 cm of edge each) and three on the window side (58.3 each, from the sofa's arm to the cabinet). The coffee table stays where it is; a seventh chair fits at the far end if you bring one.

### The dining cabinet open for six

From above, to scale. Three on the stair side, three on the window side, all in comfort; the coffee table stays where it is.



Open for six in the 3D room: three a side, bar 2 turned up over it by the table rule.

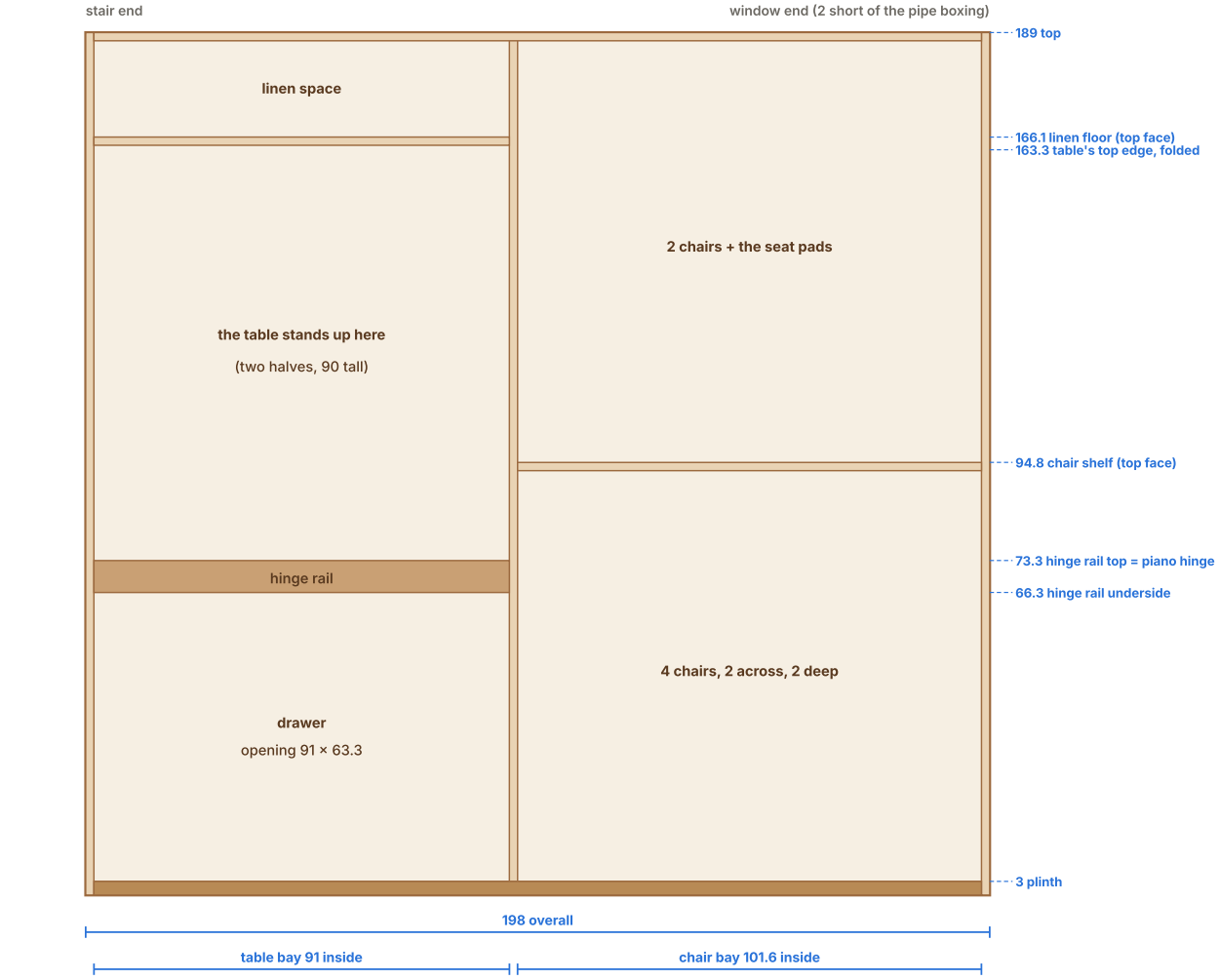
| Table | Seats | How | Notes |
|---|---|---|---|
| The dining cabinet (the plan) | 6 in comfort | 3 on the stair side, 3 on the window side | 58.5 and 58.3 of edge each; a seventh at the far end with a spare chair |
| B: the flip-top by the stairs (the alternative) | seven at a squeeze | three a side (53 each) and one at the end | five in comfort: two a side and the end |
| A: the flip-top on the chimney breast | six at a squeeze | three a side, 53 each | four in comfort; no end seat: the sofa's in the way |
| A fourth chair a side at the cabinet | eight round it | 175.4 and 175 of edge | 43.8 each: under the 55 squeeze; v26's longer table did it, with the coffee table moved and all eight chairs in a bigger bay |
| A longer table in the cabinet | halves over 113.9 | they must stand up under your shelf | 113.9 is the most that folds away; 90 leaves the linen flap's space and keeps each half to about 16 kg |

### How it's built

Step 11 of the build guide builds it in order, with every size: the carcass, fixing it to the wall, the hinge rail, the two halves with their aprons and legs, the joint, hanging the table, then the drawer, the linen flap and the doors. The two drawings below are the ones to have open on the bench.

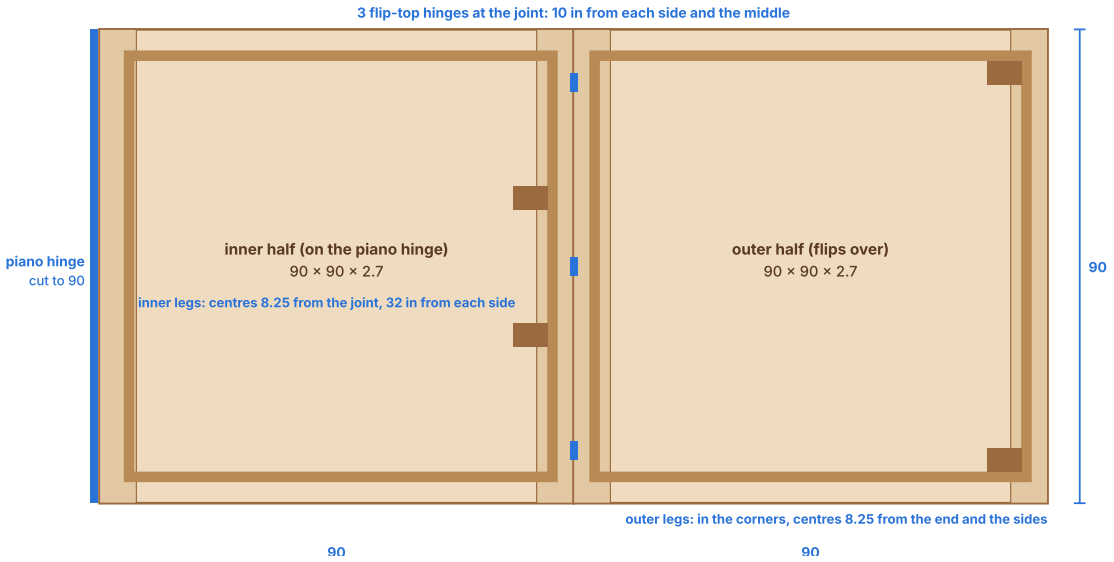
### The dining cabinet's carcass, from the front

To scale (1 cm = 3.2 px), doors, drawer and table off. Heights are from the floor, to the top face of each panel; all panels 18 mm oak-faced ply, 22.2 deep.




### The table's two halves, from underneath

Laid out open, to scale. Breadboard ends 7 wide; apron frame 2 × 4.6, 4 in from the sides, 3 from the ends; legs fold inside it.




- **The carcass:** 18 mm oak-faced ply, the merchant cuts it to size. Two ends 189 tall; the top, the floor and the 6 mm back between them; a partition making the table bay 91 wide inside; the chair shelf and the linen floor

screwed in; two 1.2 cm battens under it for the plinth.

- **The hinge rail:** solid oak, 7 tall, across the table bay with its top at 73.3: it carries the piano hinge, and the table's whole weight when it's open.
- **The halves:** edge-glued oak with pegged breadboard ends, each on an apron frame 2 thick and 4.6 deep; the legs fold flat inside the frames on self-locking brackets.
- **The joint:** three brass flip-top hinges set into the meeting edges; open, the halves are flush; folded, the outer half lies flat on the inner one.

It costs about £881–1,535, the chairs included: the "6 Dining wall" rows of the buy list (section 11).

Safety: never swing it up or down with anything on it; check all four legs have clicked before you load it; let the catches take it when it's folded; the cabinet is screwed to the wall at the top.

### The alternative: the flip-top, by the stairs (B) or on the breast (A)

Until v26 the plan was a flip-top table. It stays here, drawn, checked and costed, in case you'd rather: with its chair cupboard it comes to about £666–1,265, against the cabinet's £881–1,535. It's a **flip-top gate-leg table that never leaves the wall**. A solid oak top, 110 × 80 cm, hangs on lift-off hinges on an oak backboard and rests on two folding brackets. A second 110 × 80 oak half is hinged to its front edge on three brass flip-top hinges[30]: flip it out and the table runs 160 cm into the room, its far end on a folding oak gate leg, at the same 76 cm as the worktop. Four sit with room to spare, six elbows-in; by the stairs a seventh sits at its end. Every day it folds flat against the wall, about 13.3 cm deep. Nothing comes off the wall.

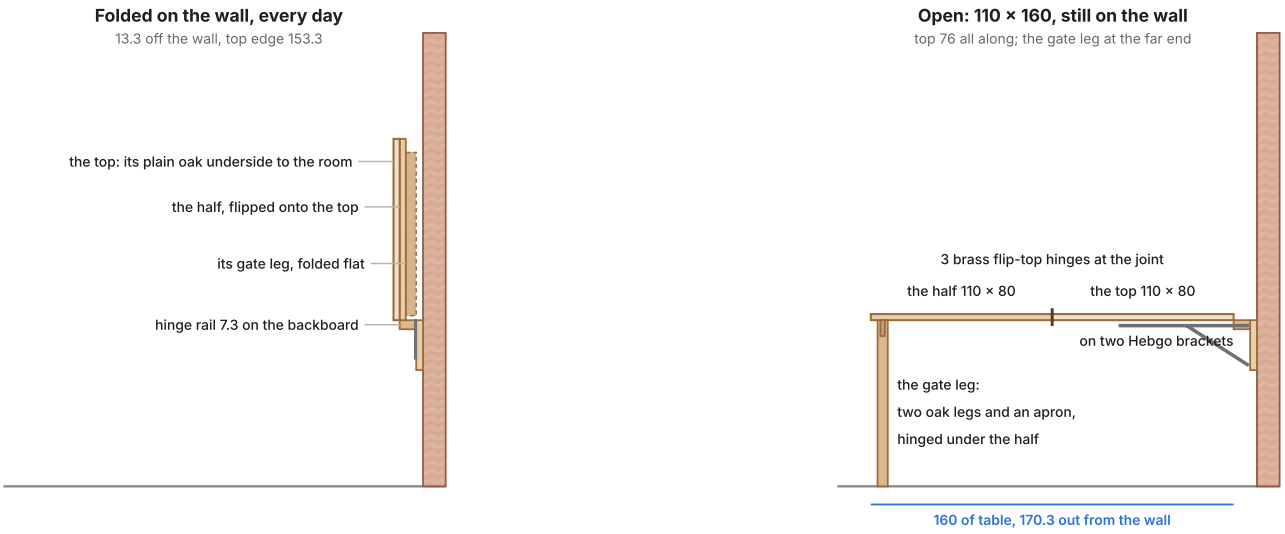
Option A, open on the breast and set for four: the top on its brackets, the half flipped out beyond it on its gate leg, two down each side. Bar 2, on the pendant point by the stairs just out of view, lights the table and the framed LPs.

Set for six, three down each side (53.33 cm each: elbows in). The camera stands back in line with the alcove's edge, so both spot bars show. The table's end stops 30.7 cm short of the sofa, so on the breast nobody sits at its end.

### The flip-top table: folded, and open

In section through its centre line, to scale, the wall on the right of each. It never comes off the wall, and there's no leaf to store.

1. To open: turn the two buttons and lower the pair onto the brackets as they swing out and lock (two people: about 37 kg).
2. Flip the half over onto the room side: it opens on its hinges to lie flush with the top. Swing the gate leg down until it stops.
3. To fold: gate leg up flat, half back over onto the top, then lift the pair up to the wall and turn the buttons.




### The flip-top's two places: under your shelf by the stairs (B), or the chimney breast (A)

You asked what it would look like on the wall going up the stairs, under your 2 m shelf. Your photos put that wall at about 268 cm from the stair box to the window corner, with the shelf about 196 cm up. The flip-top works on either

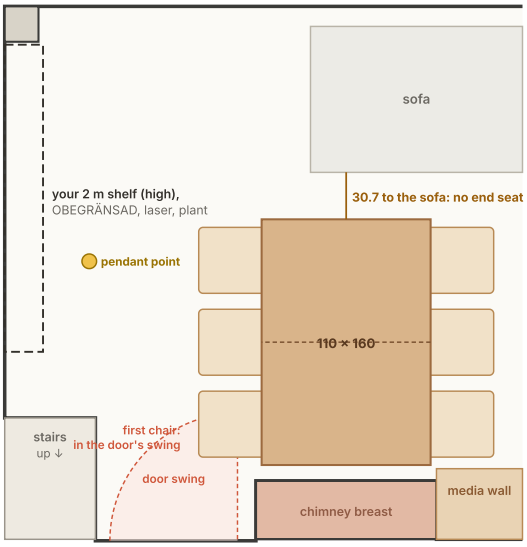
wall: only the backboard's fixings change. Both are drawn in the 3D room on the plan page, folded and open, from every angle and in each lighting scene.

### Where the dining table can live: the chimney breast or the stairs wall

Both to scale, the flip-top open at its fullest, the west half of the room from above (window at the top). It never comes off the wall: four, six or seven sit at the same tab

**A · On the chimney breast: up to six**

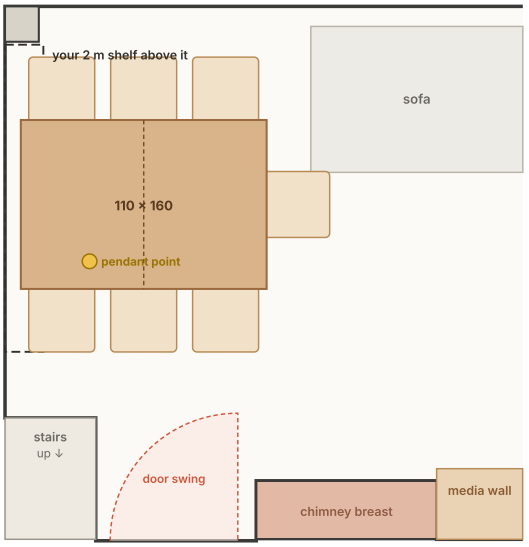
the flip-top open towards the window, three a side; folded flat every day




+ next to the media wall; the stairs wall keeps the shelf, lamp and hat
+ four sit with room to spare, six elbows-in
− no seat at the end: the sofa is 34 beyond it
− the west side's first chair sits in the door's swing

**B · Under your shelf, by the stairs: up to seven (the default)**

the flip-top open into the room, three a side and one at the end




+ seven at a squeeze, five with room; right under the pendant point
+ out of the door's swing; frees the breast for the “now playing” ledge
− the side chairs are tight: 43 to the stairs' foot, 33 to the window
− the lamp moves while it's open; check the wall before fixing the brackets

Option B in the 3D room, set for dinner: the flip-top open into the room under your shelf, bar 2 on the existing pendant point over it, the stairs behind the first chairs.

Option B folded on an everyday evening: 13.3 cm deep against the wall, 37.7 cm under the shelf with your black OBEGRÄNSAD panel, the laser (off) and the plant hanging from the window end beside the pipe boxing; your red lamp in the corner.

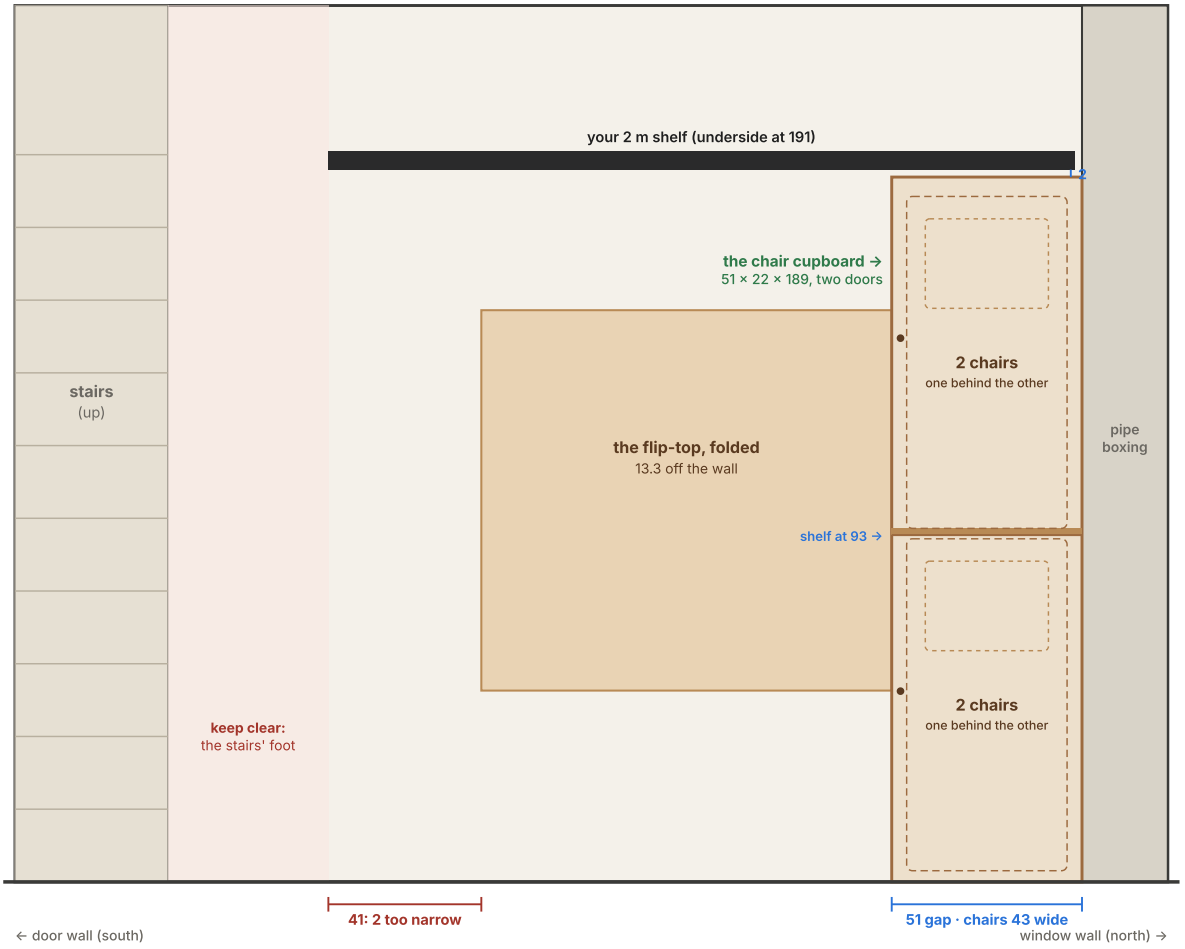
- **A, on the breast (the alternative): up to six.** It opens towards the window, 170.3 cm out, three a side. Its end stops 30.7 cm short of the sofa, so there's no seat at the end; for six, slide the coffee table towards the TV first; the west side's first chair sits in the door's swing, so keep the door shut while you eat (or rehang it to open into the hall).
- **B, by the stairs: up to seven.** It opens into the room from the stairs wall, three a side and one at the end, right under the existing pendant point and out of the door's swing; it frees the chimney breast. The side chairs are tight (43 cm to the stair box behind the stair side, 33 to the window wall), so tuck them in after a meal; your red lamp moves from the window corner while it's open. The west wall's build is unknown: knock on it before choosing (solid takes plugs; studwork means fixing the backboard into the studs).
- **What moves for B:** the dresser goes (its turntable and records already have homes in the media wall, all face-out); the sofa stays where it is.

### The flip-top's chairs: a chair cupboard built in beside it (B)

You asked for "a fully pack away setup", with "ideally 6 folding chairs but 4 is fine". So the chairs get their own oak cupboard, built in beside the table, in the 51 cm gap between the table's window end and the pipe boxing. The table itself doesn't change or move. Folded, the table and the cupboard read as one run of plain oiled oak under your shelf, with nothing left out. The chairs drawn are IKEA FRÖSVI, TERJE's successor at the same size[39]: they fold to about 43 × 9 × 89.

## Keeping the dining chairs: the stairs wall from the room, all packed away

To scale (1 cm = 2.55 px). The table doesn't change or move; an oak cupboard beside it holds four folded chairs (43 × 9 × 89); doors drawn see-through.




**1 · The chair cupboard (the plan): four folded chairs packed away beside the table, 22 deep like the pipe boxing; about £120–240.**

2 · All six inside: build it 31 deep, three chairs deep (9 proud of the boxing). Or keep chairs five and six in the garage.

3 · Cheaper: two 200 mm storage hooks in the same gap hold the same four on show, £8–15.

In the 3D room, doors open: two folded chairs stand on the cupboard's floor and two on its shelf, each pair one behind the other. Doors shut, it's a plain oak column flush with the pipe boxing.

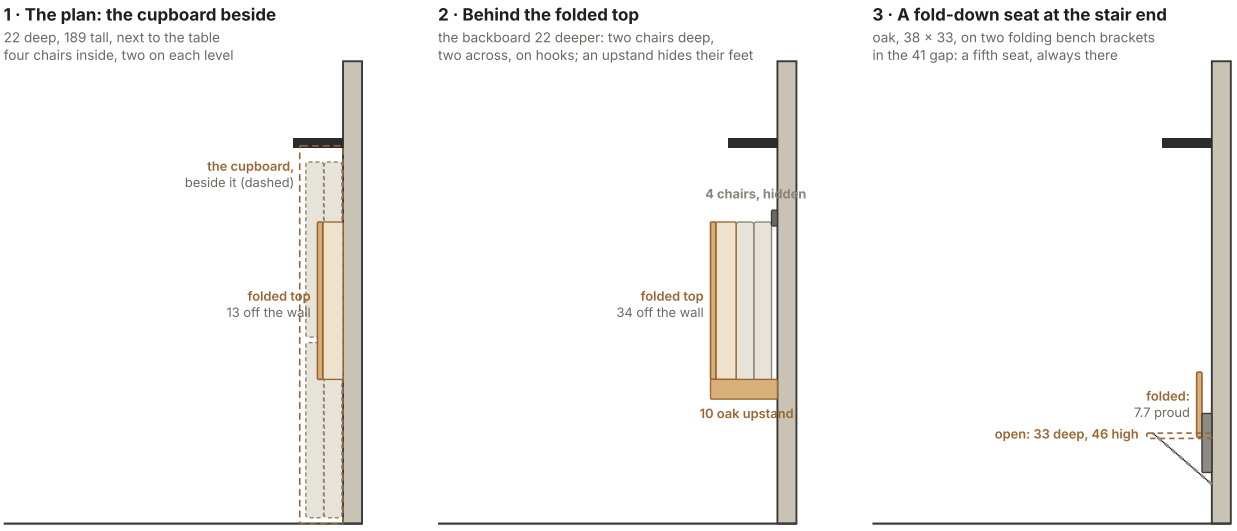
- **The cupboard:** 51 wide × 22 deep × 189 tall, in oak (or oak-faced ply) like the table. At 22 deep it stands no further out than the pipe boxing beside it, and it stops 2 cm under your shelf. A fixed shelf at 93 has a 3 cm lip. Two doors, upper and lower, are hinged on the table's side. About £120–240 with the hinges and latches. It's in the cutting list.
- **Four inside:** two chairs stand on the floor and two on the shelf, each pair one behind the other: 2.2 cm a side and 1.6 cm to spare in the depth, so they can't rattle. The floor takes their weight; two screws through the top rail into the wall stop the cupboard tipping.
- **Six inside, if you'd rather:** build it 31 deep and the chairs go three deep, so all six pack away. It then stands 9 cm proud of the boxing's face. With the 22-deep one, chairs five and six go in the garage (under the stairs is a brick side wall with plasterboard beneath, so there's no way in from the room).
- **Using it:** open the doors and take the chairs out, then open the table. With the table open, the window side's first chair starts where the cupboard ends. That changes nothing for four; six are elbows-in on that side.
- **The cheaper option:** two 200 mm storage hooks[40] in the same gap hold the same four, two on each, for £8–15. The chairs stay on show.
- **Not at the stair end:** that gap is 41 cm, too narrow for a 43 cm chair, and the foot of the stairs has to stay clear. It takes a fold-down seat, though (below).

### More ways to keep the chairs with the flip-top (B)

You asked us to "explore any further ways we can build the chairs into the table build". With the flip-top, the cupboard is the way; these two go with it or instead of it, and both are in the fit proof (section 3).

**Building the chairs into the table: three ways, cut through the stairs wall**

To scale (1 cm = 1.45 px), the wall on the right. The plan stays as 1; 2 and 3 are options that need your say.




**1 · the plan: about £120–240; six inside if it's built 31 deep**
2 · about £40–80 (hooks, a deeper oak backboard and an upstand); the table stands 34 off the wall, 22 further into the room, open too
3 · about £50–90 (two brackets and an oak seat); goes with 1 or 2: four or six chairs put away, and a fifth seat that's always there

Option 3 in the 3D room (Layout, "Fold-down seat at the stair end"): the seat down beside the folded table.

- **Behind the folded top:** make the table's backboard 22 cm deeper and four folded chairs hang behind the folded top on two hooks, two across and two deep, with a 10 cm oak upstand under the top hiding their feet. Folded, nothing shows; open the table and they're right there. The cost: the table stands 34 cm off the wall instead of 13, 22 cm further into the room, folded and open. With the cupboard as well, all six chairs (or eight) pack away. About £40–80.
- **A fold-down seat at the stair end:** an oak seat 38 × 33 on two folding bench brackets[41] (330 mm, 150 kg a pair) in the 41 cm gap. Folded it's 7.7 cm proud, flatter than the table beside it; down, it's a seat at 46 at the table's south end, so a fifth place is always there without a chair to fetch. About £50–90; knock the wall first for the fixings. In the 3D room: Layout, "Fold-down seat".
- **Ones we looked at and set aside:** stools stored in the gate leg's frame (the frame is only 4.6 cm deep); a storage bench under the table (it would stop the table folding flat); chairs under the stairs (a brick side wall).

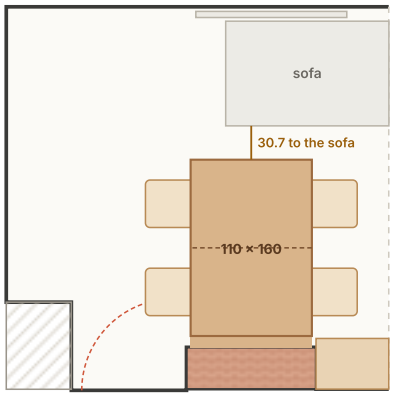
### How the flip-top works

- **The top and the half:** solid oak with **pegged breadboard ends**, 7 cm oak strips across each end on a tongue, pinned with three oak pegs in slotted holes so the boards can move. They stop the panels cupping when they live upright, and they frame them like the uprights frame the TV. Rounded 5 cm corners on the half's free end, 1 cm chamfers underneath so it reads slim.
- **Backboard and hinge rail:** oak, 110 × 22 × 3 from the worktop's offcut, screwed to the wall with plugs every 20 cm; along its top, a 110 × 4 × 7.3 oak hinge rail stands it out so the folded half and its gate fit behind the top. The rail's top is flush with the open table, at 73.3 under the top.
- **Brackets:** two Häfele Hebgo heavy-duty folding brackets (287.43.428: 580 mm arm, 500 kg a pair[18]), 80 cm apart on the backboard's face, under the top. Raised, they lock on their own; a light lift on the arm releases them.
- **Hinges:** three 100 mm lift-off hinges[19] along the rail's front edge carry the top; three brass flip-top hinges, 1.27 × 7 cm, set into the meeting edges of the top and the half, open to 180°, so the half lies flat on the top to fold and flush with it open.
- **The gate leg:** two tapered oak legs (4.5 to 3 cm, 73.3 long[20]) joined by a 7 cm oak apron, hinged under the half 3 cm in from its far edge on two butt hinges, with a folding stay each side to lock it upright. It folds flat inside the half, along its 80 cm.
- **Folded:** two oak turn buttons on the wall, plus a magnetic catch, hold the top edge at 153.3. The top's underside faces the room: plain oiled oak between the pegged breadboard ends, so folded up it reads as a calm oak panel under the two framed LPs. It's about 37 kg to swing up or down: two people, one at each corner.
- **Chairs:** drawn at IKEA TERJE size (43 × 49, seat 48[22]), now sold in the UK as FRÖSVI at the same size[39]; it folds flat and hangs on a hook (above). Any dining chair with a 45–48 cm seat works.

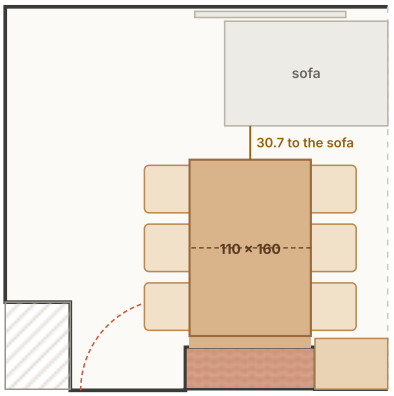
#### The flip-top table for four, six or seven

To scale, the west half of the room from above, window at the top. Red: the door's swing. Blue: the clearances that matter

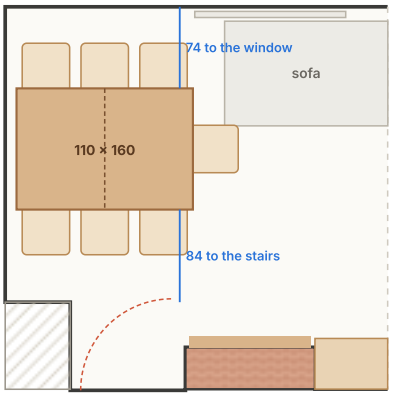
**Four, on the breast**
the flip-top open: two down each side




**Six, on the breast**
three down each side, elbows in




**Seven, by the stairs**
three a side and one at the end




- Per diner: 60 cm of table edge (55 at a squeeze). Table edge to a wall: 60 cm is a tight minimum, 75 to aim for, 100+ to walk behind someone seated.
- The flip-top never leaves the wall; the dashed line is the joint where the half flips out on its gate leg. Red: the door's swing.
- Chairs drawn at IKEA TERJE size (43 × 49, seat 48): 28 cm from seat to table top, the usual 25–30.

| Seats | Where | Per diner | Clearances |
|---|---|---|---|
| Four | Either wall, open: two down each side | 80 each | On the breast its end is 30.7 from the sofa; by the stairs the side chairs tuck in |
| Six | Either wall, open: three down each side | 53.33 each (55 is a squeeze) | On the breast slide the coffee table towards the TV and keep the door shut |
| Seven | By the stairs only: three a side and one at the end | 53.33 each; the end has the full 110 | The end chair clears the sofa's front; the lamp moves |

Clearances from a UK planning guide: 60 cm of table edge per diner, 60 cm from the table to a wall as a tight minimum, 75 to aim for, 100–110 to walk behind someone seated[23].

### The flip-top's cut list and hardware

| Part | Material | Size (cm) | Qty |
|---|---|---|---|
| Table top · the half | Solid oak, edge-glued staves | 96 × 80 × 2.7 centre + two 7 × 80 breadboard ends (110 overall), each | 1 · 1 |
| Breadboard pegs | 10 mm oak dowel, slotted holes in the tongue | 3 per end | 12 |
| Gate legs · apron | Solid oak, the legs tapered on two inside faces | 73.3 × 4.5 tapering to 3 · 94 × 7 × 2 | 2 · 1 |
| Backboard · hinge rail | Solid oak, from the worktop offcut | 110 × 22 × 3 · 110 × 4 × 7.3 | 1 · 1 |
| Turn buttons | Oak offcut | 6 × 2.5 × 2.2, on a single screw | 2 |
| Folding brackets | Häfele Hebgo 287.43.428 | arm 58, height 22 | 2 |
| Lift-off hinges | Union PowerLoad, 100 × 88 | one 3-pack, same hand | 3 |
| Flip-top hinges | Lee Valley 40574, solid brass | 1.27 × 7 | 3 |
| Gate hinges · stays | Steel butt hinges, 75 mm · folding (locking) stays | — | 2 · 2 |

Safety: never fold it up with anything on it, keep both turn buttons turned when it's up, check the brackets have locked (a click each side) and the gate's stays are locked before you load it.

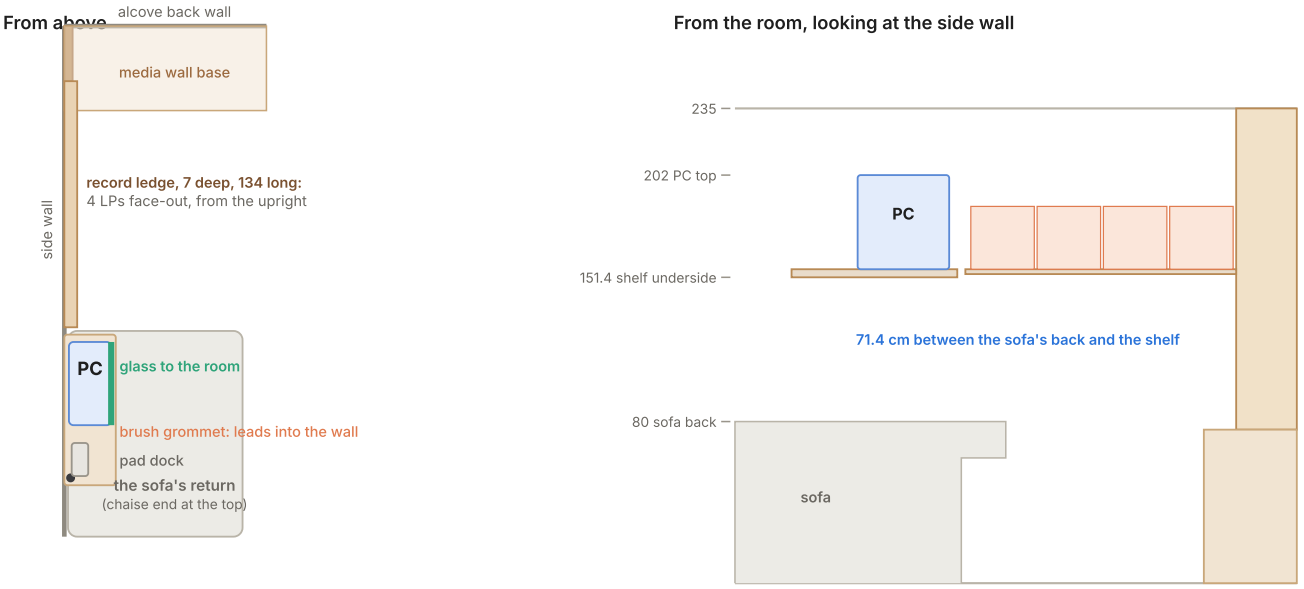
# 6 · The side wall

One oak line at 155.4 cm runs from the ledge over the TV, round the corner on the left upright, and along the side wall. The side wall's part is **one run**, 4 cm thick all along: a 7 cm record ledge with four more LPs face-out that **sweeps out in one smooth curve** to the 28 cm shelf the PC sits on, ending in a 5 cm radius like the table's corners. A warm LED strip in a groove under its front edge washes the wall below it, on the uprights' driver and dimmer. Two separate shelves at two depths become one piece of furniture, and it moves the PC out of the TV's sightlines.

The side wall from the room: the oak run round the corner with its four LPs, sweeping out under the PC, the keyboard tray's housing underneath and the LED's glow below it, the curtains at the window, and nothing else on the wall: the leads run inside it, the sofa below (drawn from your photo), and the media wall end-on at the right. The chimney breast is cut away so the wall reads end-on; its outline is dashed, with the depths marked: the alcove is 38 deep, so 38 of the 46 cm base sits inside it and 8 stands proud (picked out in gold), and the 30 cm uprights and shelves sit wholly inside.

### The side wall: the ledge wraps the corner, the PC sits above the sofa

One oak line at 155.4 cm: the record ledge from the upright, then the PC shelf over the end of the sofa, clear of the TV's sightlines




- **The run:** 220 cm long, 4 thick, cut from one 225 × 30 × 4 oak board to a full-size template: 7 deep along the LPs (with a 1.5 cm lip), easing out to 28 over 14 cm just past the last LP. It sits on an oak cleat screwed to the side wall, dowelled to the left upright's front edge.
- **The PC end:** 82 × 28 cm, on two concealed heavy-duty brackets (40 kg+ a pair) as well as the cleat, 71.4 cm above the sofa's back. The case sits with its glass to the room and its front towards the alcove. The ports face the window and stay open to the air (no screen, so nothing traps the exhaust); from the sofa you look at the glass side, not the ports. The pad dock sits beyond the case.
- **Cables:** through a brush grommet at the shelf's back corner straight into a 25 mm conduit chased into the side wall (it's solid, as you confirmed), down to skirting height and behind the skirting to the base's plinth; you chase it yourselves in step 3, with a second, smaller conduit beside it for the PC's mains lead, then it's filled and painted, so nothing shows. A draw wire left in it pulls the next lead through. The HDMI is an active optical cable (about 5 m), "source" end at the PC.
- **Power:** the PC plugs into the 10-gang behind the TV: its lead (a 5 m C13 lead with a rewireable plug, fed down bare and the plug fitted at the bottom) runs in its own 25 mm conduit beside the data one. The 10-gang's load check, with the PC, is in section 9. No new socket.
- **Keyboard tray (back from v6.1):** the shelf sits on a 7 cm oak housing with a tray on 300 mm light-duty runners that rolls 25 cm out towards the room. It holds a full-size wireless keyboard and mouse: pull it out, lift the keyboard onto your lap, park it again. The housing stops 12 cm short of the sofa end so the cables drop past it; its underside is 144.4, 64.4 above the sofa's back.
- **Headroom:** the shelf's underside is at 151.4, and it covers only the 28 cm of the sofa nearest the wall.

**The PC end, looked over again**

- **Air:** a mid-tower draws in at the front (towards the alcove) and blows out at the back and top. Both stay open: the case's top is at 202, 33 clear of the ceiling; a magnetic dust filter on the top exhaust keeps the dust out while it's off.
- **Reach from the sofa:** the PC's own power button is on its top, behind your head. A remote power button (a small card on the motherboard's power pins and a push button under the wrap ledge by the sofa end) starts it from the seat, and a powered USB 3 hub under the run's front edge puts pads, the keyboard's dongle and a stick within reach.
- **Lit, and seen:** the glass side faces the room, so its fans and lights read as part of the wall at night (the 3D view's "PC lit" button shows it). Keep its lights on a slow single colour if they pull the eye from the TV.
- **Leads:** labelled at both ends (step 13), all three down the chased conduits; nothing hangs below the shelf.

# 7 · The oak joinery

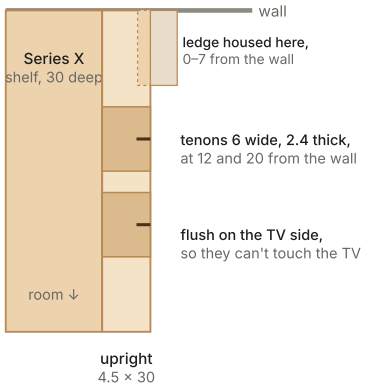
Above the worktop the wall is furniture rather than cladding: two solid oak uprights, shelves carried on joints you can see, and the Card Room Green wall showing between them.

- **Uprights:** solid oak, 4.5 × 30 cm, 159 cm long, each on two 10 mm dowels into the worktop, scribed to the ceiling and held to the back wall with concealed brackets; the left one also screws to the side wall.
- **Wedged through-tenons:** the Series X shelf (top 134) passes through the right upright on two tenons, 6 cm wide and 2.4 thick, 12 and 20 cm from the wall. They're wedged and planed flush: standing proud, they'd touch the TV.
- **Housed and pegged:** the high shelf (two lengths, top 196) and the record ledge (top 155.4) are housed 1.2 cm into the uprights and pinned with 10 mm oak pegs, which show as dark dots.
- **LED grooves:** a routed groove on each upright's inner face holds a smart colour strip that washes the wall either side of the TV: warm white day to day, any colour for a scene.
- **Hidden cables:** a 2 × 1.5 cm groove down the back of the right upright carries Deco B's 12 V lead and patch lead, the Series X's HDMI and power, and the vinyl lead.
- **Finish:** hardwax oil throughout; lightly limed or smoked oak avoids the bright orange "new oak" look.

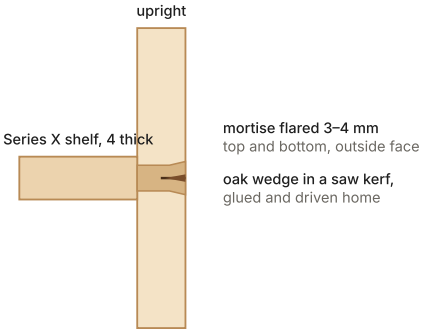
### Artisan oak: how the joints work

Drawn to scale at 7 px per cm. The joints carry the load and are left on show, finished in hardwax oil

**A · Wedged through-tenons, from above**

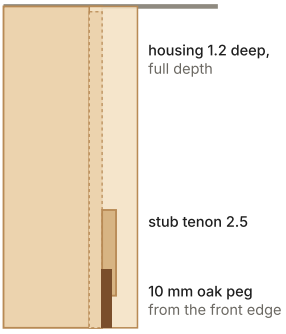



**B · The same joint from the front**




The kerf runs flat, so the wedge spreads the tenon up and down, along the upright's grain, and can't split it.

**C · Housed shelf, pegged**




high shelf / ledge
seen from above

• Through-tenons (A, B): the Series X shelf (top 134) passes through the right upright on two tenons, 12 and 20 cm from the wall, wedged and planed flush: proud, they'd touch the TV

• Housed and pegged (C): the three lengths of the high shelf (top 196) and the record ledge (top 155.4). The peg shows as a dark dot on the upright's face.

• Uprights: 159 long, worktop to ceiling, each on two 10 mm dowels into the worktop, scribed to the ceiling, with two concealed brackets. The left one also screws to the side wall, and

### Joinery cut list

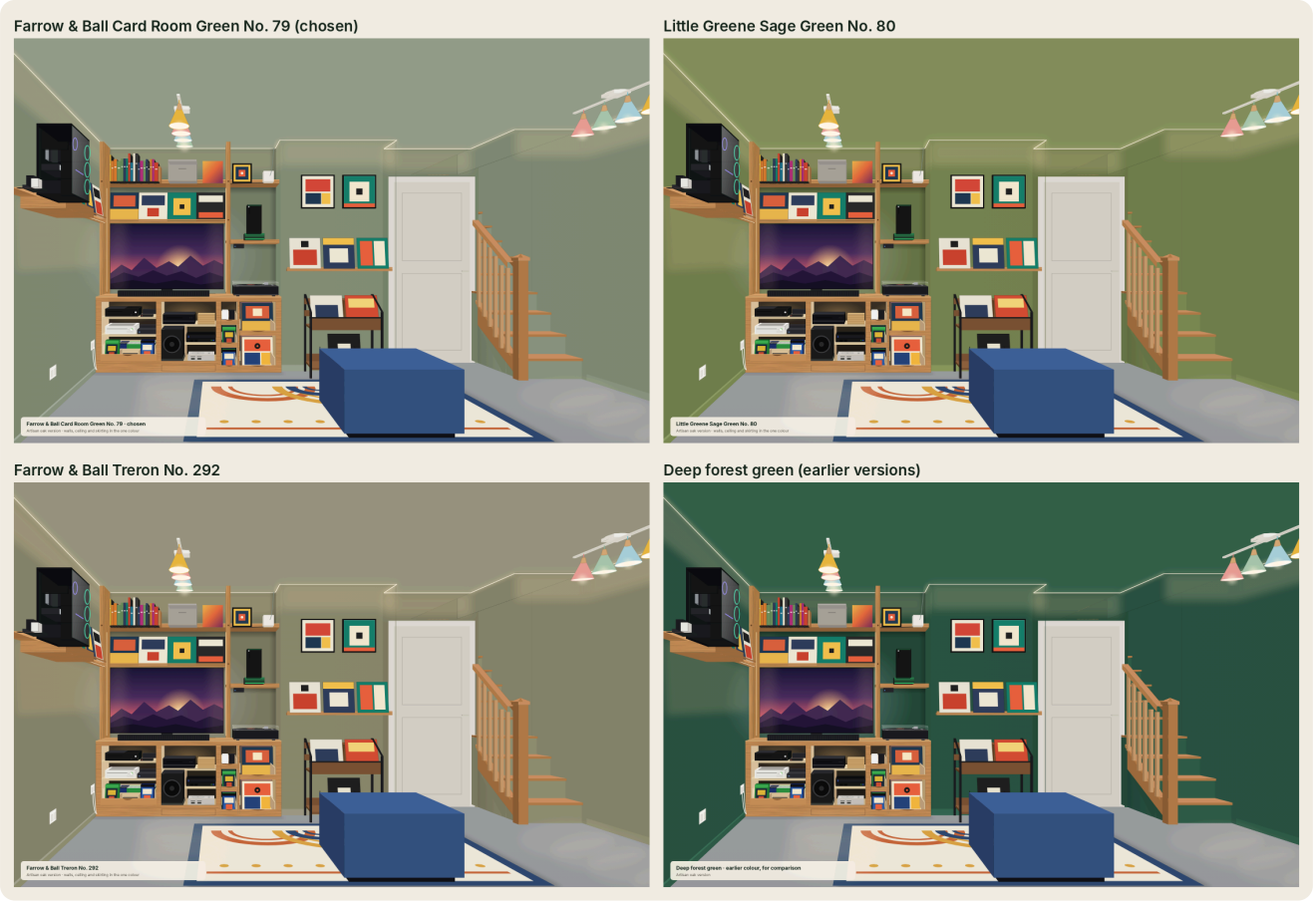
| Part | Material | Finished size (cm) | Joints | Qty |
|---|---|---|---|---|
| Uprights | Solid oak, quarter-sawn if you can | 159 × 30 × 4.5 (buy 165 and scribe) | Housings for the high shelf and ledge; mortises for the Series X shelf in the right one; LED groove inside; cable groove at the back of the right one | 2 |
| High shelf | Solid oak | 128.9 × 30 × 4 and 51.7 × 30 × 4 | Housed 1.2 into the uprights, pegged; the short length on a cleat on the reveal | 2 |
| Record ledge | Solid oak | 128.9 × 7 × 2.4; lip 0.9 × 1.5 | Housed and pegged | 1 |
| Series X shelf | Solid oak | 55 × 30 × 4 (50.5 + 4.5 through) | Two through-tenons 6 × 2.4, wedged flush; cleat on the reveal | 1 |

| | | | | |
|---|---|---|---|---|
| Side-wall run | Solid oak, one 225 × 30 × 4 board | 220 long, 7 deep sweeping to 28, 4 thick; lip 0.9 × 1.5 along the 7 cm part | Cleat, two concealed brackets under the PC end, two dowels to the left upright; LED groove under the front edge | 1 |
| Worktop and backboard | One solid oak worktop, 300 × 62 | 186 × 46 × 3 and 110 × 22 × 3 | Chamfered fronts; cut edges sealed | 1 board |
| Face frame | Oak, 20 mm | 186 × 73, stiles about 5 wide | Glued and pinned; end stiles scribed | 1 set |
| Wedges · pegs · dowels | Oak · 10 mm oak dowel | 6 × 3, tapering · 5 long | Glued | 2 + spares · 6 · 8 |
| Keyboard tray | Oak housing sides 7 × 2, ply tray, oak lip | 70 × 28 × 7; tray 54 × 26 | Housing screwed under the shelf; 300 mm runners | 1 |

# 8 · Paint and light

### Paint: Farrow & Ball Card Room Green No. 79

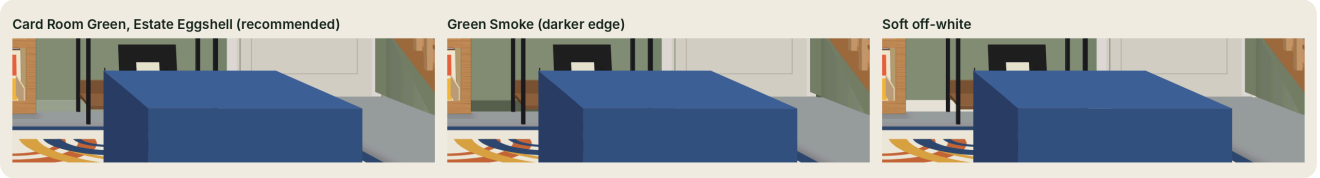
The room is colour-drenched in Card Room Green: walls, ceiling, alcove, chimney breast, skirting and the radiator. The door and its frame stay white, as they are: a crisp edge to the green, like the window. There's no trunking left to paint. It's a soft grey-green that lets the oak and the records lead.




The wall in Card Room Green (chosen) beside Sage Green, Treron and the earlier deep green. Screen colours are approximate and the renders are lit as in the evening: judge from painted samples.

- **Finishes:** Estate or Modern Emulsion on walls, ceiling, alcove and breast; Estate Eggshell in the same colour on skirting and radiator (the door stays white) (Modern Eggshell if the skirting takes knocks).
- **Order:** paint before anything is fixed, so nothing needs cutting in round the uprights, shelves or the dining wall.

### Skirting: Card Room Green too




Recommended: Card Room Green in Estate Eggshell, which completes the drench and hides scuffs far better than white. Darker option: Green Smoke. Over the white gloss: wash, key, woodwork primer, two coats of eggshell.

### Light, in three warm layers

- **The LED strip on the ceiling** (as now): round the room's edge, into the alcove and across the breast; 2700K on a dimmer.

- **Built into the oak (smart colour):** an RGB and tunable-white strip in each upright's groove and one under the side-wall run's front edge[37], in the same app as the bars' bulbs, so each scene sets them too; on their plug-in power on the 10-gang (no fused spur). The 6500K bias light on the back of the TV stays plain white: it's there for the picture, not the mood.
- **Your red lamp:** the three-headed floor lamp you have, its colour bulbs green at the top, pink in the middle and purple at the bottom. A deep green room needs a low layer as well as the light from above; it's also the one to leave on for a film. It can stand in any of the three spots in the 3D room.
- **Two spot bars** in the style of Dunelm's Wolston (£49 each; 84 × 16 cm, 22.4 cm drop; 4 × E14, 6 W max[14]) on your two existing ceiling points, **both running from the TV towards the window**. Bar 1 hangs on the rose straight out from the TV unit, halfway to the sofa; bar 2 on the pendant point by the stairs, along the west wall, so its end no longer comes near that wall. **Smart colour bulbs**: eight E14 bulbs under the bar's 6 W, the WiZ Colour P45 (4.9 W, 4.5 × 10.1 cm) or, if it shows below a shade, the shorter Philips Hue lustre (5.1 W, 4.5 × 7.7 cm)[38]. Smart bulbs need full power, so **no dimmers**: the wall switches stay as they are and stay on; level and colour come from the app (or an optional smart button).

**Where the heads point:** all eight heads straight down, evenly spread along each bar, so their pools overlap into one even light over the room: that's the **Bright** preset, the default. Tilt a head only if its pool lands badly (in the TV's screen from the sofa, or in someone's eyes). Set the bulbs to warm white (3000K) for Bright, 2700K for Cosy, and save colours for the presets you want them in.

### Three lighting presets

Three presets instead of six buttons: **Bright** (the default), **Cosy** and **Film**, each set on the strip's controller, the inline dimmer on the uprights' LED driver, and the bars' and the lamp's smart bulbs in their app. Save the three as scenes in the app (and on the smart button if you get one). In the 3D room the Lights tab has the three tiles, the evening or daytime switch, and every light's slider folded under them.

| Light | Control | Bright | Cosy | Film |
|---|---|---|---|---|
| Ceiling LED strip | its own controller | 70% | 20% | 10% |
| Uprights' grooves and the side shelf | the strips' app (smart colour) | 50% | 55% | 15% |
| TV bias light | comes on with the TV | off | off | on |
| Bar 1, between the TV and the sofa | smart bulbs: the app | 60% | 45% | off |
| Bar 2, by the stairs | smart bulbs: the app | 60% | 30% | 10% |
| Your red lamp | its colour bulbs: the app | 50% | 75% | 60% |

Levels are brightness in the app for the smart bars and the lamp, and roughly how far up the strip's controller and the uprights' inline dimmer go. A preset changes how bright the heads are (and their colour), not where they point.

**Two rules**

- The table's open, so bar 2 lights it: on top of whichever preset is on, bar 2 comes up to at least 80, bar 1 down to 15 or less, and the TV's bias light goes off.
- Evening or daytime is a switch of its own: daytime is the sun through the window, with any preset.

**Bright**

Everything on at a comfortable, even level: tidying, chatting, finding the remote. The default. *Colour: bars warm white 3000 K, every head straight down.*

### Lighting preset: Bright

Everything on at a comfortable, even level: tidying, chatting, finding the remote. The default.

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

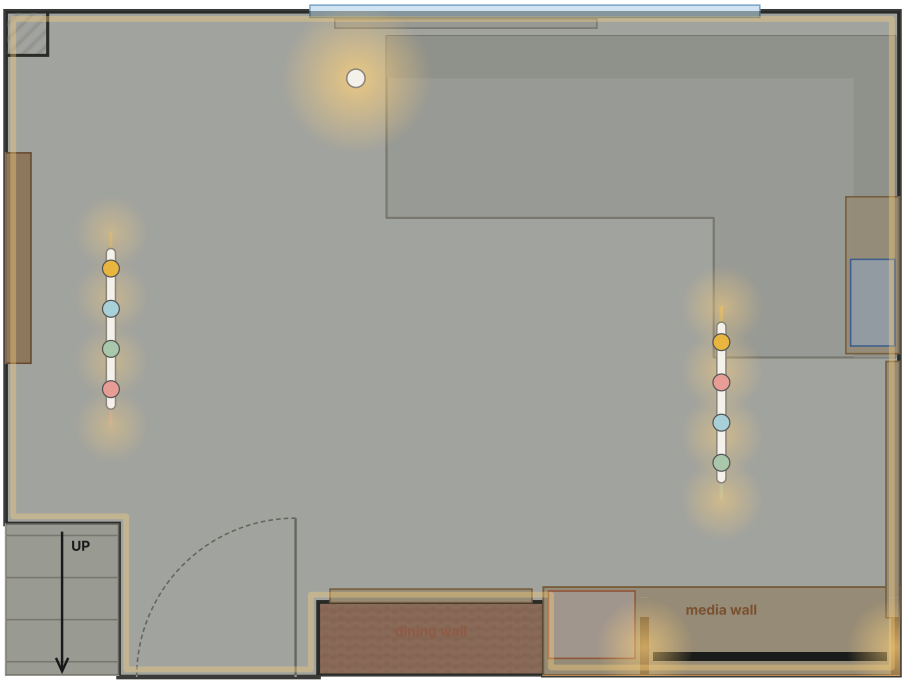
- All eight heads straight down, spread a little along each bar, so the pools overlap.
- The ceiling strip and the uprights fill the edges; the lamp adds a low warm layer.
- Turn the strip down first as the evening goes on: it's the flattest light in the room.

### Cosy

Low and warm, for records and quiet evenings: light on the deck and the shelves, the lamp's colours on. *Colour: bars 2700 K, the lamp's three colour bulbs on (green, pink, purple).*

### Lighting preset: Cosy

Low and warm, for records and quiet evenings: light on the deck and the shelves, the lamp's colours on.




**Ceiling LED strip**
20% · its own controller

**Uprights' grooves and the side shelf**
55% · the strips' app (smart colour)

**TV bias light**
off · comes on with the TV

**Bar 1**
45% · smart bulbs: the app

**Bar 2, by the stairs**
30% · smart bulbs: the app

**Your red lamp**
75% · its colour bulbs: the app

From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- Bar 1's pink head on the turntable and the Series X, blue on the record ledge.
- The uprights' grooves up, washing the wall either side of the TV.
- The lamp up for reading sleeves on the sofa; the ceiling strip nearly off.

### Film

Films and games: the screen is the brightest thing in the room, with a little warm light round it so your eyes don't strain. *Colour: the bias light cool white behind the TV, the lamp low.*

### Lighting preset: Film

Films and games: the screen is the brightest thing in the room, with a little warm light round it so your eyes don't strain.

**Ceiling LED strip**
10% · its own controller

**Uprights' grooves and the side shelf**
15% · the strips' app (smart colour)

**TV bias light**
on · comes on with the TV

**Bar 1**
off · smart bulbs: the app

**Bar 2, by the stairs**
10% · smart bulbs: the app

**Your red lamp**
60% · its colour bulbs: the app

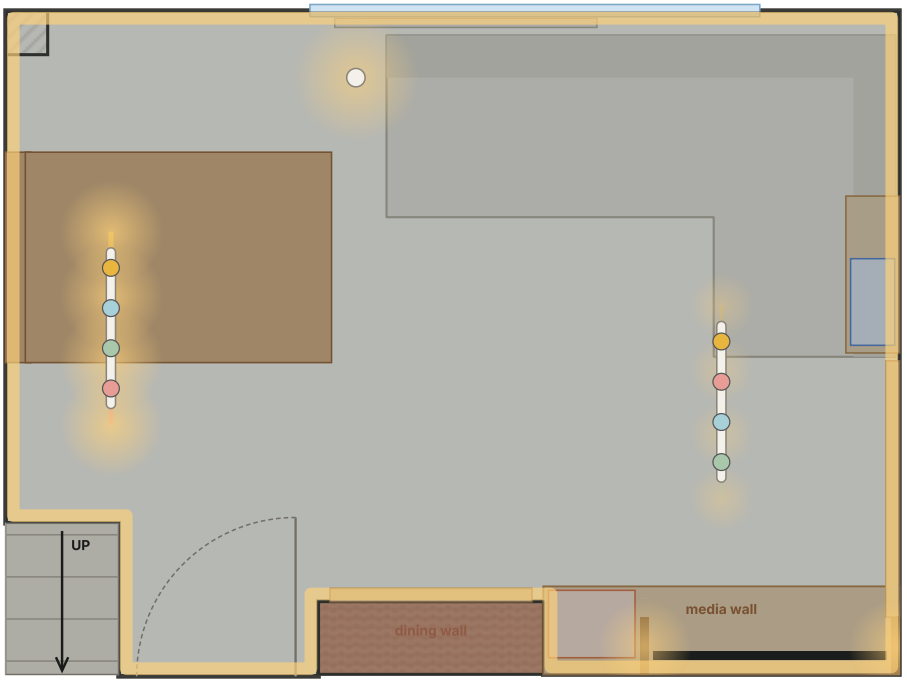
From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

- Bar 1 off: its heads hang in front of the screen, and even tilted away they'd catch it.
- The bias light and your red lamp do the work.
- Bar 2 just glowing on the framed LPs, so the far end of the room isn't a black hole.

## With the table open

### Lighting: Bright, with the table open

The table rule on top of the preset: bar 2 at least 80, bar 1 at most 15, the TV's bias light off




**Ceiling LED strip**
70% · its own controller

**Uprights' grooves and the side shelf**
50% · the strips' app (smart colour)

**TV bias light**
off · comes on with the TV

**Bar 1**
15% · smart bulbs: the app

**Bar 2, by the stairs**
80% · smart bulbs: the app

**Your red lamp**
50% · its colour bulbs: the app

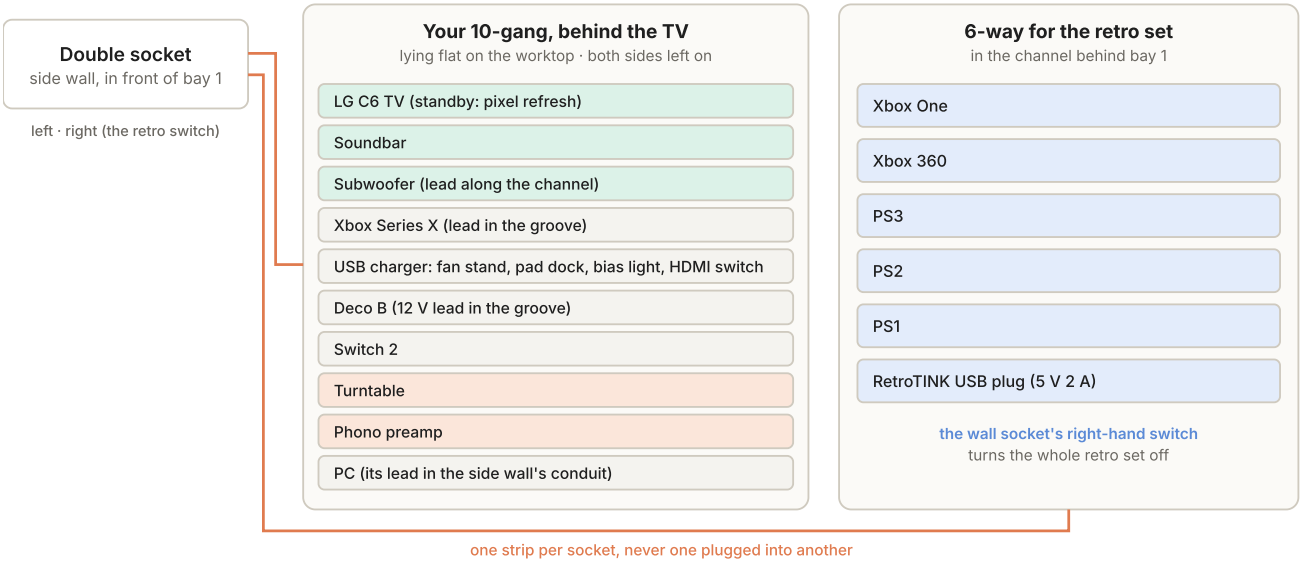
From above, to scale: the window at the top, the media wall bottom right. Glows show each light's level.

# 9 · Power, cables and network

## Power

### Power map: your 10-gang (now with the PC) plus a 6-way for the retro set

Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall




- The socket on the side wall is a single today: you swap it for a double at the same spot (like for like, your own job: step 3).
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- Rear speakers: your double socket behind the sofa's short side. Your lamp and the grow lights: the double behind the plant stand.
- The sub plugs into the 10-gang: its lead drops through the worktop's slot to the back of bay 2. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 1.55 kW with everything on, PC included (850 W at most); the 6-way up to 0.8 kW. Each under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

- **The socket:** the single on the side wall just in front of bay 1, about 30 cm up, becomes a double at the same spot. The left outlet feeds the 10-gang; the right one feeds the retro 6-way, so its own switch is now the retro switch. The 10-gang's switches sit behind the TV, 2 cm from the corner post, where you can't reach them any more.
- **10-gang:** lying flat on the worktop behind the TV, both sides left on: TV, soundbar, sub, Series X, USB charger, Deco B, Switch 2, the deck, the preamp and now the PC. With everything running that's about 1.55 kW (the PC's 850 W at most), under the strip's 13 A (3 kW).
- **Retro 6-way:** in the channel behind bay 1 (bay 2's back is the sub's port): Xbox One, Xbox 360, PS3, PS2, PS1, RetroTINK.
- **OLED care:** leave the TV in standby so it runs its pixel refresh.

### Socket by socket

| Socket | Feeds | On or off |
|---|---|---|
| Side wall, double, left outlet | The 10-gang behind the TV: TV, soundbar, sub, Series X, USB charger (fan stand, pad dock, bias light, HDMI switch), Deco B, Switch 2, the turntable, the preamp, the PC and the uprights' LED driver | Always on: about 1.55 kW with everything running, PC included |
| Side wall, double, right outlet | The retro 6-way in the channel: Xbox One, Xbox 360, PS3, PS2, PS1, the RetroTINK's USB plug | The wall switch is the retro switch: off between sessions; up to about 0.8 kW with a launch PS3 |

| | | |
|---|---|---|
| Behind the sofa's short side, double (yours) | The rear speakers; a spare outlet for a phone charger or a lamp by the sofa | On |
| Window wall behind the plant stand, double (yours) | Your red lamp and the plant stand's grow lights: both within reach of it, layout A | On; the grow lights on a plug-in timer |
| Ceiling points 1 and 2 | The two spot bars, with smart colour bulbs | Switches left on; level and colour from the app |

One strip per outlet, never one plugged into another, and each under its 13 A (3 kW) limit: about 1.55 kW on the 10-gang and up to 0.8 kW on the retro 6-way, about 10 A together.

### Electrics: your own job

You're doing the electrics yourselves. In England, Part P of the Building Regulations decides what needs an electrician or building control.

- **Notifiable (not for you):** a new circuit, a replacement consumer unit, or any change to a circuit round a bath or shower.
- **Not notifiable (yours):** the socket swapped for a double, and the two spot bars in place of the old fittings. Like for like, so no electrician and no building control.
- **Still to the wiring regulations** (BS 7671), every time.
- **The light switches stay:** the bars' smart bulbs need full power, so there are no dimmers to fit.
- **Not mains work at all:** the Cat6 and the conduits.

**The safe routine, every time**

1. **Isolate:** switch off that circuit's breaker at the consumer unit (the lighting circuit for the bars, the socket circuit for the socket). If the labels aren't clear, switch off the main switch.
2. **Lock off:** fit the MCB lock-off and padlock, keep the key in your pocket, and tape a note on the board.
3. **Prove dead:** check the tester lights on its proving unit, test between every pair of terminals (live–neutral, live–earth, neutral–earth), then check the tester on the proving unit again.
4. **Photograph** the wiring before you disconnect anything, and connect the new one wire for wire: brown (or red) to L, blue (or black) to N, green-and-yellow to earth, sleeved, no copper showing, every screw tight.
5. **Test:** power back on, then the socket tester in both outlets; each bar's four bulbs come on and join the app.

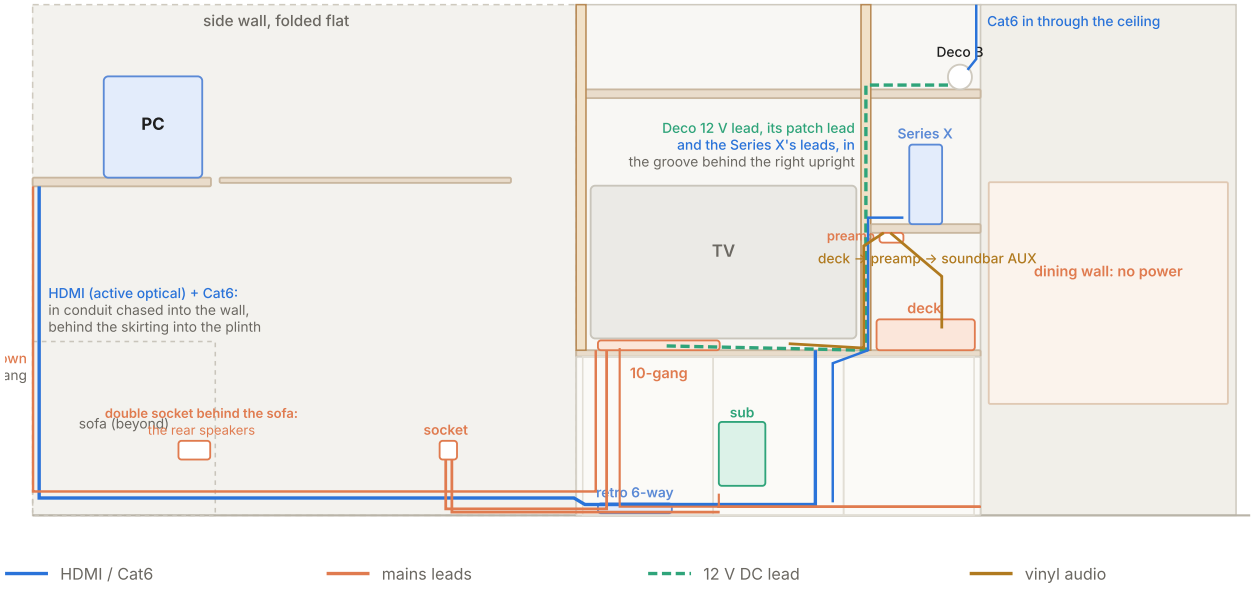
**Stop and call an electrician if…**

- the tester shows anything live after you've isolated;
- the wiring is old (rubber or fabric sheath, or red and black with no earth), or there's no earth at a ceiling point (the spot bars need one);
- there are more wires than you can match to your photo (two-way switching, a loop in a ceiling rose you can't follow), or scorch marks, or a loose, hot or crumbling fitting;
- the consumer unit has rewireable fuses or no RCD;
- a job grows into a new circuit, consumer unit work, or anything by the bath or shower.

## Cable routes

### Cable routes: everything ends in the channel behind the base

Front view, schematic, with the side wall folded flat to the left so the PC's route shows; the sub sits in bay 2

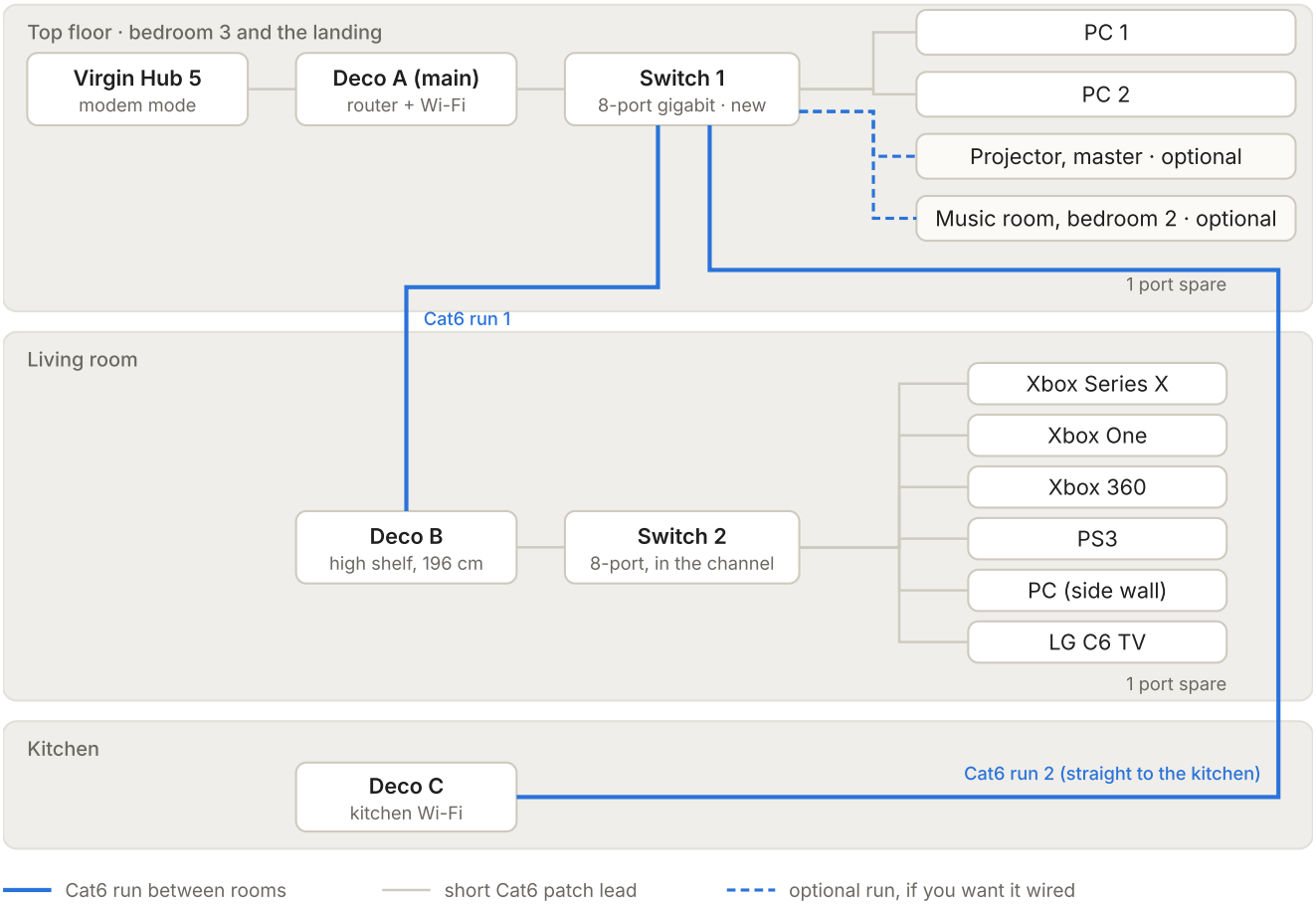



- **The channel:** 4 cm behind the frames, flat two-compartment trunking: mains on one side, HDMI, Cat6 and audio on the other. Switch 2 and the retro 6-way stand on edge in it.
- **The groove behind the right upright:** Deco B's 12 V and patch leads, the Series X's HDMI and power, and the vinyl lead, down to the worktop's slot.
- **The PC:** through the shelf's brush grommet into two conduits chased into the side wall (the HDMI and Cat6 in a 38 mm one, its mains lead in a 25 mm one), behind the skirting and into the plinth notch; the mains lead goes on to the 10-gang.
- **The sub:** its mains lead from the 10-gang drops through the slot behind the TV to the back of bay 2, about 1 m. Keep the channel behind the sub clear: its port needs the air.
- **Airflow:** console bays stay open-backed; the Series X has 15 cm each side and 21 above.

## Network

### Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room




- **Bedroom 3:** the Hub 5 (in modem mode), Deco A and Switch 1 sit together with the gaming PCs. Switch 1's eight ports: the uplink, two PCs, the two Cat6 runs down to Deco B and Deco C, and, if you want them, the projector in the master and the music room in bedroom 2.
- **The route down:** across the landing to the airing cupboard, which sits over the alcove's corner, and down through its floor. Run 1 comes through the living room ceiling in the alcove's right corner to Deco B; run 2 follows the right upright's groove down and through the floor to the kitchen, where Deco C goes on the outside of the brick cupboard, up high.
- **Sloped ceilings:** bedroom 2 and the bathroom slope down to the back wall; keep any Deco or desk in bedroom 2 on the high side.

### What each part does

| Part | Where | Job | Why it's needed |
|---|---|---|---|
| Virgin Hub 5, in modem mode | Bedroom 3 | Brings the internet in, nothing else | Two routers fighting (double NAT) breaks Xbox online play; its Wi-Fi switches off |
| Deco A (main) | Bedroom 3 | The router: Wi-Fi, parental controls, HomeShield | HomeShield and parental controls only work when the Deco is the router |
| Switch 1, 8-port | Bedroom 3 | Turns Deco A's spare port into seven | The two PCs, both Cat6 runs down, and the optional projector and music-room runs |
| Deco B | Living room, high shelf at 196 | Wi-Fi for the living room, and the wired point for the wall | Fixes the living room's weak 5 GHz side |

| | | | |
|---|---|---|---|
| Switch 2, 8-port | Standing on edge in the channel behind the base | Turns Deco B's spare port into seven | The Series X, Xbox One, Xbox 360, PS3, PC and TV each get a cable, one port spare |
| Deco C | Kitchen, high on the outside of the brick cupboard | Wi-Fi for the kitchen | The kitchen gets poor signal on every band today |
| Cat6 runs 1 and 2 | Bedroom 3 → airing cupboard → alcove corner, and on to the kitchen | The wired link between Decos ("Ethernet backhaul") | Without it the Decos relay each other over the 5 GHz band the phones use, losing speed at each hop |

**Stage 1: the mesh up today, before any cables**

1. Unplug everything from the hub. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its settings page moves to 192.168.100.1.
2. Connect a hub port to Deco A and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another port and restart the hub.
3. Plug Switch 1 into Deco A's other port, then the PCs into Switch 1.
4. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi.
5. Check the Xbox's network settings show NAT: Open.

**Stage 2: after your Cat6 runs (step 3)**

1. Plug run 1 into Deco B's free port (Switch 2 stays in the other) and run 2 into Deco C.
2. In the Deco app, check Deco B and Deco C show as "Ethernet".
3. Plug the Series X, Xbox One, Xbox 360, PS3, PC and TV into Switch 2.
4. Under QoS, give the Series X and the PC high priority; turn on guest Wi-Fi and automatic firmware.

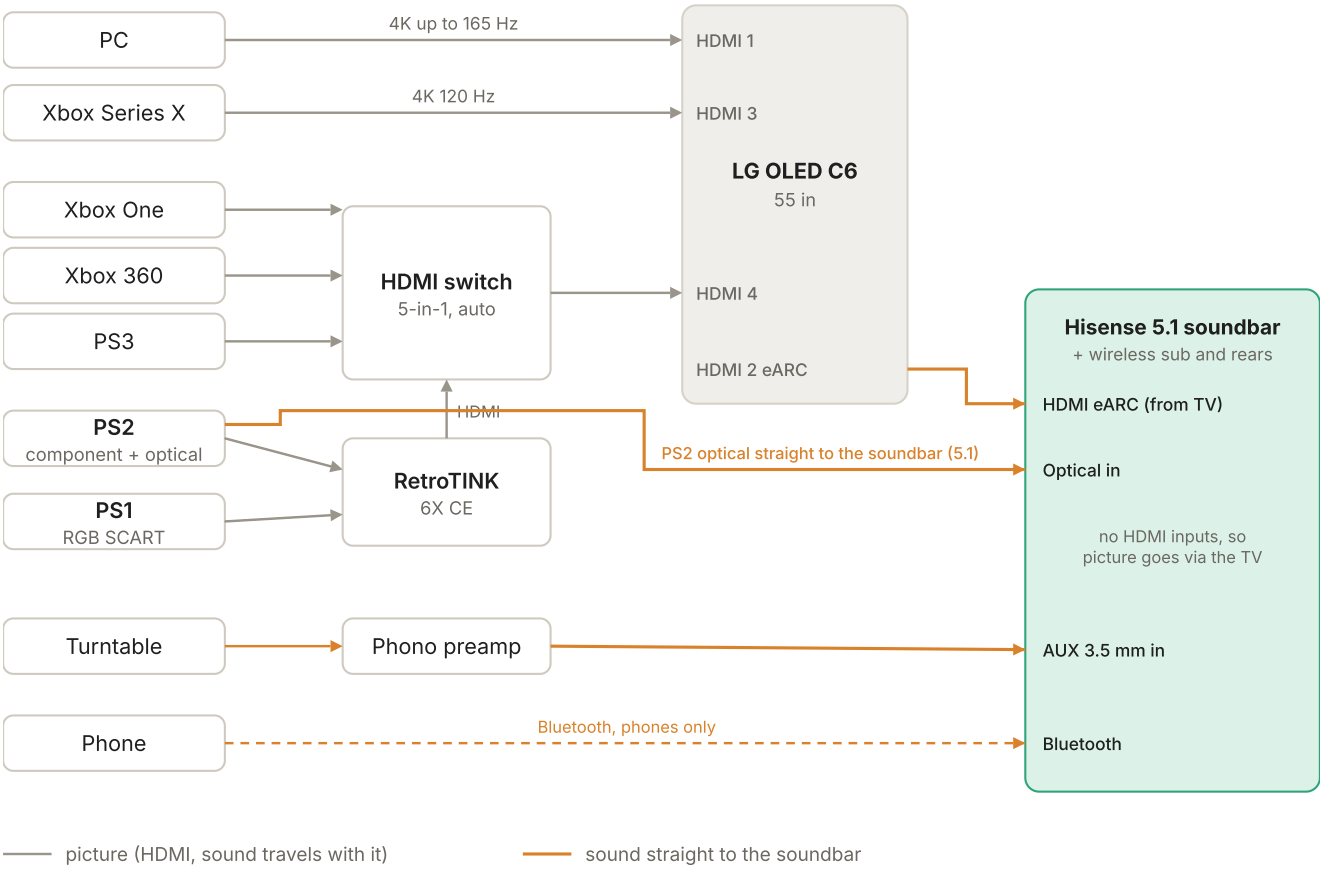
**Cabling rules**

- Solid pure-copper Cat6, round; not CCA or flat cable for the long runs. Measure the real route and add 2–3 m.
- Cross mains at right angles; never staple through a cable; no kinks; label both ends. Never two cables between the same two devices.
- **Done when:** Deco B and C show "Ethernet", each PC shows 1.0 Gbps, the Xbox shows NAT: Open, and phones hold a strong signal in the kitchen and across the living room.

## AV: everything sounds through the Hisense

### Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



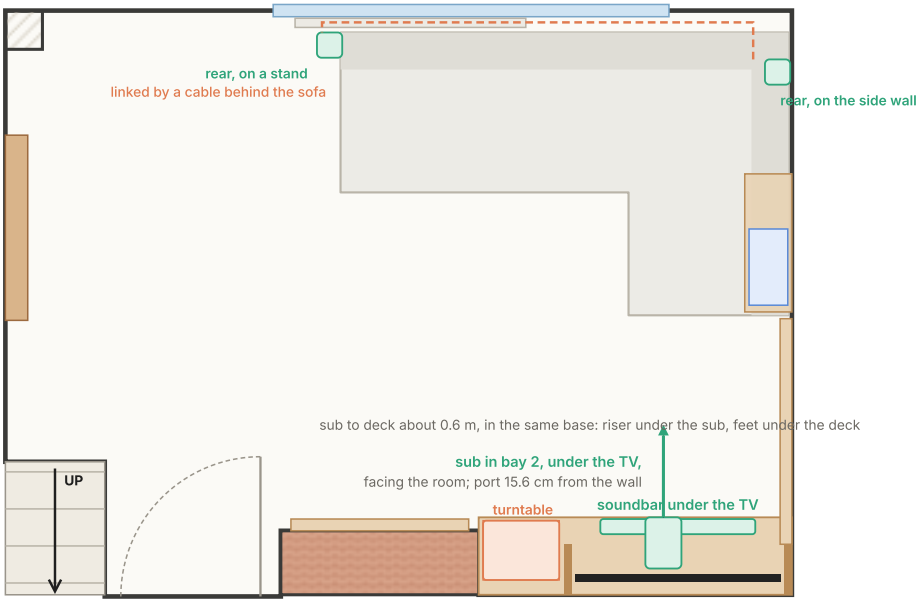

| Source | Picture | Sound |
|---|---|---|
| PC · Xbox Series X | TV HDMI 1 · HDMI 3 | Via the TV over eARC (5.1 or Atmos) |
| Xbox One, Xbox 360, PS3 | HDMI switch → TV HDMI 4 | Via the TV over eARC, 5.1 |
| PS2 | Component → RetroTINK → switch → TV | Optical straight to the soundbar |
| PS1 | RGB SCART → RetroTINK → switch → TV | Stereo through the RetroTINK and eARC |
| Turntable · phone | — | Preamp → AUX · Bluetooth |

TV settings: rename the inputs (HDMI 4 "Retro"), Deep Colour on HDMI 1, 3 and 4, Game Optimiser on the console inputs, sound out to HDMI eARC. RetroTINK-6X CE: a 5 V 2 A USB plug on the retro 6-way; the PS2's Component Video Out set to Y Cb/Pb Cr/Pr; RGB SCART for the PS1.

## Sound layout

### Sound: the sub on the front wall with the TV, the rears behind the sofa

To scale, north at the top. The sub is wireless: on the front wall it blends with the soundbar, out of the way of feet




- **Sub:** wireless, in bay 2 under the TV, its driver facing the room and its rear port 15.6 cm from the wall through the open back (15–30 is advised[25]). Next to the soundbar, its bass blends with the bar instead of coming from behind you, and it's off the floor where feet and drinks are. If the bass booms, turn it down rather than moving it.
- **Rears:** yours stand behind the sofa (from your note; your photo will fix their stands and spots): the two on slim floor stands in the 13 cm gap behind the sofa's long side, near each end, at about seated ear height (96–110.4); linked by a cable along the skirting behind the sofa, powered from your double socket behind the sofa's short side. In the 3D room they move in Edit room.

# 10 · Build guide, for a first-time builder

For anyone who's hung shelves and built flat-pack but not cut a joint. Each step: what to do, what to check, and how long it takes. Steps 3 and 12 are the electrics, with the point to stop and call an electrician. Finished sizes, in cm.

**Before you start: five rules**

- **Check every wall before you drill.** Run the cable and pipe detector over the spot, and never drill straight above, below or beside a socket or switch: cables run vertically and horizontally from them.
- **Measure twice, mark once, cut once.** Walls are rarely square: measure the alcove at the floor, 76, 134, 155, 196 and the ceiling, and cut to the smallest.
- **Two people for anything over 15 kg**: the uprights, the worktop, the table top, the TV.
- **The electrics are yours, within limits.** Swapping the socket for a double and the two ceiling fittings are like-for-like jobs, not notifiable under Part P in England, as long as they meet the wiring regulations. Isolate, lock off and prove dead every time. Anything new (a new circuit, the consumer unit, work by a bath or shower) is an electrician's, and so is anything on the stop list in section 9.
- **Dry-fit everything** (assemble without glue or screws) before the final fix.

### Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

| Tool | What it's for | Get it | About £ |
|---|---|---|---|
| **MEASURE AND MARK** | | | |
| Tape measure (5 m) and a steel rule | every cut and every height | Buy | 10–15 |
| Combination square | squaring lines across boards and checking corners | Buy | 10–15 |
| Spirit level, 60 cm and 120 cm | levelling the base, the shelves and the table | Buy | 15–30 |
| Cross-line laser level | marking all the heights round the alcove in one go | Buy or borrow | 30–60 |
| Cable and pipe detector | checking every wall before you drill: non-negotiable | Buy | 15–30 |
| Pencil, masking tape, a marking knife | marking cuts (tape stops oak splintering) | Buy | 5 |
| **CUT** | | | |
| Track saw, or a circular saw with a straight-edge guide | the worktop, the table top and the long shelves | Hire or borrow | 25–40 a day |
| Jigsaw | notches and the cable slot | Buy or borrow | 40–70 |
| Tenon saw (Japanese pull saw is easiest) | the tenons and the wedge kerfs | Buy | 15–25 |
| Sharp chisels, 6, 12 and 25 mm, and a mallet | cleaning out mortises and housings | Buy | 30–50 |
| **DRILL AND FIX** | | | |
| Cordless drill-driver (18 V) | pilot holes and screws; most of the job | Buy or borrow | 60–120 |
| SDS or hammer drill with masonry bits | plugs into the brick walls and the chimney breast | Hire or borrow | 15–25 a day |
| Dowel jig and 10 mm brad-point bit | the dowels under the uprights and at the wrap ledge | Buy | 20–35 |

| | | | |
|---|---|---|---|
| Clamps: four 60 cm sash or F-clamps, four small | glue-ups and holding work while you fix it | Buy | 40–60 |
| Stud finder (if any wall is plasterboard) | finding studs to fix into | Buy | 15 |
| **SHAPE AND FINISH** | | | |
| Router (1/4 in is enough) with a straight bit and a guide | the housings, the LED grooves and the cable groove | Hire, borrow or buy | 60–120 |
| Random-orbit sander, 120 and 180 grit | smoothing the oak before oiling | Buy or borrow | 40–70 |
| Hardwax oil, lint-free cloths, a brush | finishing the oak | Buy | 25–35 |
| **ELECTRICS (YOUR OWN: STEPS 3 AND 12)** | | | |
| Two-pole voltage tester to GS38, with its proving unit | proving every wire dead before you touch it: non-negotiable | Buy | 35–70 |
| MCB lock-off and a small padlock | keeping the circuit off while you work on it | Buy | 8–15 |
| Plug-in socket tester | checking the new double socket is wired the right way round | Buy | 8–15 |
| Insulated (VDE) screwdrivers, wire strippers, side cutters | the socket and the ceiling fittings | Buy | 20–40 |
| Wall chaser with dust extraction (or a bolster and club hammer) | the PC's two conduits in the side wall | Hire | 25–45 a day |
| Cat6 punch-down tool and a cable tester | terminating and checking the Cat6 runs | Buy | 20–35 |
| **SAFETY** | | | |
| Safety glasses, ear defenders, FFP2 dust masks | every cut, every sanding job; oak dust is an irritant | Buy | 15–25 |
| Work gloves; a first-aid kit | handling boards and sheet goods | Buy | 10 |
| A sturdy step ladder | the high shelf, the ceiling and the PC shelf | Borrow | — |

### Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.
2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

### The order of work

| # | Stage | What to do | Check before moving on | Time |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | PREP<br>**Measure and count** | The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest. The floor's level across the alcove, and the chimney breast's flatness with a long level. The door's swing, and the sofa: overall width, depth with the return, back height. Count the LPs and games (PS2, Xbox 360, Xbox One), and measure ten cases of each. Make a cardboard template of the alcove at worktop height: walls are rarely square. | Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first. | 2 h |
| 2 | NETWORK<br>**Network, stage 1** | Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi. Speed-test each room, and check the Xbox's network settings show NAT: Open. | Internet in every room, and NAT: Open on the Xbox. | 2 h |
| 3 | POWER & CABLES<br>NETWORK<br>**Cables and the socket, your own job** | The Cat6: two runs from bedroom 3, across the landing and down through the airing cupboard, one to the alcove's right corner for Deco B, one on to the kitchen for Deco C, a spare pulled with each; the optional runs to the master and bedroom 2 if you want them. Low voltage: no rules beyond care. The socket: swap the single on the side wall for a double at the same spot, with the circuit isolated, locked off and proved dead (the safe routine is in this step). Like for like, so not notifiable. The PC's two conduits: chase them into the solid side wall from behind the PC shelf to skirting height and behind the skirting to the base, 38 mm oval for the HDMI and Cat6, 25 mm for the PC's mains lead, a draw wire in each. Make good. Stop and call an electrician if anything on the stop list turns up (old wiring, no earth, a reading where there should be none, anything you can't match). Before you fill a chase, photograph it with a tape measure in shot, and mark every route on the plan. | The socket tester shows correct wiring on both outlets, both Cat6 runs test good end to end, and every route is photographed. | 2–3 days |
| 4 | PAINT<br>**Paint** | Paint an A4 board in Card Room Green and look at it by day and with the LED strip on. Wash the skirting and door with sugar soap, sand to a key, and prime them. Two coats of emulsion on the walls, ceiling, alcove and chimney breast. Two coats of eggshell in the same green on the skirting and radiator (the door and its frame stay white). Leave it a week to cure before fixing anything to it. | Two even coats, and no white showing at the edges or behind the radiator. | 2–3 days, then a week to cure |
| 5 | PREP<br>**Mark the heights** | With the laser level: 73 and 76 for the worktop, 130 and 134 for the Series X shelf, 153 and 155.4 for the ledges, 192 and 196 for the high shelf. The uprights at 0–4.5 and 131–135.5 cm from the side wall. The dining wall's backboard at 51.3–73.3, centred on the breast. Run the cable detector along every line you will drill, and mark anything it finds. | Every line level end to end, and the uprights' lines plumb. | 1 h |
| 6 | MEDIA WALL<br>**Build the base** | Order the oak and ply from the cutting list (below the steps), a week or two ahead: most merchants cut to size for a small charge. Build the 9 cm plinth for bays 1–2, with the open bay for the hidden drawer under bay 2. Level it with shims. Assemble the two BESTÅ frames with their back panels left off and stand them on the plinth. Build bay 3 from 18 mm ply, 60 × 40 × 73, and bolt it to bay 2. Screw the 4 cm spacer blocks to the wall, push the base against them and fit the anti-tip brackets. Lay the flat two-compartment trunking in the channel behind (mains one side, data the other) and run the cables in. | The tops level within 1 mm across 186 cm, and nothing rocks. | 1 day |

| | | | | |
|---|---|---|---|---|
| 7 | MEDIA WALL<br>**Fit the worktop** | Cut the worktop and the dining wall's backboard from the one 300 × 62 oak board, using the template from step 1. Scribe the ends to the walls; cut the TV slot over the channel and the grommet hole behind the deck. Oil it on every face, undersides and cut ends included. Screw it up through the frame tops in slotted holes, so the oak can move. | Tight to both walls, and the slot sits over the channel. | ½ day |
| 8 | MEDIA WALL<br>**Fit out the base** | Shelves on their pins at the fit-proof heights for the consoles. The game drawer in bay 1: runners on the frame's sides, then the box (its cut list is with this step). Bay 3: the 12 mm divider first (17.8 cm in from the left side), then the LP crates' runners at 2.3 and 37.0 cm on the right and the three slim drawers' at 2.3, 25.1 and 47.9 on the left, then each box. The hidden plinth drawer under bay 2. Bay 2: the divider and the part shelf, then the sub on its riser, its front flush with the face frame. | Every drawer runs smoothly, the two crates clear each other by 2 cm, and the slim drawers by 2.8. | 1–2 days |
| 9 | MEDIA WALL<br>**Uprights and shelves** | On the bench: cut the housings and the mortises in both uprights, and the LED groove and cable groove. Dry-fit each upright with its shelves before any glue. Lay Deco B's leads in the groove behind the right upright. Stand the uprights on their dowels and fix the concealed brackets to the back wall. Fit the high shelf and the ledge, then the Series X shelf: wedge its tenons, and peg the housings. On the chimney breast: the “now playing” ledge, its top at 104, on two concealed brackets, 6 cm in from each side of the breast (with the table by the stairs). | Uprights plumb, shelves level, the TV slides in with 2 cm each side, and the “now playing” ledge is level with an LP standing on it. | 2–3 days |
| 10 | MEDIA WALL<br>**The side-wall run and the PC end** | Cut the run from its 225 × 30 × 4 board: mark the front edge from the full-size template (7 cm deep along the LPs, sweeping out to 28 under the PC, a 5 cm radius on the end), jigsaw just outside the line and rout to it with a bearing-guided bit; rout the LED groove under the front edge. Screw the oak cleat along the side wall and fit the concealed brackets for the PC end. Lift the run on, dowel its end to the left upright's front edge, and screw it down to the cleat from above (plugged). Screw the keyboard tray's housing under it and fit its runners. Fit the brush grommet and pull the PC's leads down the two chased conduits with the draw wires: data in the big one, its mains lead in the small one. The PC's extras: screw the powered USB hub under the run's front edge below the PC, fit the remote power button's card and stick its button under the wrap ledge by the sofa, and lay the magnetic filter on the top exhaust. | One level line with the ledge over the TV, the sweep runs smooth with no flat spots, and the PC end holds your weight leaning on it. | 1 day |
| 11 | DINING TABLE<br>**The dining cabinet: the table and the six chairs in one** | Check the gap under your shelf (191 or more), mark the footprint and oil every oak part first. Build the carcass on the floor: the two ends, top and floor, the partition, the chair shelf and the linen floor, then the 6 mm back to square it. Stand it on its marks, shim it level and screw it to the wall at the top; fit the oak hinge rail across the table bay at 73.3. On the bench: the two halves with their breadboard ends, the apron frames, the legs on their folding brackets, and the three flip-top hinges at the joint. Hang the table on the piano hinge, fit the catches, the drawer, the linen flap and the chair bay's doors; level the table and trim the legs; load the six chairs. | The table swings down and up without touching the bay's sides, all four legs click, the top is level at 76 both ways, the catches hold it folded, and both doors close with an even 3 mm gap over the six chairs. | 3–4 days |

| | | | | |
|---|---|---|---|---|
| 12 | LIGHTING<br>**Lighting: smart spot bars and the LED strips, your own job** | Isolate the lighting circuit at the consumer unit, lock it off and prove it dead at each ceiling point. Take down the two old fittings. The light switches stay as they are: the smart bulbs need full power, so no dimmers. Hang bar 1 on the rose between the TV and the sofa and bar 2 on the pendant point by the stairs, earth first, both running from the TV towards the window; fit the smart colour bulbs and point every head straight down, evenly spread; tilt one only if its pool lands badly. The LED strips in the uprights' grooves and under the side-wall shelf on their plug-in driver (on the 10-gang), the ceiling strip through its notches, the bias light on the TV; your red lamp in its spot. Power back on, switches on: add the eight bulbs to the app, group them as bar 1 and bar 2, and save the three presets (Paint and light): Bright, Cosy and Film. With the table open, bar 2 comes up to 80 and bar 1 down to 15 (the table rule). | Every bulb answers the app, the presets recall, and from the sofa no spot head shows in the screen or shines in your eyes. | 1 day |
| 13 | AV<br>NETWORK<br>**Connect and finish** | TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK). The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa. | Everything on the network's "done when" list, and every source plays through the soundbar. | 1 day |

Roughly 16–20 working days spread over a few weekends, the electrics and the Cat6 included, plus the paint drying. Steps 6–11 can be split: the base and the dining cabinet first, the oak joinery once you're confident.

### Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the LP crates before the anti-tip brackets go in.** A full crate weighs about 30 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

### The cutting list

Every made part in one list, grouped by what it's cut from, for the timber merchant. Sizes are finished sizes in cm, length × width × thickness. The cut lists in the other sections say how each part joins; this list is for ordering.

#### Solid oak, hardwax oil

Ask for it planed all round; scribe the uprights and worktop on site

| Part | Size (cm) | Qty | Notes | Section |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Uprights | 159 × 30 × 4.5 | 2 | Buy 165 long and scribe to the ceiling; housings, mortises, LED and cable grooves cut on the bench | Media wall |
| High shelf, long length | 128.9 × 30 × 4 | 1 | Housed 1.2 into both uprights and pegged | Media wall |
| High shelf, short length | 51.7 × 30 × 4 | 1 | Housed into the right upright; cleat on the reveal | Media wall |
| Series X shelf | 55 × 30 × 4 | 1 | Two through-tenons 6 × 2.4, wedged flush | Media wall |
| Record ledge | 128.9 × 7 × 2.4 | 1 | Housed and pegged | Media wall |
| Side-wall run | 220 long × 7 sweeping to 28 × 4, from one 225 × 30 × 4 board | 1 | On an oak cleat, concealed brackets under the PC end, dowelled to the left upright; LED groove under the front edge | Media wall |
| Ledge lips | 128.9 × 1.5 × 0.9 · 126 × 1.5 × 0.9 | 1 · 1 | Glued along the ledge's and the run's front edges | Media wall |
| Keyboard tray housing | 70 × 28 × 7, sides 7 × 2 | 1 | Screwed under the PC shelf | Media wall |
| Worktop | 186 × 46 × 3 | 1 | From one 300 × 62 full-stave worktop; chamfered front | Media wall |

| | | | | |
|---|---|---|---|---|
| Face frame | 186 × 73 overall, 2 thick, stiles about 5 wide | 1 set | End stiles scribed to the walls | Media wall |
| Drawer fronts | game drawer 56.8 × 6.5 × 2 · plinth drawer 54 × 6 × 2 | 1 · 1 | Finger pull cut underneath | Media wall |
| LP crate lips · slim drawer lips | 36.3 × 5 × 1.5 · 16.5 × 5 × 1.5 | 2 · 3 | Cut to the face frame's openings, less 3 mm | Media wall |
| “Now playing” ledge | 105.5 × 7 × 2.4, lip 1.5 | 1 | On the breast at 104, two concealed brackets (with the table by the stairs) | Media wall |
| Follower blocks · Xbox One lift block | 19 × 12 × 2 · 15 × 12 × 2 | 2 · 1 | Offcuts, felted underneath | Media wall |
| Wedges · pegs · dowels | 6 × 3, tapering · 10 mm dowel, 5 long · 10 mm dowel | 2 + spares · 6 + 6 · 8 | Glued | Media wall |
| Table halves, centres | 81 × 90 × 2.7 | 2 | Edge-glued staves: 76 between the shoulders, plus a 2.5 tongue each end | Dining table |
| Breadboard ends | 90 × 7 × 2.7 | 4 | Pegged with three 10 mm pegs each, in slotted holes | Dining table |
| Apron rails | outer half 84 and 78 · inner half 82.4 and 78, all × 4.6 × 2 | 2 + 2 · 2 + 2 | A frame under each half, held on figure-8 fasteners | Dining table |
| Table legs | 6.5 × 4.5, cut 70 long | 4 | Trimmed to the floor once the table's hung and level | Dining table |
| Hinge rail | 91 × 23.4 × 7 | 1 | Across the table bay, top at 73.3, glued and screwed | Dining table |

### Oak-faced ply, 18 mm (the dining cabinet)

Ask the merchant to cut it to size and edge the visible edges with oak lipping

| Part | Size (cm) | Qty | Notes | Section |
|---|---|---|---|---|
| Cabinet ends | 189 × 22.2 | 2 | Full height, floor to top | Dining table |
| Cabinet top · floor | 194.4 × 21.6 | 1 · 1 | Between the ends; the floor's top face at 3 | Dining table |
| Cabinet partition | 184.2 × 21.6 | 1 | Makes the table bay 91 inside | Dining table |
| Cabinet chair shelf · linen floor | 101.6 × 21.6 · 91 × 21.6 | 1 · 1 | Top faces at 94.8 and 166.1 | Dining table |
| Cabinet doors | 51.5 × 185.7 | 2 | The chair bay, on three 170° hinges each | Dining table |
| Cabinet drawer front · linen flap | 90.7 × 62.7 · 90.7 × 25.1 | 1 · 1 | Under and over the folded table | Dining table |

### Birch ply, 18 mm

| Part | Size (cm) | Qty | Notes | Section |
|---|---|---|---|---|
| Bay 3 sides | 73 × 40 | 2 | The record carcass, standing on the floor | Media wall |
| Bay 3 top and bottom | 56.4 × 40 | 2 | Between the sides; back left open to the channel | Media wall |
| Plinth rails | 120 × 9 | 2 | Front and back of the ladder frame under bays 1–2 | Media wall |
| Plinth cross-pieces | 34.4 × 9 | 3 | At both ends and under the bay 1–bay 2 joint, so the bay under bay 2 stays clear for the drawer | Media wall |

### Birch ply, 12 mm

| Part | Size (cm) | Qty | Notes | Section |
|---|---|---|---|---|
| Drawer box (the cabinet) | sides 20 × 54.7 · back and front 86 × 54.7; a 6 mm base 88.4 × 20 | 2 · 2 · 1 | Under the folded table | Dining table |
| Game drawer sides | 35 × 15.5 | 2 | | Media wall |
| Game drawer back · front panel | 51.9 × 15.5 · 51.9 × 6 | 1 · 1 | | Media wall |
| LP crate bases · slim drawer bases | 34.9 × 40 · 15.3 × 40 | 2 · 3 | The sides and ends stand on them | Media wall |
| Bay 3 divider | 69.4 × 38 | 1 | 17.8 in from the left side, the full height inside | Media wall |
| Plinth drawer sides · ends | 34 × 4.4 · 49.6 × 4.4 | 2 · 2 | A 52 × 34 × 5 box | Media wall |
| Bay 2 divider · part shelf | 35.3 × 38 · 32.2 × 36 | 1 · 1 | The sub's column on the left; the PS2 over the PS1 on the right | Media wall |
| Keyboard tray | 54 × 26 | 1 | Oak lip on its front edge | Media wall |

### Birch ply, 9 mm

| Part | Size (cm) | Qty | Notes | Section |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Game drawer base | 52.5 × 34.4 | 1 | In a 6 mm groove | Media wall |
| LP crate sides · ends | 40 × 8 · 33.1 × 8 | 4 · 4 | Two of each per crate: a 33 cm trough | Media wall |

### Birch ply, 6 mm

| Part | Size (cm) | Qty | Notes | Section |
|---|---|---|---|---|
| Cabinet back | 194.4 × 187.8 | 1 | Glued and pinned across the back, between the ends | Dining table |
| Game drawer cross divider | 51.9 × 9 | 1 | 13 cm back from the front | Media wall |
| Game drawer bin walls · bin floors | 12.6 × 9 · 16 × 12.6 | 2 · 2 | Floors raked so the cases lean about 13° | Media wall |
| Plinth drawer base | 52 × 34 | 1 | Screwed under the box | Media wall |
| Slim drawer sides · backs and fronts | 40 × 10 · 14.1 × 10 | 6 · 6 | Three boxes 15.3 wide, 14.1 inside | Media wall |

### Hardware

Runners are sold in pairs

| Part | Size (cm) | Qty | Notes | Section |
|---|---|---|---|---|
| LP crate runners · slim drawer runners | 400 mm full extension, 68–70 kg a pair · 400 mm light duty | 2 pairs · 3 pairs | Into bay 3's sides and the divider | Media wall |
| Game drawer runners | 350 mm full extension, soft-close | 1 pair | Accuride DZ3832EC or Screwfix 964VN | Media wall |
| Plinth drawer and keyboard tray runners | 300 mm full extension · 300 mm light duty | 1 pair · 1 pair | | Media wall |
| Flip-top hinges | Lee Valley 40574, solid brass, 1.27 × 7.0 | 3 | Routed into the meeting edges of the two halves, 10 in from each side and the middle | Dining table |
| Piano hinge | Altro continuous, 1800 × 50, cut to 90 | 1 | The table's foot, on the hinge rail's top face | Dining table |
| Leg brackets · tabletop fasteners | folding, self-locking · figure-8 | 4 · 16 | The legs fold inside the aprons; the fasteners hold the aprons to the tops | Dining table |
| 170° concealed hinges | clip-on, 35 mm cup | 6 | Three a door, the chair bay | Dining table |
| Lid stays · butt hinges · drawer runners | for the linen flap · 50 mm · 200 mm side-mount | 2 · 2 · 1 pair | | Dining table |
| Sub isolation riser | dense foam or sorbothane, about 22 × 28 × 3 | 1 | Under the sub in bay 2 | Media wall |
| Concealed brackets | for the uprights; two for the PC shelf, 40 kg+ a pair | 1 set | | Media wall |

# 11 · Buy list and tracker

Every purchase and job, in build order, comes to about £3,438–6,733, with no electrician on top: the electrics are your own job, and the tools for them are on the list. Each phase carries its part's coloured label. Estimates are approximate UK prices; changed and new items are marked. Tick items off and write in what you paid.

| Phase | Estimate (£) | Biggest item |
|---|---|---|
| NETWORK 1 Network, now | 51–70 | two 8-port switches |
| POWER & CABLES NETWORK 2 Cables and power | 265–545 | the electrical safety kit and Cat6 |
| PAINT 3 Paint | 220–540 | paint |
| MEDIA WALL 4 Base unit | 457–892 | record trays and the BESTÅ frames |
| MEDIA WALL 5 Oak joinery | 745–1,355 | the solid oak worktop |
| DINING TABLE 6 Dining wall | 911–1,600 | the oak table top |
| LIGHTING 7 Lighting | 288–795 | the eight smart bulbs |
| AV 8 TV, consoles and AV | 288–420 | RetroTINK-6X CE |
| AV 9 Vinyl kit | 80–190 | phono preamp |
| AV NETWORK 10 Finishing | 133–326 | the rug |
| **Total** | **3,438–6,733** | **no electrician: the electrics are your own job (section 9)** |

Already owned: Virgin Hub 5, three Deco X1500s, two TL-SG105S switches (both become spares), LG OLED C6 55" and its stand, a 10-gang, the Hisense AX5100G 5.1 soundbar, the Series X fan stand, the turntable, the L-sofa and all the consoles.

| Status | Item | Qty | Estimate (£) | Paid (£) |
|---|---|---|---|---|
| NETWORK **1 NETWORK, NOW** | | | | |
| ☐ To do | Put the Virgin Hub 5 into modem mode | | | |
| ☐ To do | Set up Deco A, Switch 1 and the PCs in bedroom 3 | | | |
| ☐ To do | Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox | | | |
| ☐ To buy | Cat6 patch leads, 0.5–2 m | 11 | 15–20 | |
| ☐ To buy | TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3 | 2 | 36–50 | |
| POWER & CABLES NETWORK **2 CABLES AND POWER** | | | | |
| ☐ To do | Cat6, your own job: two runs from bedroom 3, across the landing and down through the airing cupboard, one to the alcove's right corner (Deco B) and one on to the kitchen (Deco C), a spare pulled with each, and the optional runs to the master and bedroom 2 **(changed: no electrician)** | | | |
| ☐ To buy | Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones | 2–4 runs | 30–60 | |
| ☐ To buy | Cat6 tools: a punch-down tool, RJ45 keystones and two surface boxes (or plugs and a crimper), a cable tester, a draw-rod set **(new: you run the Cat6 yourselves)** | 1 set | 30–60 | |
| ☐ To do | Swap the single socket on the side wall for a double at the same spot: like for like, not notifiable; step 3 has the safe way to do it **(changed: your own job)** | | | |
| ☐ To buy | A double socket in the same finish, and a deeper back box if the old one is shallow **(new)** | 1 | 5–15 | |
| ☐ To buy | Electrical safety kit: a two-pole voltage tester to GS38 with its proving unit, an MCB lock-off and padlock, a plug-in socket tester **(new: for doing it yourselves)** | 1 set | 45–90 | |
| ☐ To buy | Insulated (VDE) screwdrivers, wire strippers and side cutters **(new)** | 1 set | 20–40 | |
| ☐ To buy | 6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK) **(changed: switched at the wall)** | 1 | 20–35 | |
| ☐ To buy | 4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light) | 1 | 10–20 | |
| ☐ To buy | USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way | 1 | 5–10 | |
| ☐ To buy | Cable kit: brush grommets, velcro ties, labels | 1 | 20–30 | |
| ☐ To buy | Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m | 1 | 10–20 | |
| ☐ To do | Chase two conduits into the solid side wall for the PC, from behind the PC shelf down to skirting height and behind the skirting to the base's plinth: 38 mm oval for the HDMI and Cat6, 25 mm oval for the PC's mains lead; make good and paint **(changed: your own job; the PC now runs from the 10-gang)** | | | |
| ☐ To buy | Conduit: 38 mm and 25 mm oval, about 4 m each, two draw wires, one oak-coloured brush grommet; a 5 m C13 lead with a rewireable 13 A plug for the PC **(changed)** | 1 | 15–35 | |
| ☐ To buy | For the chase: a wall chaser with dust extraction for a day (or a bolster chisel and club hammer), one-coat filler **(new)** | 1 | 20–45 | |
| ☐ To buy | PC tidy kit: braided sleeve, shelf-notch grommet | 1 | 5–15 | |
| ☐ To buy | A powered USB 3 hub, 4-port with its own adapter, screwed under the side run's front edge below the PC: pads, the keyboard and a stick within reach of the sofa | 1 | 15–30 | |
| ☐ To buy | A remote power button for the PC: a wireless switch on the motherboard's power header (card in a spare slot), its push button on the wrap ledge | 1 | 10–30 | |
| ☐ To buy | A magnetic mesh dust filter cut to the PC's top exhaust | 1 | 5–10 | |
| PAINT **3 PAINT** | | | | |
| ☐ To do | Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on | | | |
| ☐ To buy | Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting and radiator (the door and its frame stay white); primer and undercoat | | 200–500 | |
| ☐ To buy | Woodwork primer, sanding pads and sugar soap for the white skirting and door | 1 | 20–40 | |

| Status | Item | Qty | Estimate (£) | Paid (£) |
|---|---|---|---|---|
| ☐ To do | Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell | | | |
| **MEDIA WALL** | **4 BASE UNIT** | | | |
| ☑ Done | Check the alcove back wall for fixings (solid) | | | |
| ☐ To do | Measure the alcove's width at the floor, 76, 134, 155 and 196 and the ceiling, and the floor's level across it | | | |
| ☑ Done | Check the side wall where the ledge and PC shelf go, and the chimney breast where the dining wall goes: solid (you confirmed) **(new)** | | | |
| ☐ To buy | Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2 | 2 | 70–200 | |
| ☐ To buy | Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims | 1 | 15–30 | |
| ☐ To buy | Hidden plinth drawer: oak front 54 × 6 with a finger-pull groove underneath, 12 mm ply box 52 × 34 × 5, 300 mm full-extension runners **(changed: no push-to-open at floor level, as v6.1 said)** | 1 | 30–50 | |
| ☐ To buy | Bay 3 carcass: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts, and a 12 mm birch-ply divider 38 × 69.4 that splits it into the slim game column and the LP column **(the divider)** | 1 | 50–90 | |
| ☐ To buy | Two face-out LP crates, 34.9 × 40 (a 33 cm trough): 12 and 9 mm birch ply cut to size, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg **(snug crates, no dead space)** | 2 | 120–200 | |
| ☐ To buy | Three slim drawers for the game column, 15.3 × 40: 6 mm birch ply sides and backs, 12 mm bases, oak lips, felt, three pairs of 400 mm runners (light duty is enough); a 2 m USB-C lead and a grommet for the pads drawer | 3 | 70–130 | |
| ☐ To buy | BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2 | 4 | 12–32 | |
| ☐ To buy | The sub in bay 2: an isolation riser for it (dense foam or sorbothane, about 22 × 28 × 3), 12 mm birch ply for the divider (35.3 × 38) and the part shelf (32.2 × 36), four shelf pins, felt pads **(the sub moves into the media wall)** | 1 | 25–45 | |
| ☐ To buy | Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3 | 1 set | 15–25 | |
| ☐ To do | Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer | | | |
| ☐ To buy | Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt | 1 | 50–90 | |
| ☐ To do | Build the plinth and bay 3, fix everything to the wall, fit the shelves, drawers and trays; place the TV and soundbar | | | |
| **MEDIA WALL** | **5 OAK JOINERY** | | | |
| ☐ To buy | Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible | 2 | 150–300 | |
| ☐ To buy | Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, the side-wall run from one board 225 × 30 × 4 (cut to its sweep), lips **(one run for the ledge and the PC shelf)** | 1 set | 230–380 | |
| ☐ To buy | Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair | 1 | 30–60 | |
| ☐ To buy | One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46) and the dining wall's backboard (110 × 22) from the offcut, chamfered fronts **(changed)** | 1 | 250–450 | |
| ☐ To buy | Oak face frame, 20 mm, 186 cm for the base | 1 | 60–120 | |
| ☐ To buy | Keyboard tray for the PC shelf: 7 cm oak housing under it, 12 mm ply tray 54 × 26 with an oak lip, 300 mm light-duty runners **(new: back from v6.1)** | 1 | 25–45 | |
| ☐ To do | Cut the joints and fit the uprights, shelves, ledges and face frame; notch the uprights for the ceiling strip | | | |
| **DINING TABLE** | **6 DINING WALL** | | | |
| ☐ To buy | Solid oak for the table: two halves 90 × 90 × 2.7, each a centre 76 × 90 of edge-glued staves with two pegged breadboard ends (the cutting list) **(the dining cabinet)** | 1 set | 260–450 | |

| Status | Item | Qty | Estimate (£) | Paid (£) |
|---|---|---|---|---|
| ☐ To buy | The dining cabinet: 18 mm oak-faced ply cut to size for its two ends, top, floor, partition, chair shelf, linen floor, two doors, the drawer front and the linen flap; a 6 mm ply back; 12 mm ply for the drawer box (the cutting list) | 1 set | 270–490 | |
| ☐ To buy | Oak for the cabinet's hinge rail (91 × 23.4 × 7), the eight apron rails and the four legs (6.5 × 4.5) | 1 | 35–70 | |
| ☐ To buy | Altro continuous piano hinge, 1800 × 50, cut to 90: the table's foot | 1 | 16–20 | |
| ☐ To buy | Folding, self-locking table-leg brackets, a set of four, and sixteen figure-8 tabletop fasteners for the aprons | 1 | 60–75 | |
| ☐ To buy | Six 170° clip-on concealed hinges for the chair bay's doors, a pair of 200 mm drawer runners, two 50 mm butt hinges and two lid stays for the linen flap, four magnetic catches, four knobs, two steel angle brackets for the wall | 1 | 65–125 | |
| ☐ To buy | Folding chairs, IKEA FRÖSVI (TERJE's successor, the same size) or similar, six, and six seat pads; they live in the cabinet (skip any you already have: seat about 45–48) **(six)** | 6 | 150–260 | |
| ☐ To buy | Three Lee Valley 40574 narrow flip-top hinges (solid brass, 1.27 × 7.0 cm) for the joint | 3 | 25–45 | |
| ☐ To buy | Two LP frames for the chimney breast, hung from 165 cm | 2 | 20–40 | |
| ☐ To buy | The “now playing” ledge on the breast, with the table by the stairs: oak 105.5 × 7 × 2.4 with a 1.5 lip (from the shelf board's offcut), two concealed brackets | 1 | 10–25 | |
| ☐ To do | Build the dining cabinet: the carcass, stand it and fix it to the wall, the hinge rail, the two table halves on their aprons and legs, hang the table on its piano hinge, then the drawer, the linen flap and the chair bay's doors (step 11) | | | |
| ☐ To do | Open the table, level it at 76 and trim the legs to the floor; load the six chairs (four on the floor, two on the shelf) and the seat pads | | | |
| LIGHTING | **7 LIGHTING** | | | |
| ☐ To do | Swap both ceiling fittings for the spot bars, turned to run from the TV towards the window: like for like, your own job; step 12 has the safe way. The light switches stay as they are (smart bulbs need full power) **(no dimmers)** | | | |
| ☐ To buy | Eight smart colour E14 bulbs for the bars: WiZ Colour P45 E14, 4.9 W (Wi-Fi, no hub), or Philips Hue White and Colour E14 lustre, 5.1 W, if a WiZ shows below its shade; both under the bar's 6 W | 8 | 100–360 | |
| ☐ To buy | Smart colour strips for the upright grooves and under the side-wall run, in the same app as the bulbs: about 4.5 m of RGB and tunable-white strip (Philips Hue Lightstrip Plus, or WiZ's strip kits, cut at their marks), three aluminium channels with opal diffusers, plug-in power on the 10-gang (no fused spur); a 6500K bias light for the TV stays plain white **(smart colour, so the whole wall changes with a scene)** | 1 | 90–220 | |
| ☐ To buy | Two Dunelm Elements Wolston 4-light spotlight bars, £49 each (bulbs above) **(changed: two)** | 2 | 98–110 | |
| ☐ To buy | Optional: a smart button for the bars, so a scene is one press rather than the app (WiZ remote or Hue dimmer switch) | 1 | 0–25 | |
| ☐ To do | Set up the smart bulbs in their app: group them as bar 1 and bar 2, then save the three presets (Bright, Cosy, Film; Paint and light) with their levels and colours, and the table rule; your red lamp's bulbs go in the same scenes | | | |
| ☐ To buy | Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional) | 1 | 0–80 | |
| AV | **8 TV, CONSOLES AND AV** | | | |
| ☐ To buy | 5-in-1 4K60 HDMI switch with remote | 1 | 20–35 | |
| ☐ To do | Measure the PC-to-TV cable route (about 4–5 m along the skirting) | | | |
| ☐ To buy | Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m | 1 | 20–60 | |
| ☐ To buy | High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them | 5 + 2 | 25–40 | |

| Status | Item | Qty | Estimate (£) | Paid (£) |
|---|---|---|---|---|
| ☐ To buy | RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees) | 1 | 200–240 | |
| ☐ To buy | PS2 component video cable, good quality | 1 | 10–20 | |
| ☐ To buy | Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar | 1 | 5–10 | |
| ☐ To buy | RGB SCART cable for the PS1 | 1 | 8–15 | |
| ☐ To do | Connect the TV inputs; set up the PS2 and RetroTINK | | | |
| AV **9 VINYL KIT** | | | | |
| ☐ To buy | Replacement stylus for the turntable (exact match for its model) | 1 | 15–40 | |
| ☐ To buy | External phono preamp (only if the deck has a PHONO/LINE switch) | 1 | 40–100 | |
| ☐ To buy | RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX **(changed: shorter)** | 1 | 5–10 | |
| ☐ To buy | Isolation feet for the turntable; a carbon-fibre record brush | 1 | 20–40 | |
| ☐ To do | Set the tracking weight; soundbar in Music mode | | | |
| AV NETWORK **10 FINISHING** | | | | |
| ☐ To do | Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off | | | |
| ☐ To do | Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware | | | |
| ☐ To do | Label both outlets of the double socket (left: everything, right: retro) and each HDMI switch input | | | |
| ☐ To do | Set the sub and rear speaker levels from the sofa | | | |
| ☐ To buy | Drip trays or a waterproof mat under the pots on the plant stand's slatted tiers; if you add clip-on grow lights later, a smart plug with a schedule | 1 | 10–20 | |
| ☐ To buy | Felt pads under the vinyl stand's feet, so it slides out from the breast without marking the floor | 1 | 3–6 | |
| ☐ To buy | A rug about 230 × 160 in front of the sofa: cream with deep blue, rust and ochre goes with the green, the oak and the blue sofa (the plan draws one as a guide) | 1 | 120–300 | |

# 12 · Still to confirm, later ideas, sources

### Still to confirm

- ☐ The stairs wall: solid, or plasterboard on studs? It decides how the dining cabinet's top is fixed to it.
- ☐ The electrics: you're swapping the two light fittings and the socket yourselves (as planned; no dimmers), or an electrician for the fittings?
- ☐ The sofa's return: how far it runs along the side wall (the long side is about 266 from your photo; the return is drawn at 168).
- ☐ The stairs wall's length and your shelf's height (drawn at 196: the OBEGRÄNSAD needs 198 or lower to stand on it).
- ☐ The turntable's model (the zone takes up to 45.2 × 35.2 cm) and whether it has a PHONO/LINE switch.
- ☐ The Series X fan stand's model: the shelf takes a footprint up to about 30 cm wide.
- ☐ Your rear speakers: a photo on their stands behind the sofa, to set the stands' size, height and spots.
- ☐ Your sofa's ottoman: how tall is it? (Drawn 75 square, 44 tall.)
- ☐ Records: roughly how many LPs (the wall and the vinyl stand hold about 291: 191 in the wall, all face-out, and 100 in the stand)?
- ☐ Games: roughly how many PS2, Xbox 360 and Xbox One games? The drawers hold about 92.
- ☐ Wired ports for the projector (master) and the music room (bedroom 2)?
- ☐ The 10-gang: is it surge-protected, where are its two switches, and how long is it? It now feeds the PC too.
- ☐ Console model numbers, from the labels underneath, to tighten the row heights; the plan assumes the largest version of each.
- ☐ The pendant point by the stairs wall (the photo gives about 55 cm out, a quarter of the way along the shelf).
- ☐ The window corner at floor level: from the pipe boxing to the curtains when open (the ledge's 25 cm is in; the plant stand is drawn with 4.5 cm a side).
- ☐ The RetroTINK-6X CE's size when it arrives (bay 1 allows 23 × 23 × 8).

Easiest way to answer: the viewer page has a box for each of these and takes photos; or reply in a message, a line each.

### Agreed so far

- ✓ From your v13.6 message: the legs store as part of the table (in the leaf's frame); the electrics are your own job; each step labelled with its part.
- ✓ From your v13 message: less cladding (done: no slats, spine, sub box or reeding; cables chased into the wall); the stairs go up; the drawer layouts drawn.
- ✓ From your answers: the sofa's long side ends in line with the breast's reveal; the door hinges on the breast side and opens in; the side wall and breast are solid (plugs and screws, no stud fixings); the PC is an average gaming case (allowed for at 23 × 45.3 × 46.6).
- ✓ Artisan oak, Card Room Green with matching eggshell skirting, every record face-out, the game drawer as designed.
- ✓ The TV 25.25 cm nearer the side wall, the turntable in the alcove, the Series X above it, the records below.
- ✓ The ledge wrapping round onto the side wall, with the PC above the sofa.
- ✓ A fold-down oak dining table on the chimney breast; the spot bars on the existing ceiling points; the sub back in the media wall, in bay 2.

### Later ideas

- **For the room:** a glass display case on the breast, glass plant shelves in the window, a C-table for the sofa's open end, and small ones (TV backlight that follows the picture, cushions in the rug's colours, a plant uplight, a rug pad): see the appendix “Ideas for later”, and the Ideas tab and buttons in the 3D room.
- **Dave and Emma's watch log:** a shared film and TV log, like a Spotify Blend for what you watch: log each film or episode, mark whether you watched it together, rate it, and later get suggestions for what to watch next. To be

shaped up separately.

## Sources

Numbered sources are the item sizes used in the fit proof (section 3).


1. Your hand-measured sketch of the living room, in mm (alcove 1860 × 380, chimney breast 1175, ceiling 2350)
2. LG: OLED55C6 specifications: 1222 × 703 × 45.1 mm without stand, 1222 × 757 × 230 mm with stand, stand 470 × 230 mm, 14.1/15.9 kg, VESA 300 × 200 (lg.com, OLED55C6 product pages)
3. Trusted Reviews and Expert Reviews, Hisense AX5100G reviews: soundbar 920 × 60 × 90 mm, subwoofer 214 × 294 × 304 mm (W × H × D) with a rear-firing port, rear speakers 90 × 144 × 109 mm
4. Wikipedia, Xbox Series X: 151 × 151 × 301 mm; Pure Xbox (Nov 2020) quoting Microsoft: at least 4 cm clear above the console on an open stand, 10–15 cm recommended on all sides
5. IKEA: BESTÅ frame 60 × 40 × 64 cm (10247379); BESTÅ shelf 56 × 36 cm, 20 kg maximum load
6. TP-Link: TL-SG108 specifications, 158 × 101 × 25 mm (tp-link.com)
7. Laptop Outlet: TP-Link Deco X1500 3-pack listing, 110 × 110 × 114 mm per unit
8. Audio-Technica: AT-LP120XUSB specifications, 452 × 352 × 141.6 mm, 8 kg without dust cover (audio-technica.com)
9. RetroTINK: RetroTINK-6X CE shop page (power 5 V 2 A by USB; no size published)
10. UGREEN: 5-in-1 4K60 HDMI switch CM568 (90512), 127 × 59 × 25 mm
11. Original Xbox One power supply: 17 × 7.5 × 4.8 cm (6.69 × 2.96 × 1.88 in), as listed by several replacement-PSU sellers
12. Wikipedia: Comparison of PlayStation consoles (PS1 270 × 188 × 60, PS2 302 × 183 × 78, PS3 325 × 274 × 98 mm); Xbox One (333 × 274 × 79 mm); Neowin: official Xbox 360 dimensions (309 × 258 × 83 mm)
13. Record and case sizes: LP sleeve 31.4 cm (12⅜ in); 7-inch sleeve 18.3–18.4 cm; DVD-style keep case 19 × 13.5 × 1.4 cm; Xbox One case 17 × 13.5 × 1.1–1.2 cm (videogamecasesize.info; Walvis Products; Wikipedia: Keep case)
14. Dunelm: Elements Wolston 4-light spotlight bar, £49: 84 × 16 cm, 22.4 cm drop, shades 15 cm, E14 6 W max (dunelm.com, product 1000189449)
15. Emuca heavy-duty full-extension runners, 400 mm, 68 kg per pair; Accuride DZ3832EC 350 mm, 43 kg; runner fitting allowance 12.5 mm a side
16. Xbox 360 original 203 W power supply: no Microsoft size published; Akyga's 203 W replacement for it (AK-PD-14) is 210 × 77 × 55 mm (b2b.akyga.com), so an allowance of 21 × 10 × 6 cm is used here (measure yours)
17. Turntable Lab: Audio-Technica replacement dust cover for the AT-LP120XUSB (705-140XP-1815), 17.375 × 13.625 × 2.75 in (44.1 × 34.6 × 7.0 cm)
18. Häfele UK, Technology catalogue 2023, p. 7.206: Hebgo heavy-duty folding bracket 287.43.428, arm 580 mm, height 220 mm, 500 kg per pair, locks automatically when raised; up to 800 mm between brackets (Häfele 287.43.419 listing)
19. Screwfix: Union PowerLoad lift-off hinges, 100 × 88 mm, 3-pack (5938T), 120 kg maximum door weight
20. Emuca: table legs Ø60 mm, 710 mm, height adjustable 710–730 mm, 50 kg per leg (set of 4): 710 is the standard dining leg; the plan's oak legs are made to the same 71 cm
21. RS PRO 45 mm steel spring clip (RS 175-3588): 42.1 × 56 × 12.7 mm, packs of 10 (uk.rs-online.com)
22. IKEA: TERJE folding chair, 43 W × 49 D × 78 H cm, seat height 48 cm, tested for 110 kg
23. Book a Builder UK, small dining room ideas: 600 mm of table edge per diner; table edge to a wall 600 mm tight minimum, 750 mm to aim for, 1000–1100 mm to walk behind; dining height about 750 mm
24. Your photo of the sofa (18 Oct 2026) and your note (21 Oct): a blue corduroy L-sofa, two seats then the corner on the long side, which ends about 40 cm short of the window's west end (about 266 cm long); the return's 168 is a typical corner-sofa size until measured
25. Klipsch subwoofer placement guidance, as quoted by Cloudspress and TechYorker (2026): leave about 6–12 in (15–30 cm) between a rear-firing port and the nearest obstruction
26. Xbox Support (Microsoft), About the Xbox One power supply (archived): don't block its vents, don't put it in a confined space such as a bookcase, rack or stereo cabinet unless the space is well ventilated, don't stack devices on it, and lay it flat on its four rubber pads; the Xbox 360 brick's advice is the same
27. Xbox Support (Microsoft), Xbox 360 placement tips, as quoted on Gaming Stack Exchange: an open, well-ventilated area, a hard surface, and away from high-vibrating sources such as subwoofers
28. Your two photos of the west end (20 Oct 2026): the 2 m black shelf on the wall beside the stairs, scaled from its 2 m: it starts about 41 cm past the stair box and stops about 16 cm short of the window corner, top about 200; the pendant hangs about 55 cm out from the wall, a quarter of the way along it
29. Corsair 4000D Airflow, a typical mid-tower gaming case: 453 × 230 × 466 mm (TechPowerUp review)
30. Lee Valley: 40574 narrow flip-top hinges, solid brass, 1/2 in × 2 3/4 in (1.27 × 7.0 cm), for card and flip-top tables (leevalley.com); their 40208 card-table hinge is 1 3/16 in (3.0 cm) wide, too wide for a 27 mm edge
31. Your photo (20 Oct) and note: Dunelm hessian pencil-pleat curtains; Dunelm pencil-pleat drops run 137, 182 and 228 cm with a 3 in header; Dunelm 28 mm wooden poles, 120–360 cm, rings included (one per 10 cm), finials up to 8 cm more (dunelm.com)

32. Your plant stand, from the listing you sent (20 Oct): 98 × 29.5 × 158 cm (W × D × H), slatted wood; two five-tier towers with an open middle holding two shelves
33. Your vinyl stand, from the listing you sent (20 Oct): 70 × 32 × 78 cm (W × D × H), 63 to where the frame slopes; two top bays 32.3 wide split by a wooden divider, a 10 cm front board, wire dividers 19.5 tall on the bottom shelf; black steel and rustic wood
34. IKEA: OBEGRÄNSAD LED wall lamp, black (your 905.262.63; 005.262.48 and others by country): 52 × 36 × 6 cm, 5 W, five lighting programmes, hangs landscape or portrait, runs on a 5 V USB-A adapter sold separately (ikea.com); its face is a 16 × 16 grid of white LEDs. From the Swedish House Mafia collection (IKEA press kit, Sept 2022)
35. Sony: DualSense controller 160 × 66 × 106 mm (W × H × D); Microsoft: Xbox Wireless Controller 153 × 61 × 104 mm
36. Record-shop browser crates and LP storage boxes are made about 32.5–33 cm inside for 31.4 cm sleeves (31.8–32.4 in outer sleeves), flipped by the sleeves' top edges
37. Philips Hue Lightstrip Plus V4: 2 m base kit, cuttable every 33 cm, extendable with 1 m; WiZ Wi-Fi LED strip, 2 m starter kit, cuttable; both RGB and tunable white, in the same apps as the bars' bulbs (philips-hue.com; wizconnected.com)
38. Smart colour E14 bulbs for the bars: WiZ Colour P45 E14, 4.9 W, 4.5 × 10.1 cm, Wi-Fi, no hub (Philips UK specification sheet, 8719514437333); Philips Hue White and Colour Ambiance E14 lustre, 5.1 W, 45 × 77 mm (Philips Hue; Lampmasters)
39. IKEA UK: FRÖSVI folding chair (TERJE's successor, the same 43 W × 49 D × 78 H, seat 48, tested for 110 kg); the hole in the back is made to hang it on a wall hook, hooks not included; flat-packed (folded) 89 × 43 × 9 cm (ikea.com/gb, ikea.com.tw product pages)
40. Screwfix: Smith & Locke Mid Duty Anti-Slip Storage Hook, 200 × 200 mm, steel, 60 kg maximum load, £3.59 (8882J); FRÖSVI padded weighs 3.61 kg net (ikea.com.tw)
41. Buller Ltd: Folding Heavy Duty Bracket for Bench and Tables, 330 mm long, 82 mm wide, steel, grey primed, automatic lock when raised, 50 mm proud of the wall folded, 150 kg per pair (75 kg each), brackets at most 800 mm apart (bullerltd.co.uk)
42. Client, 20 Oct 2026: the sofa's ottoman, square, blue corduroy matching the sofa, "the size of a chair on the sofa": one seat, 75 × 75; its height taken at the seat's 44 until measured
43. IronmongeryDirect: Altro continuous piano hinge, 1800 × 50 × 1.2 mm, satin 304 stainless steel, £16.00 (SKU 302829; cut to length, screwed every 10 cm) (ironmongerydirect.co.uk)
44. ManoMano UK: folding, self-locking table-leg brackets, 75 × 60 × 55 mm, set of four with screws, about £58: lock at 90°, for legs at least 2½ in (6.4 cm) at the top; seller's rating 226.8 kg a bracket (manomano.co.uk, listing 219621358)
45. Clever (Italy): Slim fold-down table in a wall unit: folded 80 × 9 × 138 cm, open 80 × 138 × 74, a 1.8 cm top (clever.it): the same idea, a top hinged at its foot that stands up into the wall
46. Blum CLIP top 170° full-overlay hinge, self-closing (71T6540; cabinetparts.com): doors open past flat, so they lie back against the cabinet's front while you take the chairs out
47. Client, 20 Oct 2026: "my surround speakers … exist standing behind the sofa" (a photo to follow): the AX5100G's two rears on floor stands; the stands' size and the spots are drawn provisionally until the photo
48. Virgin Media: Hub 5 modem mode
49. RetroTINK: Introducing the RetroTINK-6X CE (inputs including TOSLINK, deinterlacing); RetroTINK-2X Pro listed as legacy; RetroTINK-4K CE
50. Expert Reviews: LG C6 OLED review (4 × HDMI 2.1, one eARC, ports on the left); JB Hi-Fi: LG C6 specifications (165 Hz mode for PC sources)
51. Hisense AX5100G spec sheet (inputs, auto power-on); JB Hi-Fi Business: AX5100G (rears linked by cable, wireless to the soundbar)
52. TP-Link: Deco X1500 specifications (two gigabit ports per unit)
53. Glorious Vinyl Vault specifications (80 LPs and 24.5 kg per drawer), a check on the tray capacity
54. Screwfix: Essentials 964VN 350 mm runners; Metrol: Accuride DZ3832EC; IKEA: BESTÅ drawer runners
55. NIST SP 500-252 / CLIR pub121: Care and Handling of CDs and DVDs (store upright, book style)
56. Sony PS3 and Microsoft Xbox 360 S manuals: ventilation; IKEA UK: secure it
57. Ideal Home: colour drenching; The Paint Shed: colour drenching with Farrow & Ball; Livingetc: sage green shades (Card Room Green, Treron, Sage Green, Green Smoke)
58. Pontata: Card Room Green vs Little Greene Sage Green (approximate screen colours)
59. Homebase and NC Tiles: Farrow & Ball Card Room Green No. 79 in Modern Eggshell and Estate Eggshell
60. haart Medway: floor plans and listing photos for the house (Portsmouth Close)
61. ConsoleMods: PS2 Component Video Out setting
62. GOV.UK: Electrical safety, Approved Document P (2013 edition, for use in England): notifiable work is a new circuit, a replacement consumer unit, or an addition or alteration to a circuit in a special location (round a bath or shower); like-for-like replacements and additions to existing circuits elsewhere are not notifiable (gov.uk/government/publications/electrical-safety-approved-document-p)
63. IET Electrical: Part P, England and Wales (notifiable and non-notifiable work); Planning Portal: Electrics, Building Regulations (non-notifiable work must still meet the regulations, BS 7671)

# Appendix · Ideas for later (not in the plan or the costs)

Suggestions only, for later: they aren't in the plan, the buy list or the costs (all of them together would be about £400–775). The first three can be seen in the 3D room with the “Ideas (not in the plan)” buttons, all off until you turn them on.

| Idea | Why, and where | About £ |
|---|---|---|
| **A tall glass display case on the chimney breast** 90 wide × 85 tall × 12 deep, three glass shelves | For collectibles, boxed games, 7-inch singles, small plants or ceramics, lit from inside, without a cabinet on the floor: 12 deep, so it stands no further out than a picture ledge. Oak frame from the offcuts (or painted black or white), a 4 mm toughened glass front that lifts off, three 6 mm glass shelves on pins, an LED strip under the top on the strips' app. Hung on a French cleat: about 16 kg empty. It takes the place of the two framed LPs, above the “now playing” ledge. In the 3D room, try what's inside it and its frame. *On the breast, centred, from 137 to 222 (just clear of the sleeves on the “now playing” ledge, 13 under the ceiling); with the table by the stairs only.* | 90–170 |
| **Glass plant shelves across the window reveal** two 10 mm toughened shelves, 235 × 18, at 130 and 170 | Your deep sill already takes a row of plants; two glass shelves above it make the window a green screen that still lets the light through. Studs into the reveal's sides, and a slim steel rod down from the head at the middle so a 235 span can't sag. *Inside the window's 22 cm reveal, in front of the glass, clear of the curtains.* | 120–200 |
| **An oak C-shaped side table for the sofa's open end** top 45 × 30, 60 tall | The ottoman end has no arm to put a drink on. A C-table's top slides over the end of the seat and its foot sits flat on the floor beside it, so drinks and pads are at hand and it moves away when someone lies along the seat. Made from the oak offcuts. *At the sofa's open end, towards the TV.* | 25–60 |
| **TV backlight that follows the picture** a sync box on the HDMI line and a strip behind the TV (in place of the plain bias light) | The wall behind the TV glows with the colours on screen, in the same app as the bars and strips, so a film can tint the room. *Behind the TV, on the bias light's place; the box sits by the HDMI switch in bay 1.* | 60–130 |
| **Cushions and a throw in the rug's rust and ochre** two or three 45 × 45 cushions and a throw | They tie the blue corduroy sofa to the rug and the oak, and soften the ottoman end. The 3D room has three colourways to try (rust and ochre, blues, sage and cream) under Decor. *The sofa.* | 40–90 |
| **A small uplight under the plant stand's bottom tier** a plug-in spot, about 10 cm | Throws the leaves' shadows up the wall at night, so the plants are part of the evening lighting with the room lights low. *On the floor, under the stand.* | 10–20 |
| **A non-slip pad under the rug** 220 × 150, trimmed to fit | Keeps the rug flat and square on the carpet where the sofa's open end and the coffee table push it. *Under the rug.* | 15–30 |
| **A smart scene button by the door** a small wireless button, about 5 × 5 | One press for the Bright preset coming in, a long press for everything off, in the same app as the bars' bulbs and the strips. *On the wall beside the door's light switch.* | 15–25 |

| | | |
|---|---|---|
| **A motion nightlight on the stairs**<br>battery LED strips under the treads' noses, or a single motion light | A soft glow on the stairs at night with nothing to wire; it turns itself off. *The stairs.* | 15–30 |
| **A record-cleaning tray in the pads drawer**<br>a small tray, about 30 × 15 | The carbon brush, a stylus brush and a cleaning fluid kept together next to the deck. *Bay 3's pads drawer.* | 10–20 |

Idea only, from the 3D room: glass plant shelves across the window's reveal, beside your plant stand in the west corner.

Idea only: the tall glass display case (85 tall, three glass shelves) on the chimney breast, lit inside, above the “now playing” ledge: oak, with collectibles.

The same case with a black frame and a bit of everything: 7-inch singles, a cap and figures, small plants, boxed games. A white frame, and singles and tapes, are in the 3D room's Decor buttons.

### Suggested furniture and decor: each on or off in the 3D room

Not in the plan or the costs: all 18 together would be about £489–1055. Each has its own button in the 3D room, all off until you turn them on.

| Suggestion | Why, and where | About £ |
|---|---|---|

| | | |
|---|---|---|
| **A tall floor plant at the sofa's open end**<br>a fiddle-leaf fig or rubber plant, about 150 tall in a 30 cm pot | Height and green in the one corner with nothing tall in it, between the media wall and the sofa's open end. *On the floor by the side wall, clear of the media wall's drawers when they're pulled out.* | 40–90 |
| **A round oak side table at the sofa's open end**<br>40 across × 50 tall | Somewhere to put a drink down from the ottoman end, without the C-table over the seat. *Beside the sofa's open end, towards the TV.* | 40–90 |
| **A knitted pouffe on the rug**<br>50 across × 40 tall | A footrest, a spare seat, and a third colour on the rug. *On the rug in front of the media wall, beside your ottoman; it moves anywhere in Edit room.* | 30–60 |
| **A large framed print over the sofa's return**<br>80 × 55 in an oak frame | The side wall over the sofa is the one big empty wall: one print, hung 30 above the back cushions. *On the side wall over the sofa's return, past the oak run.* | 40–120 |
| **A slim wall clock over the door**<br>30 across | Fits the 37 cm between the door's head and the ceiling. *Centred over the door.* | 20–45 |
| **Coffee-table styling: an oak tray, two books and a candle**<br>a 40 × 26 tray | Keeps the remotes' landing spot tidy and gives the coffee table a finished look. *On the coffee table; it goes where the table goes.* | 20–40 |
| **Warm fairy lights along the curtain pole**<br>about 4 m of warm-white micro LEDs on copper wire, USB | A soft line of light over the window in the evening, on the same timer as the grow lights. *Wound along the curtain pole, end to end.* | 12–25 |
| **A macramé plant hanger from the pole**<br>a 75 cm hanger, a 15 cm pot, a trailing plant | Puts a plant in the window without taking sill space from the others. *From the curtain pole near the window's sofa end, clear of the curtains when they're drawn back.* | 20–40 |
| **A trailing pothos on the media wall's top shelf**<br>a 15 cm pot, trails about 60 cm | Softens the oak and the books, trailing down past the shelf's front edge. *At the right-hand end of the top shelf, 196 up.* | 12–25 |
| **A chunky knit throw over the sofa's open end**<br>about 130 × 170, cream | The ottoman end is where you put your feet up: a throw within reach, and a texture against the corduroy. *Folded over the open end of the sofa's return.* | 30–60 |
| **A mushroom lamp on the PC shelf**<br>about 22 across × 30 tall, warm glow | A low, warm light at the sofa's end for reading or gaming without the spots on. *On the PC shelf, past the PC, 155 up.* | 30–70 |
| **A seagrass basket for throws by the breast**<br>about 40 across × 40 tall | Somewhere for the throws and cushions to go when people sit at the table. *In the corner by the chimney breast, clear of the door's swing; it moves in Edit room.* | 20–40 |
| **LED pillar candles on your black shelf**<br>three, 7.5 across × 10, 12.5 and 15 tall, warm flicker, on a timer | A warm glow by the table in the evening without a flame near the shelf's things. *On your black shelf by the stairs, at the window end.* | 15–30 |
| **A linen runner and a bud vase for the table**<br>a runner about 33 × 180, a 20 cm vase | Dresses the table when it's open; folds into the drawer (option C) or the chair cupboard when it isn't. *Down the middle of the open table, whichever option.* | 20–40 |

| | | |
|---|---|---|
| **Two floor cushions by the coffee table**<br>60 × 60 × 12 each, mustard and rust | Extra seats for a film night, and two more colours from the rug. *Stacked on the rug at the TV end of the coffee table; they move in Edit room.* | 40–80 |
| **A rattan floor lantern with an LED candle**<br>about 30 across × 55 tall | A low glow by the window that the curtains can't catch. *On the floor between the sofa's arm and your plant stand; it moves in Edit room.* | 25–50 |
| **A tall arched mirror behind the sofa's corner**<br>about 50 × 90, oak frame | Bounces the window's light back into the room, in the strip of wall beside the window. *On the window wall in the corner behind the sofa's return, its foot above the back cushions.* | 60–120 |
| **A warm LED strip under your black shelf**<br>2 m warm-white strip, USB or plug-in | Lights the table (or option C's cabinet) from above, softly, for an everyday supper. *Under the shelf's front edge, the full 2 m.* | 15–30 |

All six suggestions switched on (3D room, “Suggested furniture and decor”): the floor plant and side table at the sofa's open end, the pouffe, the print over the sofa, the clock over the door, the coffee-table tray.

Decor to try (3D room, Decor): a sage-stripe rug, rust and ochre cushions with a throw on the arm, white linen curtains. Also: a jute or deep blue rug, blue or sage cushions, sage, blue or rust curtains.

# Appendix · Earlier rounds

**v22: the plan page is quicker to use.** The first round of the optimisation plan: under the 3D room the controls are now in **five drawers** (Views, Layout, Lights, Look, Extras), one open at a time, and on a phone the picture stays in sight while you choose. **Your picks** (in Extras) lists every idea, suggestion and decor choice you've switched on with their rough cost added up, to copy as a note or save for the next round. The **build steps** open one at a time, starting at the first one not done, with a Next step button, or all at once if you'd rather read straight through: on a phone the tab is under a third as long. The **3D room** draws less each frame (about half as many draws; your plant stand alone went from 152 pieces to 7) and shows a picture while it loads; in Fast each spot bar lights with two of its heads. The shared plan page links its pictures instead of carrying them, so it opens much faster. The plan itself doesn't change.
**v21: the 3D room opens on the whole room, and suggestions you can turn on or off.** You asked for the first 3D view "at the best viewing to see most if not all the room without clipping or obstruction": it now opens on **The whole room**, from above the window side with the wall nearest you **cut away** like a dolls' house, so the media wall, the breast, the door, the stairs, the table's wall, the PC wall and the sofa are all in one picture. Every view was checked: **From the window** no longer stands behind your red lamp; the chair cupboard, the dining table, the sofa, the bird's eye and the dolls' house views were re-aimed; the chimney breast and the window wall have their own buttons; and on a phone each view now takes in as much of the room across as on a computer. **Suggested furniture and decor** (appendix), each with its own on/off button in the 3D room and all off to start: a tall floor plant and a round side table at the sofa's open end, a knitted pouffe, a large print over the sofa, a clock over the door and a tray for the coffee table. Edit room's Add list gains a rattan armchair, a blanket ladder, an oak bench and an arc floor lamp. The plan itself doesn't change.
**v20: the table packs away with its chairs, a taller glass case, and decor to try.** You asked "is it not possible to build the chair storage into the table setup. i want it to be a fully pack away setup": it is. **An oak chair cupboard** is built in at the table's window end, 51 wide × 22 deep × 189 tall, flush with the pipe boxing and under your shelf, with **four folded chairs inside** (two on its floor, two on its shelf) behind two doors on push latches (about £120–240; section 5). Fold the table and shut the doors and the whole wall is packed away. **Six** fit if it's built 31 deep; otherwise chairs five and six go in the garage. Thanks for the **25 cm** window ledge: that matches the drawing (22 into the reveal and 3 of nose). **The door and its frame are white**, as you said, in every drawing and the 3D room. No HD crashes, so the HD question is gone. **The glass display case** idea is now **85 tall** with three glass shelves (90 wide, 12 deep), and in the 3D room a new **Decor** group tries five things inside it (collectibles, retro games, 7-inch singles and tapes, plants and ceramics, a bit of each), three frames (oak, black, white), four rugs, three cushion colourways and five curtain colours; Edit room's Add list has a floor vase, a pouffe, a basket, a floor mirror and a lantern. Fit proof 129 checks, none fail.
**v19: somewhere for the chairs, every LP counted, and a better edit mode.** First, **two folded chairs on hooks** in the 51 cm gap at the table's window end (still the cheap way, about £8–15). The gap at the stair end is 2 cm too narrow; under the stairs and the garage are the other two places. The LP count now includes your vinyl stand: **about 291 LPs**, 191 in the wall, all face-out, and 100 in the stand. In the 3D room's Edit room: **undo**, things **snap to a wall** within 5 cm, **copy** what you've added, each thing shows its **size**, the **sofa moves** too, and on the shared plan page one layout can be **saved for both of you**.

**v15.** Your notes on the room as it really is: **your sofa** is drawn in blue corduroy, two seats then the corner, its long side running to about 40 cm short of the window's end, with no cushion; **your red lamp** with its colour bulbs (green on top, pink, then purple); on your shelf by the stairs, **your IKEA light box** at the left end, **your laser** in the middle (its dots on the ceiling) and the **plant hanging** down off the shelf; **your hat** on an oak peg by the stairs; **soft grey carpet** (your photo shows it already is), a **designed rug** that fits in front of the sofa (230 × 160), and **your curtains** from your photo: long hessian pencil-pleat panels to the floor on a wooden pole just under the ceiling strip (open or drawn in the 3D room). The **record trays take LPs only**: one trough the full width, with room to flip with both hands. The **side wall** looks better: the ledge and the PC shelf become **one oak run** that sweeps out in a curve to the PC, with a warm LED glow underneath. **Both spot bars now run from the TV towards the window** and take **smart colour bulbs** (E14, under the bar's 6 W), so there are no dimmers to fit; the light is set softer and aimed at walls and shelves, for less spotty light. In the 3D room every light has a **slider**, the bars and the lamp a **colour**, and the **walls, ceiling and skirting** each have their own paint colour. And a **new dining table that stays on the wall and seats more**: a flip-top gate-leg table. Its second half flips out on brass hinges to make 110 × 160, on a folding oak gate leg: four with room, six elbows-in, seven by the stairs; no lifting off, no leaf or legs to store (section 5). Your **curtains** are Dunelm hessian on a wooden pole with rings and ball finials.

**v13.** You asked for less cladding all over the place, so it's gone: no slatted cable spine, no slatted PC screen, no slat panel behind the turntable, no slatted sub box, no reeded table. The PC's leads now run in conduit **chased into the solid side wall**, so there's no trunking either; the sub stands under a **plain oak cover** (below); the folded table is a **plain oiled oak panel** between its pegged breadboard ends. The oak that's left is the joinery that does a job: the uprights, the shelves, the ledges, the worktop, the face frame and the table. The **stairs go up** to the first floor (drawn rising away from the room; tell us if they run the other way). New: **the drawer layouts**, every drawer to scale with what goes in it (section 3). The slat-wall alternative is retired.

**v13.1–13.2:** the sub gets a **plain oak cover** with no sides: spills run off it, the sub can't tip, and nothing blocks its driver or port (section 5). A new render shows **every drawer out**, and the viewer can lay **rough measurements** over any view. The **stairwell opening** now starts just after the door, over the banister, and the sofa is drawn as yours. **Six at the table** now use the leaf too: 170 × 80, 85 cm a diner on the long sides and 75 cm off the breast, every clearance checked. The **side view** now shows how deep the alcove is round the wall: 38 of the 46 cm base inside it, 8 proud, the uprights and shelves wholly inside.

**v13.3:** the build guide is now step by step, each step with what to buy before it and what to check after it; the buy list shows what each phase costs; and the network and power section lists what each part does, both setup stages and what plugs into every socket (section 9).

**v13.5:** the sub is back in the media wall. The two Xbox bricks lie flat at the back of their shelves by the power leads, the RetroTINK and the HDMI switch move into the space that frees in bay 1, and the sub takes the lower left of bay 2, under the TV, on an isolation riser beside the PS2 and PS1 (section 3). The dining wall has nothing under it now, so eight can sit down without moving anything.

**v13.7:** four **lighting scenes** (TV night, dinner, records, everyday evening), each drawn on the room plan with every light's level, so you know where to set each dimmer (section 8).

**v13.8:** the whole plan is back on your own deck (the one you can edit, with your saved answers), and the summaries now follow the fit proof and the questions list, so they can't fall out of date: the PC case is answered and off the list, and the three checks on site are the TV's plugs, the RetroTINK's size and the sub's vibration.

**v13.6:** the **leaf carries the legs**: a slim oak frame underneath holds all eight oak legs in clips, so the folded table stays a plain panel, nothing new stands in the room, and the leaf goes wherever the chairs are kept (section 5). **The electrics are your own job**: swapping the socket for a double, the two dimmers and the two spot bars are like-for-like and not notifiable in England, so there's no electrician's visit; there is a safe routine and a clear point to stop and call one (section 9). The PC now runs from the 10-gang. Every step and every buy-list phase carries a **coloured label** for its part (Network, Media wall, Dining table and so on), and one list says what we still need from you.

**v14:** a **second place for the table**: on the wall by the stairs, under your 2 m shelf (section 5 compares the two). The plan page now has a **3D room** you can walk round: switch the table's wall, fold it or set it for four

or six, change the paint, the lighting scene and where the floor lamp stands, and save any view as an HD picture.

# Appendix · Options we set aside

### The five lighting scenes

Six buttons (these five and Daytime) mixed moods, an activity that depends on the table, and the time of day. They became three presets and two rules. Where each went: General → Bright; TV night → Film; Dinner → Bright; Records → Cosy; Everyday evening → Bright, with the table rule for dinner; Daytime is now a switch.

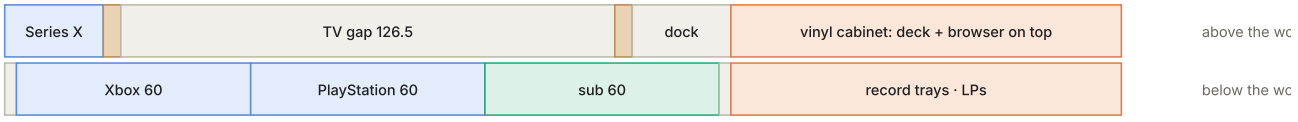
| Light | General | TV night | Dinner | Records | Everyday evening |
|---|---|---|---|---|---|
| Ceiling LED strip | 55% | 10% | 25% | 35% | 70% |
| Uprights' grooves and the side shelf | 45% | 15% | 30% | 60% | 50% |
| TV bias light | off | on | off | off | off |
| Bar 1, between the TV and the sofa | 60% | off | 15% | 60% | 55% |
| Bar 2, by the stairs | 60% | 10% | 80% | 15% | 40% |
| Your red lamp | 45% | 70% | off | 80% | 60% |

### How the media wall's layout was chosen

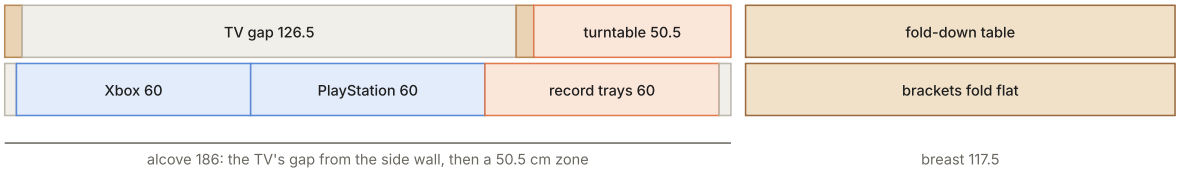
#### What moved in v11

To scale, from the side wall. Top: v10. Bottom: v11

**The TV in the middle, the vinyl cabinet on the breast**




**The TV 25.25 cm nearer the side wall, the dining wall on the breast**




- Moving the TV 25.25 cm left joins the two 25.25 cm side gaps into one 50.5 cm zone: wide enough for a 45.2 cm turntable, with 2.65 cm each side.
- The Series X moves up onto a shelf over the deck, above the lid's swing, with 15 cm of air each side. The records sit in bay 3, right under the deck.
- That frees the chimney breast for a fold-down dining table; the sub, which is wireless, moves to the floor by the side wall.

#### Why this layout, and what we ruled out

- **The TV in the middle:** left two 25.25 cm gaps either side, too narrow for a turntable, so the vinyl had to move out to the chimney breast. Moving the TV joins them into one 50.5 cm zone.
- **The turntable with the Series X beside it:** 45.2 + 20.5 cm won't go in 50.5. The Series X goes up on a shelf instead, above the lid's swing, where it gets more air than it had in v10 (15 cm each side, 21 above).
- **The vinyl cabinet on the breast as well as a table:** a table needs the breast's full width, and a cabinet in front would put the diners 44 cm further into the room.
- **A gateleg table in the recess:** it would block the door; the breast is the only clear wall.

### Options we rejected along the way

| Option | Why not |
|---|---|
| Slatted oak cladding: cable spine, PC screen, turntable panel, sub box, reeded table | You asked for less cladding: the chased conduit and plain oak do the same jobs more quietly |

| | |
|---|---|
| The slat wall alternative (v7–v12) | Cladding over the whole alcove; the oak uprights won |
| The vinyl cabinet on the chimney breast | The breast is the only clear wall for a table; the turntable fits in the alcove now |
| Hub as router, Decos as access points | Loses HomeShield and parental controls |
| All-wireless mesh · powerline | Speed drops at each hop · slower and less stable than Cat6 |
| Records spine-up in one drawer | Showed only a handful of covers; face-out trays show every one |
| A green feature wall only · white skirting | The drench ties the room together; white cuts a bright line round it |
| Bluetooth from the turntable | Compresses the sound; the AUX lead keeps full quality |
| Lifting the table off onto eight legs, with a 60 cm leaf that stores the legs (v13.1–v14) | You wanted it to stay on the wall and seat more: the flip-top's half and gate leg do both |
| Folding legs fixed under the table · a box for the legs under the dining wall · the plinth drawer | They'd show on the folded panel and stand it 5 cm off the wall · it sits where the chairs push back with six · too short for 71 cm legs. The leaf carries them instead. |
| An electrician's visit for the socket, dimmers and fittings | You're doing them yourselves: like-for-like, not notifiable; a new socket behind the sofa and a fused spur are dropped |